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CHAPTER I

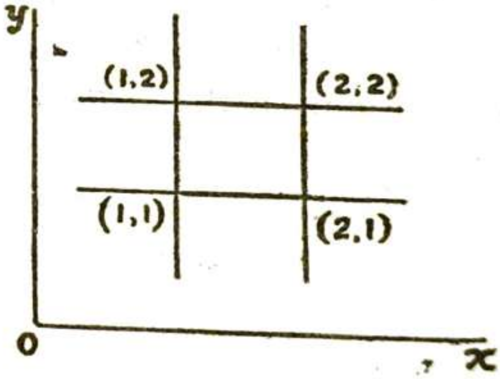
LIMITS AND CONTINUITY

1.1. Functions of Several Variables.

If an expression containing the independent variables $(x_1, x_2 \cdots x_n \cdots)$ be denoted by z and for every set of values of $(x_1, x_2 \cdots x_n \cdots)$ defined in a certain domain R under certain rules, the value of z exists as a real value, then the expression z is called a function of several variables $(x_1, x_2, \cdots x_n \cdots)$ defined in the domain R and we write $z=f(x_1, x_2, \cdots x_n \cdots)$ where z is the dependent variable. It is called a function of two variables, three varibles or more according as the independent variables are two, three or more.

1.2. Domain of the function of two variables.

Let $f(x, y)= \sqrt{(x-1)(2-x)(y-1)(2-y)}$ where the radicands are not negative. Here the two independent variables x and y can assume arbitrary chosen values independent of each other within the ranges $1 \leqslant x \leqslant 2$ and $1 \leqslant y \leqslant 2$.



Geometrically, if we draw a square with vertices at (1, 1), (2, 1), (2, 2) and (1, 2), then the arbitrary set of values of x and y must be chosen from the border or from any point within this square so as to make $f(x, y)$ a single valued function of two variables.

The aggregate of the pair of numbers (x, y) on or within this square is called the domain or region of definition of the function $f(x, y)$ and represented by 'R'. The domain or region may, however, be a rectangular, a square, a circle or any other finite figure or the whole surface according to the nature of the functions of the two independent variables x and y.

1.3. Domain of function of three variables.

Let $u = \sqrt{(x-1)(2-x)(y-1)(2-y)(z-1)(2-z)}$ where the radicands are not negative. Here the independent variables x, y, z can assume any value within the closed intervals $1 \leqslant x \leqslant 2$; $1 \leqslant y \leqslant 2$ and $1 \leqslant z \leqslant 2$, so as to make u a real valued function of x, y, z.

Geometrically, the ranges of x, y, z can be represented by a cube in three dimensional space. So long x, y, z assume the arbitrary values lying on the surface or within this cube, the value of u has a meaning and we call it a function of x, y, z of three variables defined in a domain R which is a cube here. The domain, however, may be of any solid shape and may be closed or open according to the nature of the function.

1.4. Domain of function of more than three variables.

Let $u = \sqrt{(x_1-1)(2-x_1)(x_2-1)(2-x_2)\cdots(x_n-1)(2-x_n)}$ where the radicands are not negative. Here the ranges of the independent variables $x_1, x_2 \cdots x_n$ are $1 \leqslant x_1 \leqslant 2$; $1 \leqslant x_2 \leqslant 2$ $\cdots 1 \leqslant x_n \leqslant 2$ and for every set of values of $x_1, x_2 \cdots x_n$ within these ranges, the expression u has got some meaning and we call it a function of $(x_1, x_2, \cdots x_n)$ i.e., $u = f(x_1, x_2, \cdots x_n)$.

Geometrically, though we cannot represent these ranges on a solid figure of any shape, yet we agree to say the system of n ordered variables $(x_1, x_2, \cdots x_n)$ correspond to a point in the n dimensional space. The aggregate of all such point in space constitute the domain R_n of the n dimensional space and we say $u = f(x_1, x_2, \cdots x_n)$ over the domain R_n.

1.5. Closed or open domain.

Let $f(x, y) = \sqrt{1-(x^2+y^2)}$ where the radicands are not negative. Here for values of (x, y) on or inside the circle

$x^2+y^2=1$, the function $f(x, y)$ is a single valued function of x and y. So the domain is $x^2+y^2 \leqslant 1$ and therefore closed.

Again if $f(x, y)=\dfrac{1}{\sqrt{1-(x^2+y^2)}}$ where the radicands are not negative then for all arbitrary set of values of (x, y) inside the circle $x^2+y^2=1$, the given function is a singled valued function of x, y. So here the domain is $x^2+y^2<1$ and so open.

1.6. Neighbourhood of a point.

Let $f(x, y)$ be a function of two variables x and y defined in the domain

$$R \equiv (\alpha < x < \beta,\ \gamma < y < \eta)$$

If (a, b) be a point within this domain, then $R' \equiv (a-\delta<x<a+\delta,\ b-\delta<y<b+\delta)$ which lies entirely in R is called the neighbourhood of the point (a, b). The domain R may be rectangular but the neighbourhood R' is usually taken as a square. Since $a-\delta<x<a+\delta$ is the same as $0<|x-a|<\delta$, the neighbourhood of (a, b) defines the square domain $0<|x-a|<\delta$, $0<|y-b|<\delta$.

Similarly, if $f(x, y)$ be a function of x, y defined in a circular domain R of radius δ and if (a, b) be a point within R then $0<(x-a)^2+(y-b)^2<\delta^2$ defines a circular neighbourhood of (a, b).

1.7. Double or Simultaneous limits of the functions of two variables. Analytical definition :—

A function $f(x, y)$ defined in a certain domain R is said to have a limit l as (x, y) tends to (a, b) which lie entirely in R, when any positive number ϵ having been chosen, as small as we please, there is a positive number δ such that

$$|f(x, y)-l|<\epsilon$$

for all values of (x, y) for which

$$|x-a| \leqslant \delta,\ |y-b| \leqslant \delta$$

and $$0<|x-a|+|y-b|$$

Thus, it follows that $| f(x, y)-l |$ must be less than ϵ for all points in the square, centre at (a, b) whose sides are parallel to the co-ordinate axes and length of whose side is 2δ, the centre of the square being excluded from the domain.

The square domain may, however, be replaced by a circle with centre at (a, b). The definition of the limit would then be as follows :

A function $f(x, y)$ defined in a certain domain R is said to have a limit l as (x, y) tends to (a, b) which lie entirely in R, if to the arbitrary positive numher ϵ however small, there corresponds a positive number δ depending on ϵ such that

$$| f(x, y)-l | <\epsilon$$

for all values of (x, y) for which

$$0< \sqrt{(x-a)^2+(y-b)^2} \leqslant \delta.$$

Note : The limit l is also called the Double limit or Simultaneous limit of $f(x, y)$ as (x, y) tends to (a, b) and is usually written as

$$\underset{(x, y)\to(a, b)}{Lt} f(x, y) \quad \text{or} \quad \underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)=l.$$

Ex. 1. Verify $\underset{\substack{x\to 0\\ y\to 0}}{\mathbf{Lt}} \mathbf{f(x, y)=0}$

where $\mathbf{f(x, y)=x \sin \frac{1}{x}+y \sin \frac{1}{y}, x>0, y>0,}$

for $\mathbf{0<x<\epsilon/2}$ **and** $\mathbf{0<y<\epsilon/2}$

$$\left| f(x, y)-0 \right|=\left| x \sin \frac{1}{x}+y \sin \frac{1}{y} \right|$$

$$<\epsilon/_2 \left| \sin \frac{2}{\epsilon}+\sin \frac{2}{\epsilon} \right|$$

$$i.e., \quad <\epsilon \left| \sin \frac{2}{\epsilon} \right|$$

$$i.e., \quad <\epsilon \qquad \because \left| \sin \frac{2}{\epsilon} \right| \leqslant 1.$$

Thus $| f(x, y)-0 | <\epsilon$ for all points (x, y) within the square whose sides are along the coordinate axes and of side $\epsilon/_2$.

Hence $\underset{\substack{x\to 0\\ y\to 0}}{Lt} f(x, y)=0.$

Ex. 2. Verify $\underset{\substack{x\to 0\\ y\to 0}}{\mathbf{Lt}} \mathbf{f(x, y)=0}$

where $\mathbf{f(x, y)=(x+y) \sin \frac{1}{x} \sin \frac{1}{y}, x>0, y>0}$

for $\mathbf{0<x<\epsilon/_2}$ **and** $\mathbf{0<y<\epsilon/_2}$

$$\left| f(x, y)-0 \right| <\epsilon \left| \sin \frac{2}{\epsilon} \sin \frac{2}{\epsilon} \right|$$

$$i.e., \quad <\epsilon \left| \sin \frac{2}{\epsilon} \right| \left| \sin \frac{2}{\epsilon} \right|$$

$$i.e., \quad <\epsilon \qquad \because \left| \sin \frac{2}{\epsilon} \right| \leqslant 1$$

Thus $| f(x, y)-0 | <\epsilon$ for all points (x, y) within the square whose sides are along the co-ordinate axes of side equal to $\epsilon/2$.

Hence $\underset{\substack{x\to 0\\ y\to 0}}{Lt} f(x, y)=0.$

1.8. Analytical definition of the limit of a function of three variables.

A function $f(x, y, z)$ defined in a certain three dimensional domain R is said to tend to the limit l as (x, y, z) tends to (a, b, c) which lie entirely within R, if given any pre-assigned positive number ϵ, however small, we can find a positive number δ depending on ϵ such that

$$| f(x, y, z)-l | <\epsilon$$

for all values of (x, y, z) for which

$$| x-a | \leqslant\delta, | y-b | \leqslant\delta, | z-c | \leqslant\delta.$$

and $0< | x-a | + | y-b | + | z-c |$

In other words, $| f x, y, z)-l | < \epsilon$ for all points in the cube, centre at (a, b, c) whose sides are parallel to the co-ordinate axes and of length 2δ. The centre of the square being excluded from the domain.

The cube may, however, be replaced by a sphere and the definition of the limit would then be as follows :

A function $f(x, y, z)$ is said to tend to the limit l as (x, y, z) tends to (a, b, c) if given any pre-assigned positive number ϵ however small, we can find a positive number δ depending on ϵ such that

$$| f(x, y, z)-l | < \epsilon$$

for all values of (x, y, z) for which

$$0< \sqrt{(x-a)^2+(y-b)^2+(z-c)^2} \leqslant \delta.$$

Ex. 1. Verify $\underset{(x, y, z)\to(0, 0, 0)}{\text{Lt}} f(x, y, z)=0$

where $f(x, y, z)=x \sin \frac{1}{x}+y \sin \frac{1}{y}+z \sin \frac{1}{z}$.

For $0<x<\epsilon/3, 0<y<\epsilon/3, 0<z<\epsilon/3$

$$| f(x, y, z)-0 | < \left|\frac{\epsilon}{3} \sin \frac{3}{\epsilon}+\frac{\epsilon}{3} \sin \frac{3}{\epsilon}+\frac{\epsilon}{3} \sin \frac{3}{\epsilon}\right|$$

$$\text{i e.} < \epsilon \quad \sin \frac{3}{\epsilon}\Big|$$

$$\text{i.e.} < \epsilon \qquad \because \left|\sin \frac{3}{\epsilon}\right| \leqslant 1$$

Thus $| f(x, y, z)-0 | < \epsilon$ for all value of (x, y, z) lying within a cube whose sides lie along the co-ordinate axes of side $\epsilon/3$.

Hence $\underset{(x, y, z)\to(0, 0, 0)}{Lt} f(x, y, z)=0.$

1.9. Working rule for Evolution of limit of the function of several variables.

Suppose we are to evaluate

$$\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y$$

If a function $y=\phi(x)$ exists such that $\phi(x)\to b$ as $x\to a$

then $$\underset{\substack{x\to a\\ y\to b}}{Lt} f(x,y)=\underset{x\to a}{Lt} f\{x, \phi(x)\}$$

If now $\underset{x\to a}{Lt} f\{x, \phi(x)\}$ exists and $=l$, then the double limit is said to exist and we write

$$\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)=l.$$

On the other hand, if two functions $\phi_1(x)$ and $\phi_2(x)$ exist such that each one $\to b$ as $x\to a$ such that

$$\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)=\underset{x\to a}{Lt} f\{x, \phi_1(x)\}=l \text{ (say)}$$

$$\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)=\underset{x\to a}{Lt} f\{x, \phi_2(x)\}=m \text{ (say)}$$

If $l\neq m$, then the double limit is said to be non-existent.

Ex. 1. Evaluate $\underset{(x, y)\to(0, 0)}{Lt} \dfrac{2y^2}{\sqrt{x^2+xy}}$.

Let $y=mx$. then as $x\to 0$, $y\to 0$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} \frac{2y^2}{\sqrt{x^2+xy}}=\underset{x\to 0}{Lt} \frac{2m^2x^2}{\sqrt{x^2+mx^2}}$$

$$=\underset{x\to 0}{Lt} \frac{2m^2x}{\sqrt{1+m}}$$

$$=0 \quad \because m \text{ is finite.}$$

Hence $$\underset{(x, y)\to(0, 0)}{Lt} \frac{2y^2}{\sqrt{x^2+xy}}=0.$$

Ex. 2. Evaluate $\underset{(x, y)\to(0, 0)}{Lt} \dfrac{x^2+y^2}{x^2+xy+y^2}$.

Let $y=mx$ then as $x\to 0$, $y\to 0$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} \frac{x^2+y^2}{x^2+xy+y^2}=\underset{x\to 0}{Lt} \frac{x^2+m^2x^2}{x^2+mx^2+m^2x^2}$$

$=\underset{x\to 0}{Lt} \dfrac{1+m^2}{1+m+m^2}=\dfrac{1+m^2}{1+m+m^2}$ which is different for different finite values of m.

Hence $\underset{(x, y)\to(0, 0)}{Lt} \dfrac{x^2+y^2}{x^2+xy+y^2}$ does not exist.

1.10. Repeated Limits.

Suppose a function $u=f(x, y)$ is defined in a certain neighbourhood of (a, b) and we are to evaluate

$$\underset{x\to a,\, y\to b}{Lt} f(x, y) \quad \text{or,} \quad \underset{y\to b,\, x\to a}{Lt} f(x, y).$$

1. If $\underset{y\to b}{Lt} f(x, y)$ exist and equal to a function of x, say, $\phi(x)$, then,

$$\underset{x\to a,\, y\to b}{Lt} f(x, y)=\underset{x\to a}{Lt} \phi(x).$$

If now $\underset{x\to a}{Lt} \phi(x)$ also exists and equal to l, a finite or infinite value with a definite sign then, we say that the repeated limit of $f(x, y)$ at (a, b) is l and we write

$$\underset{x\to a,\, y\to b}{Lt} f(x, y)=l$$

where the limit for $y\to b$ is taken first and afterward for $x\to a$.

2. If $\underset{x\to a}{Lt} f(x, y)$ exists and equal to a function of y, say, $\phi(y)$, then

$$\underset{y\to b,\, x\to a}{Lt} f(x, y)=\underset{y\to b}{Lt} \phi(y).$$

If now $\underset{y\to b}{Lt} \phi(y)$ exists and equal to m, a finite or infinite value with a definite sign, then also we say that the repeated limit of $f(x, y)$ at (a, b) is m and we write

$$\underset{y\to b,\, x\to a}{Lt} f(x, y)=m$$

where the limit for $x\to a$ is taken first and afterward for $y\to b$

Note 1. If the double limit $\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)$ exists and equal to l, a finite or infinite value with a definite sign, then

$$\underset{x\to a,\ y\to b}{Lt} f(x, y)=\underset{y\to b,\ x\to a}{Lt} f(x, y)=l$$

But the converse of the above statement is not always true. Thus $\underset{\substack{x\to a\\ y\to b}}{Lt} f(x, y)$ may exist or not although the two repeated limits exist with definite values and equal.

Ex. 1. Let $f(x, y)=x+y+\dfrac{xy}{\sqrt{x^2+y^2}}$, $x>0, y>0$ then discuss the existence of double limits and repeated limits of the function at (0, 0).

Let $y=mx$, then as $x\to 0$. $y\to 0$.

$$\therefore \text{ Double } Lt=\underset{\substack{x\to 0\\ y\to 0}}{Lt} f(x, y)$$

$$=\underset{x\to 0}{Lt}\left(x+mx+\frac{mx^2}{\sqrt{x^2+m^2x^2}}\right)$$

$$=\underset{x\to 0}{Lt}\left\{1+m+\frac{m}{\sqrt{1+m^2}}\right\}x=0$$

Again $\underset{x\to 0,\ y\to 0}{Lt} f(x, y)=\underset{x\to 0}{Lt}(x)=0$

and $\underset{y\to 0,\ x\to 0}{Lt} f(x, y)=\underset{y\to 0}{Lt}(y)=0$

So double limit of $f(x, y)$ is equal to its repeated limit at (0, 0).

Ex. 2. Let $f(x, y)=\dfrac{x^2y^2}{x^2y^2+(x-y)^4}$; $x>0, y>0$. then discuss the existence of the repeated and double limits at (0, 0).

$$\underset{x\to 0,\ y\to 0}{Lt} f(x, y)=\frac{0}{0+(x-0)^4}=\underset{x\to 0}{Lt}(0)=0$$

$$\underset{y\to 0,\ x\to 0}{Lt} f(x, y)=\underset{y\to 0}{Lt}\frac{0}{0+(0-y)^4}=\underset{y\to 0}{Lt}(0)=0$$

So we find that the repeated limits exist and $=0$

For double limit let $y=mx$, then

Double $\displaystyle Lt = \underset{x \to 0}{Lt} \frac{x^2m^2x^2}{x^2m^2x^2+(x-mx)^4} = \frac{m^2}{m^2+(1+m)^4}$

which is different for different values of m.

$\therefore$ Double limit does not exist.

Thus although the repeated limits exist and $=0$, the double limit does not exist.

1.11. Important Theorems on Limits

If $\underset{\substack{x \to a \\ y \to b}}{Lt} f(x, y) = l$ and $\underset{\substack{x \to a \\ y \to b}}{Lt} \phi(x, y) = m$

then (a) $\underset{\substack{x \to a \\ y \to b}}{Lt} \{f(x, y) \pm \phi(x, y)\} = l \pm m.$

(b) $\underset{\substack{x \to a \\ y \to b}}{Lt} \{f(x, y).\phi(x, y\} = lm.$

(c) $\underset{\substack{x \to a \\ y \to b}}{Lt} \left\{\dfrac{f(x, y)}{\phi(x, y)}\right\} = \dfrac{l}{m}$, provided $m \neq 0$

Proofs are exactly the same as in the case of the function of single variable.

1.12. Continuity of a function of two variables.

A function $f(x, y)$ is said to be continuous at a given point (a, b) of its domain of definition if

$$\underset{(x, y) \to (a, b)}{Lt} f(x, y) = f(a, b)$$

Also it is said to be continuous in a domain of definition if it is continuous at every point of the domain.

Analytically : A function $f(x, y)$ is said to be continuous at $x=a$ and $y=b$, if given any pre-assigned positive number ϵ however small, we can find a $\delta > 0$ such that

$$| f(x, y) - f(a, b) | < \epsilon$$

for all values of x, y for which

$$| x-a | \leq \delta \text{ and } | y-b | \leq \delta$$

In other words, $| f(x, y)-f(a, b) | < \epsilon$ for all points in the square, centre at (a, b) whose sides are parallel to the coordinate axes and of length 2δ.

If $f(x, y)$ is continuous at every point of its domain then it is said to be continuous in the domain.

The square domain may, however, be replaced by a circle and the definition would then be as follows :

A function $f(x, y)$ is said to be continuous at (a, b) if to the arbitrary positive number ϵ however small, there corresponds a positive number δ such that

$$| f(x, y)-f(a, b) | < \epsilon$$

for all values of (x, y) for which

$$(x-a)^2+(y-b)^2 \leqslant \delta^2.$$

1.13. If f(x, y) is a continuous function of two variables at (a, b) then it will be continuous at (a, k) and (k, b) where k is a constant which may be given either to x or y. But the converse of this is not necessarily true.

Consider $f(x, y)=x \sin \frac{1}{x}+y \sin \frac{1}{y}$ where x and $y \neq 0$

$f(x, 0)=x \sin \frac{1}{x}$ when $x \neq 0$; $f(0, y)=y \sin \frac{1}{y}$ when $y \neq 0$ and $f(0, 0)=0$.

Here for $0<x<\epsilon/_2$ and $0<y<\epsilon/_2$

$$| f(x, y)-f(0, 0) | = | x \sin \frac{1}{x}+y \sin \frac{1}{y}-0 |$$

$$< \left| \frac{\epsilon}{2} \sin \frac{2}{\epsilon}+\frac{\epsilon}{2} \sin \frac{2}{\epsilon} \right|$$

$$\text{i.e. } < \epsilon \left| \sin \frac{2}{\epsilon} \right|$$

$$\text{i.e. } < \epsilon \quad \because \left| \sin \frac{2}{\epsilon} \right| \leqslant 1$$

So we see that $| f(x, y)-f(0, 0) | < \epsilon$ for every pair of values of x and y within the square whose sides lie along the co-ordinate axes and whose side is of length $\epsilon/_2$

$\therefore$ $f(x, y)$ is continuous at $(0, 0)$.

$$f(x, k)=x \sin \frac{1}{x}+k \sin\frac{1}{k}$$

$$\therefore \underset{x\to 0}{Lt} f(x, k)=\underset{x\to 0}{Lt} x \sin \frac{1}{x}+k \sin\frac{1}{k}=0+k \sin \frac{1}{k}$$

$$=k \sin \frac{1}{k}$$

Also $\quad f(0, k)=k \sin \dfrac{1}{k}$

$$\therefore \underset{x\to 0}{Lt} f(x, k)=f(0, k)$$

Hence $f(x, y)$ is continious for all values of x including $x=0$. Similarly, it can be shown that $f(x, y)$ is continuous for all values of y including $y=0$

For converse, consider the function

$$f(x, y)=\frac{2xy}{x^2+y^2} \text{ when } x\neq 0,\ y\neq 0$$

$$=0 \text{ when either } x \text{ or } y \text{ or both}=0.$$

Here $f(x, k)=\dfrac{2xk}{x^2+k^2}$

$$\therefore \underset{x\to 0}{Lt} f(x, k)=0$$

and $f(0, k)=0$

$$\therefore \underset{x\to 0}{Lt} f(x, k)=f(0, k)$$

$\therefore$ $f(x, k)$ is continuous at $x=0$. So $f(x, y)$ is continuous for all values of x including $x=0$.

Also $f(k, y)=\dfrac{2ky}{k^2+y^2}$

$$\therefore \underset{y\to 0}{Lt} f(k, y)=0$$

and $\quad f(k, 0)=0$

$$\therefore \underset{y\to 0}{Lt} f(k, y)=f(k, 0)$$

$\therefore$ $f(k, y)$ is continuous at $y=0$. So $f(x, y)$ is continuous for all values of y including $y=0$.

Again consider the continuity of f(x. y) at (0, 0).

Let $y=mx$, then as $x\to 0$, $y\to 0$

$\therefore \underset{\substack{x\to 0\\ y\to 0}}{Lt} f(x, y)=\underset{x\to 0}{Lt}\dfrac{2x.\,mx}{x^2+m^2x^2}=\dfrac{2m}{1+m^2}$ which is different for different values of m. So the limit does not exist.

Hence $f(x, y)$ is not continuous at (0, 0). However, $f(x, y)$ is continuous in any domain excepting the point (0, 0).

1.14. (a, b) is a point in the region R in which f(x, y) is continuous. If f(a, b)$\neq$0, then f(x, y) will have the same sign as f(a, b) in some neighbourhood of (a, b)

(C. H. 1963)

Proof : Since $f(x, y)$ is continuous in (a, b), corresponding to any pre-assigned positive number ϵ, however small, there exists a positive number δ, such that

$$|f(x, y)-f(a, b)|<\epsilon \text{ whenever } 0<|x-a|\leqslant\delta \text{ and } 0<|y-b|\leqslant\delta.$$

In other words,

$$f(a, b)-\epsilon<f(x, y)<f(a, b)+\epsilon\cdots\cdots\cdots(i)$$
$$\text{whenever } 0<|x-a|\leqslant\delta \text{ and } 0<|y-b|\leqslant\delta.$$

Case I. Let $f(a, b)>0$

If we now choose ϵ such that $\epsilon<f(a, b)$ then $f(a, b)-\epsilon$ is positive.

Also $f(a, b)+\epsilon$ is always positive.

So from (i) we find that for every point (x, y) in the square domain $R'(a-\delta\leqslant x\leqslant a+\delta\,;\ \ b-\delta\leqslant y\leqslant b+\delta)$, the function $f(x, y)$ lies between two positive quantities and so itself positive.

$\therefore$ $f(x, y)$ has the same sign as $f(a, b)$

Case II Let $f(a, b)<0$

Then $-f(a, b)>0$

If we now choose $\epsilon<-f(a, b)$

Then $f(a, b)+\epsilon<0$

Also $f(a, b)-\epsilon<0 \quad \because f(a, b)<0$

So from (i) we find that for every point (x, y) in the square domain R' the function $f(x, y)$ lies between two negative quantities and so itself negative.

$\therefore f(x, y)$ has the same sign as $f(a, b)$. Hence the theorem.

1.15. Some Important theorems on continuity of functions of two variables, proofs of which are exactly the same as in the case of a single variable.

1. The sum, difference and product of two functions each continuous at a given point or in a given domain is continuous at the point or in the domain.

Thus if $f(x, y)$ and $\phi(x, y)$ are both continuous at (x_0, y_0) then $f(x, y)\pm\phi(x, y)$ and $f(x, y)\,\phi(x, y)$ are also continuous at (x_0, y_0).

Also if $f(x, y)$ and $\phi(x, y)$ are both continuous in a domain R, then $f(x, y) \pm \phi(x, y)$ and $f(x, y)\,\phi(x, y)$ are also continuous in the domain R.

2. The quotient of two functions both of which are continuous at a point or in a domain is continuous at the point or in the domain except at points where the denominators vanish.

Thus if $f(x, y)$ and $\phi(x, y)$ are both continuous at (x_0, y_0) or in a given domain R and $\phi(x, y)\neq 0$ at (x_0, y_0) or any point in R, then $f(x, y)/\phi(x, y)$ is continuous at (x_0, y_0) or in R except at points in R where $\phi(x, y)$ vanish.

3. A continuous function of one or more continuous function is a continuous function.

In particular let $u=\phi(x, y)$, $v=\psi(x, y)$ be continuous at (x_0, y_0) and let $u_0=\phi(x_0, y_0)$ and $v_0=\psi(x_0, y_0)$.

If now $z=f(u, v)$ be continuous in (u, v) at (u_0, v_0) then

$z=f\{\phi(x, y), \psi(x, y)\}$ is also continuous in x, y at (x_0, y_0).

1.16. Properties of a continuous function of two variables.

(*a*) A function continuous in a closed domain is bounded and attains its bounds at least once in the domain.

(*b*) If $f(x, y)$ is continuous in a closed domain then it must assume at least once every value between its upper and lower bounds.

(*c*) If a function of two variables is continuous at every-point of a closed domain, it is uniformly continuous in the domain.

In other words, corresponding to any pre-assigned positive number ϵ however small, there exists a positive number δ such that

$$| f(x', y') - f(x'', y'') | < \epsilon$$

where (x', y') and (x'', y'') are any two points in the domain for which $\sqrt{(x'-x'')^2+(y'-y'')^2} \leqslant \delta$.

Illustrative Examples :

Ex. 1. If $f(x, y)=\dfrac{2xy}{x^2+y^2}$, $x^2+y^2 \neq 0$ and $f(0, 0)=0$ **show that f(x, y) is not continuous at (0, 0)** (C. H. 1962)

Let $y=mx$. Then $x \to 0$ as $y \to 0$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y) = \underset{x\to 0}{Lt} \frac{2mx^2}{x^2+m^2x^2} = \frac{2m}{1+m^2}$$ which is different for different values of m.

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y) \text{ does not exist}$$

but $f(0, 0)=$ exists which is zero

$\therefore$ Lt $f(x, y) \neq f(0, 0)$ at $(0, 0)$

Hence $f(x, y)$ is not continuous at $(0, 0)$

Ex. 2. If $f(x, y)=\dfrac{x^2}{x^2+y^2-x}$, $x^2+y^2 \neq 0$ **and** $f(0, 0)=0$, **examine whether f(x, y) is continuous at (0, 0)** (C. H. 1962)

Suppose we proceed along the curve $y^2-x=mx^2$. Then as $x \to 0$, $y \to 0$.

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y)=\underset{x\to 0}{Lt} \frac{x^2}{x^2+mx^2}=\frac{1}{1+m}$$ which is different for different values of m

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y) \text{ does not exist}$$

But $f(x, y)$ is defined at $(0, 0)$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y)\neq f(0, 0)$$

Hence $f(x, y)$ is not continuous at $(0, 0)$.

Ex. 3. Examine whether

$$f(x, y)=\begin{cases} xy & \text{if } |x| \geqslant |y| \\ -xy & \text{if } |x| < |y| \end{cases}$$

is continuous at the origin. (C. H. (old) 1963)

Let $y=mx$, then $y\to 0$ as $x\to 0$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y)$$
$$=\underset{x\to 0}{Lt}\, mx^2 \text{ if } |x| \geqslant |y|$$
$$=0$$

and $$\underset{(x, y)\to(0, 0)}{Lt} f(x, y)=\underset{x\to 0}{Lt} -(mx^2) \text{ if } |x|<|y|$$
$$=0$$

Also $f(0, 0)=0$

Thus $$\underset{(x, y)\to(0, 0)}{Lt} f(x, y)=f(0, 0)$$

Hence $f(x, y)$ is continuous at $(0, 0)$.

*****Ex. 4. Examine the continuity of the function**

$$f(x, y)=\frac{x^3+y^3}{x^2+y^2},\ x^2+y^2\neq 0$$ (C. H. 1961)

Let $y=mx$, then $y\to 0$ as $x\to 0$

$$\therefore \underset{(x, y)\to(0, 0)}{Lt} f(x, y)=\underset{x\to 0}{Lt} \frac{x^3+m^3x^3}{x^2+m^2x^2}$$
$$=\underset{x\to 0}{Lt} \frac{1+m^3}{1+m^2}\cdot x=0.$$

but $f(0, 0)$ is not defined in the question.

$\therefore \quad \underset{(x,\, y)\to(0,\, 0)}{Lt} f(x, y) \neq f(0, 0)$

$\therefore \quad f(x, y)$ is not continuous at $(0, 0)$

Again quotient of $\dfrac{x^3+y^3}{x^2+y^2}$ is a polynomial and so it is continuous for all values of x excepting at $x=0,\ y=0$.

Hence $f(x, y)$ is continuous everywhere excepting at $(0, 0)$.

Exercise 1

1. Show that $\underset{(x,\, y)\to(0,\, 0)}{Lt} \dfrac{x^3y}{x^4+y^4}$ does not exist.

2. Shew that $\underset{(x,\, y)\to(0,\, 0)}{Lt} \dfrac{(xy+y^2)+(x+y)^3}{y-(x+y)^2}$ does not exist.

3. Let $f(x, y)=\dfrac{x^2y^2}{x^2y^2+(x-y)^2}$, show that the two repeated limits of $f(x, y)$ exist and are equal.

Also show that $\underset{(x,\, y)\to(0,\, 0)}{Lt} f(x, y)$ does not exist.

4. Show that a functio..

$f(x, y)=xy\dfrac{x^2-y^2}{x^2+y^2}$, $f(x, y)\neq(0, 0)$

and $f(0, 0)=0$ is continuous at $(0, 0)$.

5. Let a function of two variables in x and y is defined as

$f(x, y)=e^{-|x-y|/(x^2-2xy+y^2)}$, when $(x, y)\neq(x, x)$

and $f(x, x)=0$, show that $f(x, y)$ is continuous at $(0, 0)$.

6. Show that a function defined as

$f(x, y)=\dfrac{x^3}{x^3+y^2-2x}$ when $(x, y)\neq(0, 0)$ and $f(0, 0)=0$

is not continuous at $(0, 0)$.

7. If $\underset{(x,\, y)\to(a,\, b)}{Lt} f(x, y)=l$, then show that

$$\underset{x\to a}{Lt} f(x, b)=\underset{y\to b}{Lt} f(a, y)=l.$$

8. Let $f(x, y)=(x+y)\sin\dfrac{1}{x}\sin\dfrac{1}{y}$, $x>0,\ y>0$.

Discuss the existence of the repeated and double limits at $(0, 0)$.

9. Show that the function $f(x, y)=y^2\sin\dfrac{1}{x}$, $x\neq0$

and $f(0, 0)=0$ is continuous at the origin.

CHAPTER II

PARTIAL DERIVATIVES

2.1. Let $z=f(x, y)$

Then $\underset{h\to 0}{Lt}\dfrac{f(a+h, b)-f(a, b)}{h}$ if exist is called the partial derivative of $f(x, y)$ with respect to x at (a, b) and it is denoted by $\left(\dfrac{\partial z}{\partial x}\right)_{(a, b)}$ or $f_x(a, b)$.

Thus $f_x(a, b)=\underset{h\to 0}{Lt}\dfrac{f(a+h, b)-f(a, b)}{h}$

If, however, $f(x, y)$ possesses partial derivative with respect to x, at every point of its domain of definition, then at any point (x, y) of the domain

$$f_x(x, y)=\underset{h\to 0}{Lt}\frac{f(x+h, y)-f(x, y)}{h}$$

Again $\dfrac{f(a, b+k)-f(a, b)}{k}$ if exist is called the partial derivative of $f(x, y)$ with respect to y at (a, b) and it is denoted by $\left(\dfrac{\partial z}{\partial y}\right)_{(a,b)}$ or $f_y(a, b)$.

Also if $f(x, y)$ possesses partial derivative with respect to y at every point of its domain of definition, then at any point (x, y) of its domain

$$f_y(x, y)=\underset{k\to 0}{Lt}\frac{f(x, y+k)-f(x, y)}{k}$$

Illustrations :

Ex. 1. If $f(x, y)=0$ when $x=0$ or $y=0$, and $f(x, y)=1$ when $x\neq 0$, $y\neq 0$, find f_x and f_y at $(0, 0)$.

$$f_x(0, 0)=\underset{h\to 0}{Lt}\frac{f(0+h, 0)-f(0, 0)}{h}=\underset{h\to 0}{Lt}\frac{0-0}{h}=0.$$

$$f_y(0, 0)=\underset{k\to 0}{Lt}\frac{f(0, 0+k)-f(0, 0)}{k}=\underset{k\to 0}{Lt}\frac{0-0}{k}=0.$$

Hence both f_x and f_y exist at $(0, 0)$ and $=0$.

2.2. Partial Derivatives and Continuity.

Let $z=f(x, y)$

Then the partial derivatives of z with respect to x and y at (a, b) are respectively

$$\left(\frac{\partial z}{\partial x}\right)_{(a,b)}=\underset{h\to 0}{Lt}\frac{f(a+h, b)-f(a, b)}{h}$$

$$\text{and } \left(\frac{\partial z}{\partial y}\right)_{(a,b)}=\underset{k\to 0}{Lt}\frac{f(a, b+k)-f(a, b)}{k}$$

Thus in forming partial derivatives we vary x along the line $x=a$ in the first case and y along the line $y=b$ in the second case. So $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ exist at (a, b) only when the function $f(x, y)$ exist for points on the lines $x=a$ and $y=b$. For a point outside these two lines plays no roll on partial derivatives. But for continuity of $f(x, y)$ at (a, b) we have to consider the existence and values of the function not only on the lines $x=a$ and $x=b$ but at every point on the neighbourhood of (a, b). *Thus a function may possess partial derivatives without being continuous at the point.* If, however, the partial derivatives are bounded in the domain of definition of the function, then the function must be continuous in that domain.

Theorem :

If a function f(x, y) defined in a domain R have partial derivatives f_x and f_y in R and further if they are bounded in R, then f(x, y) is continuous in R. **(C. H. 1962)**

Let us write

$$f(x+h, y+k)-f(x, y)=\{f(x+h, y+k)-f(x, y+k)\}$$
$$+\{f(x, y+k)-f(x, y)\} \quad \cdots \quad (1)$$

Since f_x and f_y are bounded in R, they are finite in R
$\therefore$ $f(x, y)$ is derivable in R with respect to x and y.
Hence applying mean value theorem in (1)

$$f(x+h, y+k)-f(x, y)=hf_x(x+\theta h, y+k)$$
$$+hf_y(x, y+\theta' k) \quad \cdots \quad (2)$$

$$\text{for } 0<\theta<1$$
$$\text{and } 0<\theta'<1.$$

Again if M be the greatest of the upper bounds of f_x and f_y

Then we must have

$$|f_x| \leqslant M \text{ and } |f_y| \leqslant M$$

So from (2)

$$|f(x+h, y+k)-f(x, y)| \leqslant |hf_x(x+\theta h, y+k)| + |hf_y(x, y+\theta' k)|$$

i.e., $\leqslant M\{|h|+|k|\}$

i.e., $< M.\dfrac{\epsilon}{M}$

$\because$ we can choose $|h|+|k|<\dfrac{\epsilon}{M}$

i.e, $<\epsilon$

So it follows that $f(x, y)$ is continuous at (x, y) in the domain R.

Cor. If f_x and f_y are continuous at (x, y) in R, then f(x, y) is continuous in R.

Since f_x and f_y are continuous at (x, y) in R, they must be bounded in R and so $f(x, y)$ is continuous in R (by the last theorem).

Illustrations :

Ex. 1. Let $f(x, y)=\dfrac{xy}{x^2+y^2}$, $(x, y)\neq 0$

$=0$, otherwise

Prove that $f_x(0, 0)$ and $f_y(0, 0)$ both exist but $f(x, y)$ is discontinuous at (0, 0). **(C. H. 1966)**

$$f_x(0, 0)=\underset{h\to 0}{Lt}\ \frac{f(h, 0)-f(0, 0)}{h}$$

$$=\underset{h\to 0}{Lt}\ \frac{0-0}{h}=0.$$

$$f_y(0, 0) = \underset{k \to 0}{Lt} \frac{f(0, k) - f(0, 0)}{k}$$

$$= \underset{k \to 0}{Lt} \frac{0 - 0}{k} = 0.$$

Thus both f_x and f_y exist at $(0, 0)$ and their value $= 0$.

Again let us suppose that (x, y) approaches $(0, 0)$ along the line $y = mx$.

Then $\underset{(x, y) \to (0, 0)}{Lt} f(x, y) = \underset{(x, y) \to (0, 0)}{Lt} \frac{mx^2}{x^2 + m^2 x^2}$

$= \frac{m}{1 + m^2}$ which is different for different values of m.

Hence $f(x, y)$ is discontinuous at $(0, 0)$.

Ex. 2. If $f(x, y) = 0$ when either $x = 0$ or $y = 0$ and $f(x, y) = 1$ when $x \neq 0$, $y \neq 0$.

Show that $f(x, y)$ is discontinuous at $(0, 0)$ but the partial derivatives exist at $(0, 0)$.

Since $| f(x, y) - f(0, 0) | = | 1 - 0 | = 1 \nless \epsilon$, $f(x, y)$ is not continuous at $(0, 0)$.

Again $f_x(0, 0) = \underset{h \to 0}{Lt} \frac{f(0 + h, 0) - f(0, 0)}{h} = \underset{h \to 0}{Lt} \left(\frac{0}{h}\right) = 0$

$$f_y(0, 0) = \underset{k \to 0}{Lt} \frac{f(0, 0 + k) - f(0, 0)}{k} = \underset{k \to 0}{Lt} \left(\frac{0}{k}\right) = 0$$

Thus partial derivatives of $f(x, y)$ with respect to x and y exist at $(0, 0)$ although $f(x, y)$ is discontinuous at $(0, 0)$.

Ex. 3. Prove that the function $f(x, y)$ defined by

$f(x, y) = \frac{2xy}{x^2 + y^2}$, $x^2 + y^2 \neq 0$ and $f(0, 0) = 0$ has partial derivatives everywhere. Is it continuous every where ?

(C. H. 1982)

$$f_x = \underset{h \to 0}{Lt} \frac{f(x + h, y) - f(x, y)}{h}$$

$$= \underset{h \to 0}{Lt} \frac{1}{h} \left[\frac{2(x + h)y}{(x + h)^2 + y^2} - \frac{2xy}{x^2 + y^2} \right]$$

$$=\underset{h\to 0}{Lt}\ \frac{2}{h}\ \frac{\{(xy+hy)(x^2+y^2)-xy(x^2+y^2+h^2+2xh)\}}{(x^2+y^2)(x^2+y^2+h^2+2xh)}$$

$$=\underset{h\to 0}{Lt}\ \frac{2(y^3-x^2y-xyh)}{(x^2+y^2)(x^2+y^2+h^2+2xh)}=\frac{2y(y^2-x^2)}{(x^2+y^2)^2}$$

which exists for all values of x other than $x=0,\ y=0.$

Similarly, it can be shown that f_y exists for all values of y other than $x=0,\ y=0.$

$$\text{Again } f_x(0, 0)=\underset{h\to 0}{Lt}\ \frac{f(h, 0)-f(0, 0)}{h}$$

$$=\underset{h\to 0}{Lt}\ \frac{0-0}{h}=0.$$

Similarly, $f_y(0, 0)=0.$

Thus we find that $f(x, y)$ has derivatives every where.

For continuity of the function see Art 1·13, page 12.

2.3. Differentiability of a function f(x, y).

A function $u=f(x, y)$ is said to be differentiable at a point (x, y) of its domain of definition, if the increment δu of u, corresponding to any arbitrary assigned increment δx and δy of x and y can be expressed in the form

$$\delta u=f(x+h, y+k)-f(x, y)$$

$$=Ah+Bk+h\phi(h, k)+k\psi(h, k) \qquad \dots \quad \dots \quad (1)$$

where A and B are constant independent of h and k; $\phi(h, k)$ and $\psi(h, k)$ are functions of (h, k) such that

$$\underset{(h, k)\to(0, 0)}{Lt}\ \phi(h, k)=0$$

$$\text{and}\quad \underset{(h, k)\to(0, 0)}{Lt}\ \psi(h, k)=0.$$

The part of δu in (1) which is linear in h and k *i.e.*, the part $Ah+Bk$ is called the total differential of u and is denoted by du.

So we write $du=Ah+Bk$

$$=A\delta x+B\delta y \qquad \dots \quad \dots \quad (2)$$

$\because$ h and k are increment or $x+y$

$$h=\delta x,\ k=\delta y$$

Now From (1)

$$\delta u=A\delta x+B\delta y+\delta x\phi(\delta x, \delta y)+\delta y\psi(\delta x\ \delta y) \ \cdots \quad \cdots \quad (3)$$

Suppose we now regard y as fixed and x as variable, then $\delta y = 0$

$\therefore$ from (3)

$$\delta u = A\delta x + \delta x.\phi(\delta x, 0)$$

or, $\dfrac{\delta u}{\delta x} = A + \phi(\delta x, 0).$

or. $\underset{\delta x \to 0}{Lt} \dfrac{\delta u}{\delta x} = A$

or, $\dfrac{\partial u}{\partial x} = A \quad \because \; y$ is regarded as a constant.

Similarly, if we regard x as fixed and y as variable, then $\delta x = 0$ and we shall get

$\dfrac{\partial u}{\partial y} = B \quad \because \; x$ is regarded as a constant.

Hence from (2) we get

$$du = \frac{\partial u}{\partial x}\delta x + \frac{\partial u}{\partial y}\delta y \quad \cdots (4)$$

But $\dfrac{\partial u}{\partial x}\delta x$ and $\dfrac{\partial u}{\partial y}\delta y$ are called the partial differentials of u with respect to x and y respectively. So it follows that *the total differential of u is the sum of its partial differentials.*

If now we put $u = f(x, y) = x$

Then $\dfrac{\partial u}{\partial x} = 1$ and $\dfrac{\partial u}{\partial y} = 0$

$\therefore$ From (4) we get $dx = \delta x$.

Similarly, if we put $u = f(x, y) = y$, we shall get $dy = \delta y$.

Hence (4) can be written as

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy.$$

Similarly, for n number of variables $x, y, z \cdots$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz + \cdots$$

From this it follows that a function is differentiable at a point if it possess a total differential at that point.

Theorem :

The necessary condition for a function f(x, y) to be differentiable at a point (a, b) of its domain is that f(x, y) should be continuous at that point.

Since $u=f(x, y)$ is differentiable at (a, b) the increment of u can be expressed in the form

$$\delta u = Ah+Bk+h\phi(h, k)+k\psi(h, k) \quad \ldots \quad (1)$$

where A and B are constants independent of h and k; $\phi(h, k)$ and $\psi(h, k)$ are both functions of h and k and both tend to zero as $(h, k)\to(0, 0)$.

But $\delta u=f(a+h, b+k)-f(a, b)$

$\therefore$ from (1)

$$f(a+h, b+k)-f(a, b)=Ah+Bk+h\phi(h, k)+k\psi(hk) \quad \ldots \quad (2)$$

The linear portion $Ah+Bk$ is the differential of $f(x, y)$ at (a, b)

$\therefore$ $Ah+Bk=d.f(a, b)=0$ in the limit.

Hence from (2)

$$\underset{(h, k)\to(0, 0)}{Lt} \{f(a+h, b+k)-f(a, b)\}=0$$

or, $$\underset{(h, k)\to(0, 0)}{Lt} f(a+h, b+k)=f(a, b)$$

$\therefore$ $f(x, y)$ is continuous at (a, b)

The Converse of the above theorem is not always true, for there are functions which are continuous at some points of their domain but are not differentiable at those points.

Illustrations :

Let $f(x, y)=\dfrac{xy}{\sqrt{x^2+y^2}}$, $(x, y)\neq(0, 0)$ **and** $f(0, 0)=0$,

Show that f(x, y) is continuous and possesses partial derivatives at (0, 0) but is not differentiable at that point.

Suppose (x, y) approaches $(0, 0)$ along the line $y=mx$ then $x\to0$ as $y\to0$

$$\therefore \quad \underset{(x,\, y)\to(0,\, 0)}{Lt} f(x,\ y) = \underset{x\to 0}{Lt} \frac{x.mx}{\sqrt{x^2+m^2x^2}}$$

$$= \underset{x\to 0}{Lt} \frac{mx}{\sqrt{1+m^2}} = 0$$

and given $f(0, 0)=0$

$$\therefore \quad \underset{(x,\, y)\to(0,\, 0)}{Lt} f(x, y) = f(0, 0)$$

$\therefore$ $f(x, y)$ is continuous at $(0, 0)$.

Again $\quad f_x(0, 0) = \underset{h\to 0}{Lt} \dfrac{f(0+h, 0)-f(0, 0)}{h.}$

$$= \underset{h\to 0}{Lt} \frac{\dfrac{h\times 0}{\sqrt{h^2+0}}-0}{h} = \underset{h\to 0}{Lt} \frac{0-0}{h} = 0.$$

$$f_y(0,0) = \underset{k\to 0}{Lt} \frac{f(0, 0+k)-f(0, 0)}{k}.$$

$$= \underset{k\to 0}{Lt} \frac{\dfrac{0\times k}{\sqrt{0+k^2}}-0}{k} = \underset{k\to 0}{Lt} \frac{0-0}{k} = 0.$$

Now the condition for differentiability of the function at $(0, 0)$ is the validity of the relation

$$f(0+h, 0+k)-f(0, 0) = Ah+Bk+h\phi(h, k)+k\psi(h, k) \quad \cdots (1)$$

Where $\quad A = f_x(0, 0) = 0$

$\qquad\qquad B = f_y(0, 0) = 0$

And $\quad \underset{(h,\, k)\to(0,\, 0)}{Lt} \phi(h, k) = 0, \quad \underset{(h,\, k)\to(0,\, 0)}{Lt} \psi(h, k) = 0.$

$\therefore$ from (1)

$$f(h, k)-f(0, 0) = h\phi(h, k)+k\psi(h, k)$$

$$= h\left\{\phi(h,\ k)+\frac{k}{h}\psi(h, k)\right\}$$

or, $\quad \dfrac{k}{\sqrt{h^2+k^2}} = \phi(h, k)+\dfrac{k}{h}\psi(h, k).$

or, $\quad \underset{(h,\, k)\to(0,\, 0)}{Lt} \dfrac{k}{\sqrt{h^2+k^2}}$

$$= \underset{(h,\, k)\to(0,\, 0)}{Lt} \left\{\phi(h,\ k)+\frac{k}{h}\psi(h, k)\right\} = 0. \quad \cdots (2)$$

To evaluate the $L.H.S.$ let us suppose that (h, k) approaches $(0, 0)$ along the line $k=mh$

Then $L.H.S=\underset{h\to 0}{Lt} \dfrac{mh}{\sqrt{h^2+m^2h^2}}=\dfrac{m}{\sqrt{1+m^2}}$

So from (2) we get

$$\frac{m}{\sqrt{1+m^2}}=0 \text{ which is impossible.}$$

$\because$ m is a non zero positive quantity, $\dfrac{m}{\sqrt{1+m^2}}$ can never be zero.

So the relation (1) is not valid and hence $f(x, y)$ is not differentiable at $(0, 0)$.

Theorem :

The necessary condition for a funetion $f(x, y)$ to be differentiable at (x, y) is that the function should possess partial derivatives f_x and f_y at that point.

Let $u=f(x, y)$ be a function of x, y defined in the domain R.

Then the function $f(x, y)$, is said to be differentiable at (x, y) if it possess total differential du at that point such that

$$du=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}.dy$$

So for existence of du, the existence of both $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ are essential.

Hence if $u=f(x, y)$ is differentiable at (x, y), then it must possess partial derivatives f_x and f_y at that point.

But the converse of this theorem is not true for there are functions which have partial derivatives but are not differentiable at that point.

Illustrations :

For the function $f(x, y)=(|xy|)^{\frac{1}{2}}$ both f_x and f_y exist at (0, 0) but is not differentiable at (0, 0).

$$f_x(0, 0)=\underset{h\to 0}{Lt}\ \frac{f(h, 0)-f(0, 0)}{h}=\underset{h\to 0}{Lt}\ \frac{0-0}{h}=0.$$

$$f_y(0, 0)=\underset{k\to 0}{Lt}\ \frac{f(0, k)-f(0, 0)}{k}=\underset{k\to 0}{Lt}\ \frac{0-0}{k}=0$$

Thus f_x and f_y both exist at (0, 0).

Again for differentiable at (0, 0) we have the relation

$$f(0+h, 0+k)-f(0, 0)=Ah+Bk+h\phi(h, k)+k\psi(h, k)\cdots\cdots(1)$$

where $A=f_x(0, 0)=0$

$B=f_y(0, 0)=0$

and $\underset{(h, k)\to(0, 0)}{Lt}\ \phi(h, k)=0,\ \underset{(h, k)\to(0, 0)}{Lt}\ \psi(h, k)=0.$

$\therefore$ from (1) $f(h, k)-f(0, 0)=h\phi(h, k)+k\psi(h, k)$

$$\text{or, } \underset{(h, k)\to(0, 0)}{Lt}\ \left\{f(h, k)-f(0, 0)\right\}$$

$$=\underset{(h, k)\to(0, 0)}{Lt}\ \left\{h\phi(h, k)+k\psi(h, k)\right\}$$

$$\text{or, } \underset{(h, k)\to(0, 0)}{Lt}\ \left\{|h.k|\right\}^{\frac{1}{2}}=\underset{(h, k)\to(0, 0)}{Lt}\ \left\{h\phi(h, k)+k\psi(h, k)\right\}$$

$$\text{or, } \underset{(h, k)\to(0, 0)}{Lt}\ \frac{\{|h, k|\}^{\frac{1}{2}}}{h}=\underset{(h, k)\to(0, 0)}{Lt}\ \left\{\phi(h, k)+\frac{k}{h}\psi(h, k)\right\}$$

$$=0\cdots\cdots\cdots(2)$$

Now $L.\ H.\ S.=\underset{h\to 0}{Lt}\ \frac{\{h.\ mh\}^{\frac{1}{2}}}{h}=\sqrt{m}$ putting $k=mh$.

$\therefore$ from (2) $\sqrt{m}=0$ which is absurd as $\sqrt{m}$ may have any value not necessarily zero.

$\therefore$ The relation (1) is not true and hence $f(x, y)$ is not differentiable at (0, 0).

Theorem :

A sufficient condition for the differentiability of a function $u = f(x, y)$, at a given point (a, b) of its domain, is that (i) $f_x(a, b)$ exists and (ii) $f_y(x, y)$ is continuous at (a, b).

(C. H. 1968)

Since $f_y(x, y)$ is continuous at (a, b), $f_y(x, y)$ exists in a certain neighbourhood of (a, b). Let $(a+h, b+k)$ be any point of this neighbourhood. Then for values within this neighbourhood, we can write :

$$f(a+h, b+k)-f(a, b)=f(a+h, b+k)-f(a+h, b)+f(a+h,b)-f(a, b) \cdots\cdots (1)$$

The function $f(a+h, y)$ of y is derivable with respect to y in $(b, b+k)$. So by Mean Value Theorem

$$f(a+h, b+k)-f(a+h, b)=kf_y(a+h, b+\theta k) \cdots\cdots (2)$$

where $0<\theta<1$.

If we construct a function $\psi(h, k)$ such that

$$\psi(h, k)=f_y(a+h, b+\theta k)-f_y(a, b) \cdots\cdots (3)$$

Then $$\underset{(h, k)\to(0, 0)}{Lt} \psi(h, k)= \underset{(h, k)\to(0, 0)}{Lt} \left\{f_y(a+h, b+\theta k)-f_y(a, b)\right\}$$

$$=f_y(a, b)-f_y(a, b)$$

$$=0 \quad \because f_y \text{ is continuous at } (a, b)$$

Also since f_x exists at (a, b)

$$\underset{h\to 0}{Lt} \frac{f(a+h, b)-f(a, b)}{h}=f_x(a, b)$$

So we can write

$$\frac{f(a+h, b)-f(a, b)}{h}-f_x(a, b)=\phi(h) \cdots\cdots (4)$$

where $\phi(h)\to 0$ as $h\to 0$.

From (2) and (3)

$$f(a+h, b+k)-f(a+h, b)=k\psi(h, k)+kf_y(a, b) \cdots\cdots (5)$$

From (4)

$$f(a+h, b)-f(a, b)=hf_x(a, b)+h\phi(h) \cdots\cdots (6)$$

Putting (5) and (6) in (1) we get

$$f(a+h, b+k)-f(a, b)=hf_x(a, b)+kf_y(a, b)+h\phi(h)+k\psi(h, k)$$

which is the condition for differentiability of $f(x, y)$ at (a, b). Hence, the theorem.

Note : If $f_y(a, b)$ exist and $f_x(x, y)$ is continuous at (a, b), then also $f(x, y)$ is differentiable at (a, b). Thus the sufficient condition of differentiability at (a, b) is that only one of the partial derivatives exists at (a, b) and the other is continuous at (a, b). *But there are functions which are differentiable although no partial derivative is continuous.*

This is illustrated in the following example :—

Let $f(x, y)=x^2 \sin \frac{1}{x}+y^2 \sin \frac{1}{y}$, $x \neq 0$, $y \neq 0$

$f(x, 0)=x^2 \sin \frac{1}{x}$, $x \neq 0$

$f(0, y)=y^2 \sin \frac{1}{y}$, $y \neq 0$

and $f(0, 0)=0$.

Show that f_x and f_y are discontinuous at the origin but $f(x, y)$ is differentiable at the origin.

$\because\ f(x, y)=x^2 \sin \frac{1}{x}+y^2 \sin \frac{1}{y}$, $x \neq 0$, $y \neq 0$

$f_x(x, y)=2x \sin \frac{1}{x}-\cos \frac{1}{x}$, $x \neq 0$ and y remaining constant.

and $f_y(x, y)=2y \sin \frac{1}{y}-\cos \frac{1}{y}$, $y \neq 0$ and x remaining constant.

But $f_x(0, y)=\underset{h \to 0}{Lt} \dfrac{f(0+h, y)-f(0, y)}{h}$

$$=\underset{h \to 0}{Lt} \frac{h^2 \sin \frac{1}{h}+y^2 \sin \frac{1}{y}-y^2 \sin \frac{1}{y}}{h}$$

$$=\underset{h \to 0}{Lt}\ h \sin \frac{1}{h}=0.$$

and $f_y(x, 0) = \underset{k \to 0}{Lt} \dfrac{f(x, 0+k) - f(x, 0)}{k}$

$$= \underset{k \to 0}{Lt} \frac{x^2 \sin \frac{1}{x} + k^2 \sin \frac{1}{k} - x^2 \sin \frac{1}{x}}{k}$$

$$= \underset{k \to 0}{Lt}\ k \sin \frac{1}{k} = 0$$

$\therefore$ f_x and f_y is not continuous at $(0, 0)$

Now if $f(x, y)$ is differentiable at $(0, 0)$, then we must have

$$f(0+h, 0+k) - f(0, 0) = h^2 \sin \frac{1}{h} + k^2 \sin \frac{1}{k}$$

$$= Ah + Bk + h\phi(h, k) + k\psi(h, k) \cdots \quad (1)$$

where $A = f_x(0, 0) = \underset{h \to 0}{Lt} \dfrac{f(h, 0) - f(0, 0)}{h}$

$$= \underset{h \to 0}{Lt} \frac{\left(h^2 \sin \frac{1}{h} - 0\right)}{h} = 0$$

$B = f_y(0, 0) = \underset{k \to 0}{Lt} \dfrac{f(0, k) - f(0, 0)}{k}$

$$= \underset{k \to 0}{Lt} \frac{k^2 \sin \frac{1}{k} - 0}{k} = 0$$

and $\underset{(h, k) \to (0, 0)}{Lt} \phi(h, k) = \underset{(h, k) \to (0, 0)}{Lt} \psi(h, k) = 0 \cdots (2)$

So from (1) we find that

$$h^2 \sin \frac{1}{h} + k^2 \sin \frac{1}{k} = 0.\ h + 0.\ k + h\left(h \sin \frac{1}{h}\right) + k\left(k \sin \frac{1}{k}\right) \cdots \quad (3)$$

$\because$ $\phi(h, k) = h \sin \dfrac{1}{h}$ and $\psi(h, k) = k \sin \dfrac{1}{k}$

both being zero as $(h, k) - (0, 0)$

This shows that (3) is identically satisfied.

Hence, the give function is differentiable at the origin although their partial derivatives do not exist at $(0, 0)$.

2.4. Calculation of small errors by differentials.

When the values of x and y are obtained by any measurement are subject to small errors dx and dy, the corresponding error dz of z can be calculated approximately by the formula.

$$dz = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy. \qquad (1)$$

For example, consider the volume of a cone of height h and radius of the base r as

$$V = \tfrac{1}{3}\pi r^2 h.$$

Then $\dfrac{\partial V}{\partial r} = \tfrac{2}{3}\pi r h$ and $\dfrac{\partial V}{\partial h} = \tfrac{1}{3}\pi r^2$.

$\therefore$ using formula (1)

$$dV = (\tfrac{2}{3}\pi r h)dr + (\tfrac{1}{3}\pi r^2)dh$$

or, $$\frac{dV}{V} = \frac{\frac{2}{3}\pi r h}{V}dr + \frac{\frac{1}{3}\pi r^2}{V}dh$$

$$= \frac{2}{r}dr + \frac{1}{h}dh. \qquad \ldots \qquad (2)$$

So in measuring r and h, if there happens to be an error of dr and dh respectively, then by (2) it is possible to calculate $\dfrac{dV}{V}$ which measures the relative error in volume of the cone.

In particular if r and h are measured as 4 and 8 inches with the possible error of ·04 and ·08 inches then

$$\frac{dV}{V} = \frac{2}{4}\times{\cdot}04 + \frac{1}{8}\times{\cdot}08 = {\cdot}03$$

$\therefore$ Percentage error $= \dfrac{100dV}{V} = 3\%.$

Ex. 1. The side a of the triangle ABC is calculated by measuring the sides b, c and the angle A. If the corresponding error be db, dc and dA, show that the error in the calculated value of a is

$$da = (b \sin C)\,dA + (\cos C)db + (\cos B)dc.$$

We have $a^2=b^2+c^2-2bc\cos A$

$\therefore\quad 2a\,da=2b\,db+2c\,dc-2(c\cos A\,db+b\cos A\,dc-bc\sin A\,dA)$

or, $da=(b-c\cos A)\dfrac{db}{a}+(c-b\cos A)\dfrac{dc}{a}+bc\dfrac{\sin A}{a}dA$

$[\because\ b=c\cos A+a\cos C,\ c=a\cos B+b\cos A\ ]$

$$=a\cos C\frac{db}{a}+a\cos B\frac{dc}{a}+bc\frac{\sin A}{a}dA$$

$$=(\cos C)db+(\cos B)dc+bc.\frac{\sin C}{c}dA$$

$$=(\cos C)db+(\cos B)dc+(b\sin C)\,dA.$$

Ex. 2. The side a and the opposite angle A of a triangle ABC remains constant. Show that when the other sides and angles are slightly varied, then. (C. H. 1969)

$$\frac{db}{\cos B}+\frac{dc}{\cos C}=0$$

We have

$$b\cos C+c\cos B=a$$

$\therefore\quad db\cos C-b\sin C\,dC+dc\cos B-c\sin B\,dB=0.$

or, $db\cos C+dc\cos B-\{(b\sin C)dC+(c\sin B)dB\}=0$

or, $db\cos C+dc\cos B-c\sin B(dC+dB)=0$

$\because\quad \dfrac{\sin B}{b}=\dfrac{\sin C}{c}$, i.e., $b\sin C=c\sin B.$

or, $db\cos C+dc\cos B-c\sin B\,d(C+B)=0$

or, $db\cos C+dc\cos B-c\sin B\,d(\pi-A)=0.$

or, $db\cos C+dc\cos B=0\quad\because\ d(\pi-A)=0.$

or, $\dfrac{db}{\cos B}+\dfrac{dc}{\cos C}=0.$

2·5. Partial derivatives of higher order.

The partial derivatives f_x and f_y are themselves functions of x and y and so may again admit of partial differentiation with respect to x and y.

The partial derivative of $f_x(x, y)$ with respect to x is denoted by f_{xx} or, $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$ or, $\frac{\partial^2 f}{\partial x^2}$.

Thus $$f_{xx}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)=\frac{\partial}{\partial x}f_x(x, y)$$

$$=\underset{h\to 0}{Lt}\ \frac{f_x(x+h, y)-f_x(x, y)}{h}$$

Similarly, $$f_{yx}=\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)=\frac{\partial}{\partial y}f_x(x, y)$$

$$=\underset{k\to 0}{Lt}\ \frac{f_x(x, y+k)-f_x(x, y)}{k}$$

$$f_{yy}=\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)=\frac{\partial}{\partial y}f_y(x, y)$$

$$=\underset{k\to 0}{Lt}\ \frac{f_y(x, y+k)-f_y(x, y)}{k}$$

$$f_{xy}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)=\frac{\partial}{\partial x}f_y(x, y)$$

$$=\underset{h\to 0}{Lt}\ \frac{f_y(x+h, y)-f_y(x, y)}{h}$$

The four partial derivatives are the second order derivatives of $f(x, y)$ and are also denoted by

$$\frac{\partial^2 f}{\partial x^2},\ \frac{\partial^2 f}{\partial y\,\partial x},\ \frac{\partial^2 f}{\partial y^2},\ \frac{\partial^2 f}{\partial x\,\partial y}.$$

The higher order partial derivatives are defined in the same way.

2.6. Commutative property of the order of partial derivatives for a function of two variables.

We have by definition

$$f_{xy}(a, b)=\left(\frac{\partial^2 f}{\partial x.\partial y}\right)_{(a,\,b)}$$

$$=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_{(a,\,b)}, \text{ at a point } (a, b)$$

$$=\frac{\partial}{\partial x} f_y(a, b)$$

$$=\underset{h \to 0}{Lt} \frac{f_y(a+h, b)-f_y(a, b)}{h} \quad \ldots \quad (1)$$

But $f_y(a+h, b)=\underset{k \to 0}{Lt} \dfrac{f(a+h, b+k)-f(a+h, b)}{k}$

And $f_y(a, b)=\underset{k \to 0}{Lt} \dfrac{f(a, b+k)-f(a, b)}{k}$.

Putting these values in (1) we get, $f_{xy}(a, b)$

$$=\underset{h \to 0}{Lt} \underset{k \to 0}{Lt} \left\{\frac{f(a+h, b+k)-f(a+h, b)-f(a, b+k)+f(a, b)}{hk}\right\}$$

$$=\underset{h \to 0}{Lt} \underset{k \to 0}{Lt} \frac{F(h, k)}{hk} \text{ (say.)} \quad \ldots \quad \ldots \quad (2)$$

Similarly, it can be shown that

$$f_{yx}(a, b)=\underset{k \to 0}{Lt} \underset{h \to 0}{Lt} \frac{F(h, k)}{hk}. \quad \ldots \quad \ldots \quad (3)$$

From (2) and (3) we find that $f_{xy}(a, b)$ and $f_{yx}(a, b)$ are repeated limits of the same function taken in different orders. But we know that the repeated limits may or may not be equal. So it follows that $f_{xy}(a, b)$ and $f_{yx}(a, b)$ may or may not be equal.

Illustrations :

Ex. 1. Let $f(x, y)=\dfrac{xy(x^2-y^2)}{x^2+y^2}$, $(x, y) \neq (0, 0)$ and $f(0, 0)=0$. Then show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$ *(C. H. 1964)*

$$f_{xy}(0, 0)=\underset{h \to 0}{Lt} \frac{f_y(0+h, 0)-f_y(0, 0)}{h} \quad \ldots \quad (1)$$

But $f_y(h, 0)=\underset{k \to 0}{Lt} \dfrac{f(h, 0+k)-f(h, 0)}{k}$

$$=\underset{k \to 0}{Lt} \frac{f(h, k)-f(h, 0)}{k}$$

$$= \underset{k \to 0}{Lt} \frac{1}{k} \left\{ \frac{hk(h^2 - k^2)}{h^2 + k^2} - 0 \right\}$$

$$= \underset{k \to 0}{Lt} \frac{h(h^2 - k^2)}{h^2 + k^2} = \frac{h^3}{h^2} = h.$$

And $f_y(0, 0) = \underset{k \to 0}{Lt} \dfrac{f(0, 0+k) - f(0, 0)}{k}$

$$= \underset{k \to 0}{Lt} \frac{1}{k} \{f(0, k) - f(0, 0)\}$$

$$= \underset{k \to 0}{Lt} \frac{1}{k} \{0 - 0\} = 0$$

$\therefore$ From (1) we get

$$f_{xy}(0, 0) = \underset{h \to 0}{Lt} \frac{h - 0}{h} = 1$$

Again $f_{yx}(0, 0) = \underset{k \to 0}{Lt} \left\{ \dfrac{f_x(0, 0+k) - f_x(0, 0)}{k} \right\}$ ... (2)

But $f_x(0, k) = \underset{h \to 0}{Lt} \dfrac{f(0+h, k) - f(0, k)}{h}$

$$= \underset{h \to 0}{Lt} \frac{f(h, k) - f(0, k)}{h}$$

$$= \underset{h \to 0}{Lt} \frac{1}{h} \left\{ \frac{hk(h^2 - k^2)}{h^2 + k^2} - 0 \right\}$$

$$= \underset{h \to 0}{Lt} \frac{k(h^2 - k^2)}{h^2 + k^2} = -\frac{k^3}{k^2} = -k$$

And $f_x(0, 0) = \underset{h \to 0}{Lt} \dfrac{f(0+h, 0) - f(0, 0)}{h}$

$$= \underset{h \to 0}{Lt} \frac{1}{h} \{f(h, 0) - f(0, 0)\}$$

$$= \underset{h \to 0}{Lt} \frac{1}{h} \{0 - 0\} = 0.$$

$\therefore$ From (2) we get

$$f_{yx}(0, 0) = \underset{k \to 0}{Lt} \left\{ \frac{-k - 0}{k} \right\} = -1$$

Hence, $f_{xy}(0, 0) \neq f_{yx}(0, 0)$.

Ex. 2. Let $f(x, y)=\frac{2xy}{\sqrt{x^2+y^2}}$, $(x, y)\neq(0, 0)$

and $f(0, 0)=0$. Then show that

$$f_{xy}(0, 0)=f_{yx}(0, 0).$$

$$f_{xy}(0, 0)=\underset{h\to 0}{Lt}\ \frac{f_y(0+h, 0)-f_y(0, 0)}{h} \quad \ldots \quad (1)$$

But $f_y(h, 0)=\underset{k\to 0}{Lt}\ \frac{f(h, 0+k)-f(h, 0)}{k}$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\{f(h, k)-f(h, 0)\}$$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\ \frac{2hk}{\sqrt{h^2+k^2}}=\underset{k\to 0}{Lt}\ \frac{2h}{\sqrt{h^2+k^2}}=2$$

And $f_y(0, 0)=\underset{k\to 0}{Lt}\ \frac{f(0, 0+k)-f(0, 0)}{k}$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\{f(0, k)-f(0, 0)\}$$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}(0-0)=0$$

$\therefore$ From (1) we get

$$f_{xy}(0, 0)=\underset{h\to 0}{Lt}\ \frac{2-0}{h}=\infty$$

Again $f_{yx}(0, 0)=\underset{k\to 0}{Lt}\ \frac{f_x(0, 0+k)-f_x(0, 0)}{k} \quad \ldots \quad (2)$

But $f_x(0, k)=\underset{h\to 0}{Lt}\ \frac{f(0+h, k)-f(0, k)}{h}$

$$=\underset{h\to 0}{Lt}\ \frac{1}{h}\left[\frac{2hk}{\sqrt{h^2+k^2}}-0\right]$$

$$=\underset{h\to 0}{Lt}\ \frac{2k}{\sqrt{h^2+k^2}}=2$$

And $f_x(0, 0)=\underset{h\to 0}{Lt}\ \frac{f(0+h, 0)-f(0, 0)}{h}$

$$=\underset{h\to 0}{Lt}\ \frac{1}{h}(0-0)=0$$

$\therefore$ From (2) we get

$$f_{yx}(0, 0)=\underset{k\to 0}{Lt}\ \frac{2-0}{k}=\infty.$$

Hence, $f_{xy}(0, 0)=f_{yx}(0, 0)$.

Ex. 3. **Let** $\mathbf{f(x, y)}=\begin{cases}\ \ \mathbf{xy} \text{ if } |\mathbf{x}| \geqslant |\mathbf{y}| \\ -\mathbf{xy} \text{ if } |\mathbf{x}| < |\mathbf{y}|\end{cases}$

Show that $\mathbf{f_{xy}(0, 0)\neq f_{yx}(0, 0)}$ *(C. H. (old) 1963)*

$$f_{xy}(0, 0)=\underset{h\to 0}{Lt}\ \frac{f_y(0+h, 0)-f_y(0, 0)}{h} \qquad \cdots \quad (1)$$

But $\quad f_y(h, 0)=\underset{k\to 0}{Lt}\ \frac{f(h, 0+k)-f(h, 0)}{k}$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\{hk-0\} \qquad \because \ h>k, \ |x|>|y|$$

$$=h$$

And $\quad f_y(0, 0)=\underset{k\to 0}{Lt}\ \frac{f(0, 0+k)-f(0, 0)}{k}$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\{0-0\}=0$$

$\therefore$ From (1) $\quad f_{xy}(0, 0)=\underset{h\to 0}{Lt}\ \frac{h-0}{h}=1.$

Again $\quad f_{yx}(0, 0)=\underset{k\to 0}{Lt}\ \frac{f_x(0, 0+k)-f_x(0, 0)}{k} \qquad \cdots \quad (2)$

But $\quad f_x(0, k)=\underset{h\to 0}{Lt}\ \frac{f(0+h, k)-f(0, k)}{h} \qquad \because \ k>h, \ |y|>|x|$

$$=\underset{h\to 0}{Lt}\ \frac{1}{h}\{-hk-0\}=-k$$

And $\quad f_x(0, 0)=\underset{h\to 0}{Lt}\ \frac{f(0+h, 0)-f(0, 0)}{h}$

$$=\underset{h\to 0}{Lt}\ \frac{1}{h}\{0-0\}=0$$

$\therefore$ From (2)

$$f_{yx}=\underset{k\to 0}{Lt}\ \frac{-k-0}{k}=-1$$

Hence, $f_{xy}(0, 0)\neq f_{yx}(0, 0)$

2.7. Reversal order of derivation : Sufficient conditions.

1. **Schwarz's Theorem. Form—I.** Let $f(x, y)$ be defined in some neighbourhood of (x, y). If its derivatives (*i*) f_x, f_y, f_{xy} and f_{yx} all exist in that neighbourhood of (x, y) and

(*ii*) f_{xy} (or f_{yx}) is continuous at the point (x, y)

then $\dfrac{\partial^2 y}{\partial x \partial y}=\dfrac{\partial^2 y}{\partial y \partial x}$.

Proof. Let us consider the points $(x+h, y)$, $(x, y+k)$ and $(x+h, y+k)$ in the neighbourhood of (x, y) so that h, k are small enough $\neq 0$. Then in this neighbourhood of (x, y) $\{f(x+h, y+k)-f(x, y+k)\}-\{f(x+h, y)-f(x, y)\}$ will be a function of (h, k) only. So we can write

$$F(h, k)=\{f(x+h, y+k)-f(x, y+k)\}-\{f(x+h, y)-f(x, y)\}$$
$$=\phi(y+k)-\phi(y) \qquad \cdots \qquad \cdots \qquad (1)$$

if we write, $\phi(y)=f(x+h, y)-f(x, y)$

Since f_y exists in some neighbourhood of (x, y), $\phi(y)$ is derivable in that neighbourhood of (x, y)

$\therefore$ $\phi'(y)=f_y(x+h, y)-f_y(x, y)$

So at the point $y=y+\theta k$

$$\phi'(y+\theta k)=f_y(x+h, y+\theta k)-f_y(x, y+\theta k) \qquad \cdots \qquad (2)$$

Now applying Mean Value Theorem in (1)

$F(h, k)=k\phi'(y+\theta k)$ where $0<\theta<1$

$$=k\{f_y(x+h, y+\theta k)-f_y(x, y+\theta k)\} \text{ from (2), } \cdots \qquad (3)$$

Since f_{xy} is continuous at the point (x, y), f_{xy} exists in some neighbourhood of (x, y)

$\therefore$ $f_y(x, y+\theta k)$ is derivable with respect to x in $(x, x+h)$

$\therefore$ Applying Mean Value Theorem

$$f_y(x+h,\ y+\theta k)-f_y(x,\ y+\theta k)$$
$$=h\ f_{xy}(x+\theta' h,\ y+\theta k)\quad \text{where}\quad 0<\theta<1$$
$$0<\theta'<1$$

$\therefore$ From (3)

$$F(h,\ k)=khf_{xy}(x+\theta' h,\ y+\theta k)$$

or, $$f_{xy}(x+\theta' h,\ y+\theta k)=\frac{F(h,\ k)}{hk} \quad \cdots \quad (4)$$

Since f_{xy} is continuous at $(x,\ y)$

$$\underset{h\to 0}{Lt}\ \underset{k\to 0}{Lt}\ f_{xy}(x+\theta' h,\ y+\theta k)=f_{xy}(x,\ y).$$

Also $\underset{h\to 0}{Lt}\ \underset{k\to 0}{Lt}\ \dfrac{F(h,\ k)}{hk}$

$$=\underset{h\to 0}{Lt}\ \underset{k\to 0}{Lt}\ \frac{1}{k}\left\{\frac{f(x+h,\ y+k)-f(x,\ y+k)}{h}-\frac{f(x+h,\ y)-f(x,\ y)}{h}\right\}$$

$$=\underset{k\to 0}{Lt}\ \frac{1}{k}\left\{\underset{h\to 0}{Lt}\ \frac{f(x+h,\ y+k)-f(x,\ y+k)}{h}-\underset{h\to 0}{Lt}\ \frac{f(x+h,\ y)-f(x,\ y)}{h}\right\}$$

$$=\underset{k\to 0}{Lt}\ \frac{f_x(x,\ y+k)-f_x(x,\ y)}{k}$$

$$=f_{yx}(x,\ y).$$

Hence from (4)

$$f_{xy}=f_{yx}$$

Schwarz's Theorem. Form II. If $f_y(x, y)$ and $f_{yx}(x, y)$ exist in a certain neighbourhood of (a, b) and if $f_{yx}(x, y)$ is continuous at (a, b) then $f_{xy}=f_{yx}$ at (a, b).

Proof. Since f_{yx} is continuous at (a, b), f_x exists in the neighbourhood of (a, b). So the given conditions imply that $f_x(x, y)$, $f_y(x, y)$ and $f_{yx}(x, y)$ exist at every point (x, y)

of a certain neighbourhood of (a, b). Let $(a+h, b)$, $(a+h, b+k)$ and $(a, b+k)$ be three neighbouring points of (a, b) where h, k are very small $\neq 0$. Then we can write

$$F(h, k)=f(a+h, b+k)-f(a+h, b)-f(a, b+k)+f(a, b) \quad \cdots (1)$$

Now denoting $\phi(x)=f(x, b+k)-f(x, b) \quad \cdots \quad \cdots (2)$

we get

$$F(h, k)=\phi(a+h)-\phi(a) \quad \cdots \quad \cdots \quad \cdots (3)$$

Since f_x exists in the neighbourhood of (a, b), the function $\phi(x)$ is derivable with respect to x in the closed interval $(a, a+h)$.

$\therefore$ applying the Mean Value Theorem to the R. H. S. of (3)

$$F(h, k)=h\phi'(a+\theta h) \text{ where } 0<\theta<1$$
$$=h[f_x(a+\theta h, b+k)-f_x(a+\theta h, b] \quad \cdots \quad (4)$$
[using (2)]

Since f_{yx} exists in the neighbourhood of (a, b), the function $f_x(a+\theta h, y)$ is derivable with respect to y in the closed interval $(b, b+k)$

$\therefore$ Applying the mean value theorem to the R. H. S. of (4)

$$F(h, k)=hk\, f_{yx}(a+\theta h, b+\theta' k) \text{ where } 0<\theta<1 \text{ and } 0<\theta'<1$$

or, $$\frac{F(h, k)}{hk}=f_{yx}(a+\theta h, b+\theta' k) \quad \cdots \quad \cdots \quad (5)$$

Now $$\frac{F(h, k)}{hk}=\frac{1}{h}\left[\frac{f(a+h, b+k)-f(a+h, b)}{k}-\frac{f(a, b+k)-f(a, b)}{k}\right]$$

$$\therefore \underset{h\to 0}{Lt}\ \underset{k\to 0}{Lt}\ \frac{F(h, k)}{hk}$$

$$=\underset{h\to 0}{Lt}\ \frac{1}{h}\left[\underset{k\to 0}{Lt}\ \frac{f(a+h, b+k)-f(a+h, b)}{k}-\frac{f(a, b+k)-f(a, b)}{k}\right]$$

$$=\underset{h\to 0}{Lt}\ \frac{f_y(a+h,\ b)-f_y(a,\ b)}{h}$$

$$=f_{xy}(a,\ b)$$

Also $\underset{h\to 0}{Lt}\ \underset{k\to 0}{Lt}\ f_{yx}(a+\theta h,\ b+\theta' k)=f_{yx}(a,\ b)$

$\because$ f_{yx} is continuous at $(a,\ b)$

Hence from (5) we get

$$f_{xy}(a,\ b)=f_{yx}(a,\ b).$$

Schwarz's Theorem. Form III.

If $f_{xy}(x,\ y)$ and $f_{yx}(x,\ y)$ are both continuous at (a, b) then

$$\mathbf{f_{xy}(a,\ b)=f_{yx}(a,\ b).}$$

Since f_{xy} and f_{yx} are both continuous at $(a,\ b)$, f_y and f_x exist in some neighbourhood of $(a,\ b)$

So the given conditions imply that f_x, f_y and f_{xy} exists at every point $(x,\ y)$ of a certain neighbourhood of $(a,\ b)$ and it is the same as form II above.

Young's Theorem.

If f_x and f_y exists in a certain neighbourhood of (a, b) and if f_x and f_y are both differentiable at (a, b), then

$$\mathbf{f_{xy}=f_{yx}\ \text{at}\ (a,\ b).}$$

Proof. The given conditions imply that $f^2{}_x$, f_{yx}, $f^2{}_y$, f_{xy} exist at $(a,\ b)$. Let $(a+h,\ b)$, $(a+h,\ b+h)$ and $(a,\ b+h)$ be three points in the neighbourhood of $(a,\ b)$ so that h is very small $\neq 0$. Then we can write

$$F(h,\ h)=f(a+h,\ b+h)-f(a+h,\ h)-f(a,\ b+h)+f(a,\ b) \qquad \cdots \qquad (1)$$

If then $\phi(x)=f(x,\ b+h)-f(x,\ b)$ $\qquad \cdots \qquad \cdots \qquad (2)$

we get

$$F(h,\ h)=\phi(a+h)-\phi(a) \qquad \cdots \qquad \cdots \qquad (3)$$

Since f_x exists in the neighbourhood of $(a,\ b)$, $\phi(x)$ is derivable with respect to x in the closed interval $(a,\ a+h)$.

So applying the Mean Value Theorem to the R. H. S. of (3)

$$F(h, h)=h\phi'(a+\theta h),\ 0<\theta<1$$
$$=h[f_x(a+\theta h, b+h)-f_x(a+\theta h, b)] \quad \ldots \quad (4)$$
[using (2)]

Now since $f_x(a, b)$ is differentiable at (a, b), we have using the condition of differentiability

$$f_x(a+\theta h, b+h)-f_x(a, b)=\theta h\, f_x{}^2(a, b)+hf_{yx}(a, b)$$
$$+\theta h\, \phi_1(h, h)+h\psi_1(h, h) \quad \ldots \quad (5)$$

and $f_x(a+\theta h, b)-f_x(a, b)=\theta hf^2{}_x(a, b)+\theta h\, \phi_2(h, h)$ ... (6)

where ϕ_1, ϕ_2 and ψ_1 all $\to 0$ as $h\to 0$.

Putting (5) and (6) in (4)

$$\frac{F(h, h)}{h^2}=f_{yx}(a, b)+\theta\phi_1(h, h)+\psi_1(h, h)-\theta\phi_2(h, h) \cdots \quad (7)$$

Similarly, writing

$\psi(y)=f(a+h, y)-f(a, y)$ and proceding as before we shall get

$$\frac{F(h, h)}{h^2}=f_{xy}(a, b)+\theta'\psi_2(h, h)+\phi_3(h, h)-\theta'\psi_3(h, h) \cdots \quad (8)$$

Since ϕ_1, ϕ_2, ϕ_3, ψ_1, ψ_2 all tends to zero as h tends to zero we get in the limits as $h\to 0$ from (7) and (8)

$$f_{yx}(a, b)=f_{xy}(a, b).$$

Exercise 2

1. If da, db, dc be the errors in measuring the sides a, b, c of the triangle ABC, then the error $d\triangle$ of the area $\triangle$ of the triangle is given by

$$d\triangle=\frac{abc}{4\triangle}(\cos A\ da+\cos B\ db+\cos C\ dc).$$

2. If da, db, dc be the errors in measuring the sides a, b, c of a triangle ABC, show that

$$dA=\tfrac{1}{2}\, a\,(da-db\cos C-dc\cos B)/\triangle$$

where $\triangle$ is the area of the triangle and verify that

$$dA+dB+dC=0.$$

3. Show that (*a*) $\sqrt{|xy|}$ is not differentiable at the origin.

(*b*) $|x|+|y|$ is continuous but not differentiable at the origin.

4. For the function

$$f(x,\ y)=xy\frac{x^2+a^2y^2}{x^2+y^2},\ (x^2+y^2\neq 0 \text{ and } a\neq 0)$$

and $f(0,\ 0)=0$

show that $f_{xy}(0,\ 0)\neq f_{yx}(0,0)$ (C. H 1960)

5. If $f(x,\ y)=x^2\tan^{-1}\dfrac{y}{x}-y^2\tan^{-1}\dfrac{x}{y}$, $xy\neq 0$

and $=0$ elsewhere

show that $f_{xy}(0,\ 0)\neq f_{yx}(0,\ 0)$

6. If $f(x,\ y)=\dfrac{x^3+y^3}{x-y}$, $x\neq y$

and $=0$ when $x=y$

show that $f_x(0,\ 0)$ and $f_y\ (0,\ 0)$ both exist but $f(x,\ y)$ is discontinuous at the origin.

7. Show that the function defined by

$$f(x,\ y)=x^2\tan^{-1}\frac{y}{x}-y^2\tan^{-1}\frac{x}{y}$$

and $f(0,\ y)=f(x,\ 0)=0$

satisfies $f_{xy}=f_{yx}$ for all point except at $(0,\ 0)$ (C. H. 1964)

8. If the mixed partial derivatives f_{xy} and f_{yx} of a function $f(x,\ y)$ are continuous in a region R, then show that

$$f_{xy}=f_{yx}$$

holds throughout the interior of the region R.

9. For the function

$$f(x,\ y)=\frac{1}{\sqrt{y}}e^{-(x-a)^2/4y}$$

show that $f_{xy}=f_{yx}$

and $f_{xx}=f_y$.

10. For the function

$$f(x,\ y)=\sqrt{x^2+y^2}\ \sin 2\phi \text{ where } f(0,\ 0)=0$$

and $\phi=\tan^{-1}\left(\dfrac{y}{x}\right)$. Show that

$$f_{xy}\ (0,\ 0)=f_{yx}\ (0,\ 0)$$

CHAPTER III

HOMOGENEOUS FUNCTIONS

3.1. A function $f(x, y)$ is said to be homogeneous of degree n if it can be expressed in the form $x^n\phi\left(\frac{y}{x}\right)$, or in the form $y^n\psi\left(\frac{x}{y}\right)$.

For example consider $f(x, y)=\dfrac{x^{\frac{1}{4}}+y^{\frac{1}{4}}}{x^{\frac{1}{5}}+y^{\frac{1}{5}}}$

$$\text{Hence}\quad f(x, y)=\frac{x^{\frac{1}{4}}\left\{1+\left(\frac{y}{x}\right)^{\frac{1}{4}}\right\}}{x^{\frac{1}{5}}\left\{1+\left(\frac{y}{x}\right)^{\frac{1}{5}}\right\}}=x^{\frac{1}{20}}\left\{1+\left(\frac{y}{x}\right)^{\frac{1}{4}}\right\}\left\{1+\left(\frac{y}{x}\right)^{\frac{1}{5}}\right\}^{-1}$$

$$=x^{\frac{1}{20}}\phi\left(\frac{y}{x}\right)$$

$\therefore$ $f(x, y)$ is a homogeneous function of degree 1/20.

Again a function $f(x, y)$ can always be found to satisfy an identity of the form

$$f(tx, ty)=t^n f(x, y)$$

for all positive values of t. So alternatively, we can say that, a function $f(x, y)$ is said to be homogeneous of degree n if it can be expressed in the form

$f(tx, ty)=t^n f(x, y)$ for *all* positive values of t.

For examples

$f(x, y)=\dfrac{x+y}{2x+3y}$ is a homogeneous function of degree zero. For

$$f(tx, ty)=\frac{tx+ty}{2tx+3ty}=t^{\circ}\ \frac{x+y}{2x+3y}=t^{\circ}f(x, y).$$

A function $f(x, y, z)$ is said to be homogeneous of degree n if it can be expressed in the form

$$x^n\phi\left(\frac{y}{x}, \frac{z}{x}\right) \quad \text{or,} \quad y^n\psi\left(\frac{x}{y}, \frac{z}{y}\right) \quad \text{or,} \quad z^n\zeta\left(\frac{x}{z}, \frac{y}{z}\right).$$

Also alternatively, a function $f(x, y, z)$ is said to be homogeneous of degree n if it can be expressed in the form

$$f(tx, ty, tz) = t^n f(x, y, z)$$

for all positive values of t.

Note. If u be a homogeneous function of degree n in x, y, z, then each of $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$ is a homogeneous function of degree $(n-1)$.

3.2. Euler's Theorem on Homogeneous functions of two variables.

If $f(x, y)$ be a homogeneous function of x and y of degree n, then

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = n.f(x, y).$$

Proof. Since $f(x, y)$ is a homogeneous function of degree n, we can write

$$f(x, y) = x^n\phi\left(\frac{y}{x}\right)$$

$$= x^n\phi(v) \text{ if we put } v = \frac{y}{x}.$$

$$\therefore \quad \frac{\partial f}{\partial x} = nx^{n-1}\phi(v) + x^n\phi'(v)\frac{\partial v}{\partial x}$$

$$= nx^{n-1}\phi(v) + x^n\phi'(v)\left(-\frac{y}{x^2}\right)$$

$$\left[\because \quad v = \frac{y}{x} \quad \frac{\partial v}{\partial x} = -\frac{y}{x^2}.\right]$$

$$\therefore \quad x\frac{\partial f}{\partial x} = nx^n\phi(v) - x^{n-1}y\phi'(v) \qquad \ldots \qquad \ldots \qquad (1)$$

Also $\dfrac{\partial f}{\partial y}=x^n\phi'(v)\dfrac{\partial v}{\partial y}$

$$=x^n\phi'(v)\cdot\frac{1}{x}$$

$$\left[\because\ v=\frac{y}{x}\quad \frac{\partial v}{\partial y}=\frac{1}{x}\right]$$

$$\therefore\quad y\frac{\partial f}{\partial y}=x^{n-1}y\phi'(v) \qquad \ldots \qquad \ldots \qquad (2)$$

Adding (1) and (2)

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nx^n\phi(v)$$

$$=nx^n\phi\left(\frac{y}{x}\right)=n.f(x,\ y).$$

Cor. If u be a homogeneous function of x and y of degree n. then

$$\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^2u=n(n-1)u$$

where $\left(x\dfrac{\partial}{\partial x}+y\dfrac{\partial}{\partial y}\right)^2u=x^2\dfrac{\partial^2u}{\partial x^2}+2xy\dfrac{\partial^2u}{\partial x\,\partial y}+y^2\dfrac{\partial^2u}{\partial y^2}.$

Since u is a homogeneous function of degree n, each of $\dfrac{\partial u}{\partial x}$, $\dfrac{\partial u}{\partial y}$ is a homogeneous function of degree $(n-1)$.

So by Euler's theorem

$$\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)\frac{\partial u}{\partial x}=(n-1)\frac{\partial u}{\partial x} \qquad \ldots \qquad \ldots \qquad (1)$$

and

$$\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)\frac{\partial u}{\partial y}=(n-1)\frac{\partial u}{\partial y}. \qquad \ldots \qquad \ldots \qquad (2)$$

Multiplying (1) by x and (2) by y and adding

$$x^2\frac{\partial^2u}{\partial x^2}+y^2\frac{\partial^2u}{\partial y^2}+2xy\frac{\partial^2u}{\partial x.\partial y}$$

$$=(n-1)\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)=(n-1)nu$$

$$\therefore\quad \left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial u}\right)^2u=n(n-1)u.$$

3.3. Composite functions.

Let $z=f(x, y)$

where $x=\phi(t), \quad y=\psi(t)$.

Then z is said to be a composite function of t.

Again, if $z=f(x, y)$

where $x=\phi(u, v), \ y=\psi(u, v)$

Then z is said to be composite function of u and v.

3.4. Differentiation of composite functions.

Let $z=f(x, y)$

possess continuous partial derivatives and

let $x=\phi(t)$

$y=\psi(t)$

possess continuous derivatives.

Then $$\frac{dz}{dt}=\frac{\partial z}{\partial x}\frac{dx}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}$$

Proof. The given conditions imply that f_x, f_y and $\phi'(t)$, $\psi'(t)$ are continuous.

When t changes to $t+\delta t$, let x and y change to $x+\delta x$ and $y+\delta y$ respectively.

$$\therefore \quad x+\delta x=\phi(t+\delta t), \quad y+\delta y=\psi(t+\delta t)$$

$$\therefore \quad z+\delta z=f(x+\delta x, \ y+\delta y)$$

$$\text{or} \quad \delta z=f(x+\delta x, y+\delta y)-f(x, y) \qquad \because \ z=f(x, y)$$

$$=[f(x+\delta x, y+\delta y)-f(x, y+\delta y)] + [f(x, y+\delta y)-f(x, y)]$$

Applying M. V. T.

$$=\delta x.f_x(x+\theta_1\delta x, y+\delta y)+\delta y f_y(x, y+\theta_2\delta y)$$

for $(0<\theta_1, \theta_2<1)$

$$\text{or} \quad \frac{\delta z}{\delta t}=\frac{\delta x}{\delta t}f_x(x+\theta_1\delta x, \ y+\delta y)+\frac{\delta y}{\delta t}f_y(x, y+\theta_2\delta y) \quad \cdots \quad (1)$$

Let $\delta t \to 0$, so that δx and $\delta y \to 0$

Then $\underset{\delta t \to 0}{Lt} f_x(x+\theta_1 \delta x, y+\delta y)$

$$= \underset{(\delta x, \delta y)\to(0,0)}{Lt} f_x(x+\theta_1\delta x, y+\delta y)$$

$$= f_x(x, y) \qquad [\because f_x \text{ is continuous}]$$

$$= \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x}.$$

and $\underset{\delta t \to 0}{Lt} f_y(x, y+\theta_2 \delta y) = \underset{\delta y \to 0}{Lt} f_y(x, y+\theta_2\delta y)$

$$= f_y(x, y) \qquad [\because f_y \text{ is continuous}]$$

$$= \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y}.$$

Hence, from (1) proceeding to the limit as $\partial t \to 0$ we get

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}.$$

Cor 1. The last result may be written as

$$\frac{dz}{dt}dt = \frac{\partial z}{\partial x}\frac{dx}{dt}dt + \frac{\partial z}{\partial y}\frac{dy}{dt}dt. \qquad \ldots \qquad (1)$$

But the differential dx, dy and dz are given by

$$dx = \frac{dx}{dt}dt, \quad dy = \frac{dy}{dt}dt, \quad dz = \frac{dz}{dt}dt.$$

Hence from (1) we get

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

Cor 2. Generally, for functions of several variables $u = f(x, y, z \cdots)$ where each of $x, y, z \cdots$ is a function of t

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt} + \cdots$$

and $$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz + \cdots$$

Cor 3. $z=f(x, y)$, let $y=\phi(x)$,

then z is a function of x alone.

$$\frac{dz}{dx}=\frac{\partial z}{\partial x}\frac{dx}{dx}+\frac{\partial z}{\partial y}\frac{dy}{dx}$$

$$=\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\cdot\frac{dy}{dx}. \quad \ldots \quad (1)$$

If in particular $z=f(x, y)=0$

Then $\frac{dz}{dx}=0$

$\therefore$ From (1), we get

$$\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\cdot\frac{dy}{dx}=0$$

$$\text{or,} \quad \frac{dy}{dx}=-\frac{\frac{\partial z}{\partial x}}{\frac{\partial z}{\partial y}}$$

This gives $\frac{dy}{dx}$ for implicit function $f(x, y)=0$.

3.5. Implicit Functions.

If from an equation $f(x, y)=0$, it is not possible to solve for y in terms of x, then $f(x, y)$ is called an implicit function.

If on the other band, it is possible to solve for y in terms of x i.e., it is possible to find a relation of the form $y=\phi(x)$, then $f(x, y)$ is called an explicit function.

As for example $x^2+y^2+2=0$ is an implicit equation for from it $y=\pm\sqrt{-2-x^2}$ and so y cannot be obtained for any value of x.

The equation $x^2+y^2=0$ gives $y=\pm\sqrt{-x^2}$ which gives $y=0$ for $x=0$. But for $x\neq 0$, y can not be obtained. So $x^2+y^2=0$ is an implicit equation for $x\neq 0$ and an explicit equation for $x=0$.

Condition for the existence of an explicit function $y=\phi(x)$ from an implicit relation $f(x, y)=0$.

Theorem :

If $f(x, y)$ be continuous together with its first partial derivatives in the neighbourhood of a point (x_0, y_0) where $f(x_0, y_0)=0$ and $f_y(x_0, y_0)\neq 0$, then there exists a function $y=\phi(x)$ continuous in the neighbourhood of x_0 satisfying $y_0=\phi(x)$ such that $f\{x, \phi(x)\}=0$.

3.6. Differentiation of Implicit Functions.

Let $f(x, y)=0$ be an equation of an implicit function where y is a differentiable function of x and the partial derivatives f_x and f_y are continuous.

Then $\dfrac{dy}{dx}=-f_x/f_y$.

$\because$ $f(x, y)=0$

$f(x+\delta x, y+\delta y)=0$

$\therefore$ $f(x+\delta x, y+\delta y)-f(x, y)=0.$

or, $f(x+\delta x, y+\delta y)-f(x, y+\delta y)+f(x, y+\delta y)$
$-f(x, y)=0$ ... (1)

Now by the Mean Value Theorem

$$f(x+\delta x, y+\delta y)-f(x, y+\delta y)=\delta x\frac{\delta}{\delta x}f(x+\theta_1\delta x, y+\delta y)$$

for $0<\theta_1<1$

and $f(x, y+\delta y)-f(x, y)=\delta y\dfrac{\delta}{\delta y}f(x, y+\theta_2\delta y)$

for $0<\theta_2<1$

So from (1)

$$\delta x.\frac{\partial}{\partial x}f(x+\theta_1\delta x, y+\delta y)+\delta y\frac{\delta}{\delta y}f(x, y+\theta_2\delta y)=0$$

or, $\dfrac{\partial}{\partial x}f(x+\theta_1\delta x, y+\delta y)+\dfrac{\delta y}{\delta x}\dfrac{\delta}{\delta y}f(x, y+\theta_2\delta y)=0$ ... (2)

Since y is a differentiable function of x, when $\delta x\to 0$,

δy also $\to 0$. Also since f_x and f_y are continuous, we get from (2) after proceeding to the limit as $\delta x \to 0$

$$\frac{\partial f}{\partial x}+\frac{dy}{dx}\cdot\frac{\partial f}{\partial y}=0$$

or $$\frac{dy}{dx}=-\frac{\dfrac{\partial f}{\partial x}}{\dfrac{\partial f}{\partial y}},\quad \frac{\partial f}{\partial y}\neq 0$$

3.7. Partial Derivatives of a function of two functions.

Let $z=f(x, y)$

possess continuous first order partial derivatives w. r. to x, y.

Let $x=\phi(u, v)$

$y=\psi(u, v)$

possess continuous first order partial derivatives. Then

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}.$$

$$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}.$$

Proof. Since $z=f(x, y)$ possess continuous first order partial derivatives

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy \qquad \cdots \qquad (1)$$

Since $x=\phi(u, v)$ and $y=\psi(u, v)$ possess continuous first order partial derivations

$$dx=\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv$$

and $$dy=\frac{\partial y}{\partial u}du+\frac{\partial y}{\partial v}dv$$

Putting these values of dx and dy in (1) we get

$$dz=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv\right)+\frac{\partial z}{\partial y}\left(\frac{\partial y}{\partial u}du+\frac{\partial y}{\partial v}dv\right)$$

$$=\left(\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}\right)du+\left(\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}\right)dv. \qquad \cdots (2)$$

Again $\because \quad z=f(x, y)$

$$=f\{\phi(u, v), \psi((u, v)\}$$

$$=F(u, v)$$

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv \quad \cdots \quad \cdots \quad (3)$$

Comparing (2) and (3), since du and dv are independent, we get

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

and $$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}$$

Note : Similarly, for more than two variables

i.e., if $z=f(x_1, x_2, x_3 \cdots)$ and $x_1=f_1(u_1, u_2, u_3 \cdots)$

$$x_2=f_2(u_1, u_2, u_3 \cdots)$$

$x_3=f_3(u_1, u_2, u_3 \cdots)$ and so on

Then $$\frac{\partial z}{\partial u_1}=\frac{\partial z}{\partial x_1}\cdot\frac{\partial x_1}{\partial u_1}+\frac{\partial z}{\partial x_2}\cdot\frac{\partial x_2}{\partial u_1}+\frac{\partial z}{\partial x_3}\cdot\frac{\partial x_3}{\partial u_1}+\cdots$$

$$\frac{\partial z}{\partial u_2}=\frac{\partial z}{\partial x_1}\cdot\frac{\partial x_1}{\partial u_2}+\frac{\partial z}{\partial x_2}\cdot\frac{\partial x_2}{\partial u_2}+\frac{\partial z}{\partial x_3}\cdot\frac{\partial x_3}{\partial u_2}+\cdots$$

$$\frac{\partial z}{\partial u_3}=\frac{\partial z}{\partial x_1}\cdot\frac{\partial x_1}{\partial u_3}+\frac{\partial z}{\partial x_2}\cdot\frac{\partial x_2}{\partial u_3}+\frac{\partial z}{\partial x_3}\cdot\frac{\partial x_3}{\partial u_3}+\cdots$$

and so on.

Cor. 1. If $f=f(x, y, z)$, $g=g(x, y, z)$ are differentiable functions of x, y, z and $x=x(p, q)$, $y=y(p, q)$, $z=z(p, q)$ are differentiable functions of p, q, then (C.H. 1969)

$$\begin{pmatrix} f_p & f_q \\ g_p & g_q \end{pmatrix}=\begin{pmatrix} f_x & f_y & f_z \\ g_x & g_y & g_z \end{pmatrix}\begin{pmatrix} x_p & x_q \\ y_p & y_q \\ z_p & z_q \end{pmatrix}$$

$$\text{L.H.S}=\begin{pmatrix} f_x x_p+f_y y_p+f_z z_p & f_x x_q+f_y y_q+f_z z_q \\ g_x x_p+g_y y_p+g_z z_p & g_x x_q+g_y y_q+g_z z_q \end{pmatrix}$$

$$=\begin{pmatrix} \frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial p}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial p}+\frac{\partial f}{\partial z}\cdot\frac{\partial z}{\partial p} & \frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial q}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial q}+\frac{\partial f}{\partial z}\cdot\frac{\partial z}{\partial q} \\ \frac{\partial g}{\partial x}\cdot\frac{\partial x}{\partial p}+\frac{\partial g}{\partial y}\cdot\frac{\partial y}{\partial p}+\frac{\partial g}{\partial z}\cdot\frac{\partial z}{\partial p} & \frac{\partial g}{\partial x}\cdot\frac{\partial x}{\partial q}+\frac{\partial g}{\partial y}\cdot\frac{\partial y}{\partial q}+\frac{\partial g}{\partial z}\cdot\frac{\partial z}{\partial q} \end{pmatrix}$$

$$=\begin{pmatrix} \frac{\partial f}{\partial p} & \frac{\partial f}{\partial q} \\ \frac{\partial g}{\partial p} & \frac{\partial g}{\partial q} \end{pmatrix}=\begin{pmatrix} f_p & f_q \\ g_p & g_q \end{pmatrix} \text{ from the given conditions.}$$

Cor. 2. If $y_1=f_1(x_1, x_2, x_3)$ and $y_2=f_2(x_1, x_2, x_3)$ are differentiable functions of x_1, x_2, x_3 in a region R and $x_1=\phi_1(t_1, t_2)$, $x_2=\phi_2(t_1, t_2)$, $x_3=\phi_3(t_1, t_2)$ are differentiable functions of t_1, t_2 in a region S, then (C. H. 1967)

$$\begin{pmatrix} \frac{\partial y_1}{\partial t_1} & \frac{\partial y_1}{\partial t_2} \\ \frac{\partial y_2}{\partial t_1} & \frac{\partial y_2}{\partial t_2} \end{pmatrix}=\begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \end{pmatrix}\begin{pmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} \\ \frac{\partial x_3}{\partial t_1} & \frac{\partial x_3}{\partial t_2} \end{pmatrix}$$

$$\text{R.H.S.}=\begin{pmatrix} \frac{\partial y_1}{\partial x_1}\cdot\frac{\partial x_1}{\partial t_1}+\frac{\partial y_1}{\partial x_2}\cdot\frac{\partial x_2}{\partial t_1}+\frac{\partial y_1}{\partial x_3}\cdot\frac{\partial x_3}{\partial t_1}, & \frac{\partial y_1}{\partial x_1}\cdot\frac{\partial x_1}{\partial t_2}+\frac{\partial y_1}{\partial x_2}\cdot\frac{\partial x_2}{\partial t_2}+\frac{\partial y_1}{\partial x_3}\cdot\frac{\partial x_3}{\partial t_2} \\ \frac{\partial y_2}{\partial x_1}\cdot\frac{\partial x_1}{\partial t_1}+\frac{\partial y_2}{\partial x_2}\cdot\frac{\partial x_2}{\partial t_1}+\frac{\partial y_2}{\partial x_3}\cdot\frac{\partial x_3}{\partial t_1}, & \frac{\partial y_2}{\partial x_1}\cdot\frac{\partial x_1}{\partial t_2}+\frac{\partial y_2}{\partial x_2}\cdot\frac{\partial x_2}{\partial t_2}+\frac{\partial y_2}{\partial x_3}\cdot\frac{\partial x_3}{\partial t_2} \end{pmatrix}$$

$$=\begin{pmatrix} \frac{\partial y_1}{\partial t_1} & \frac{\partial y_1}{\partial t_2} \\ \frac{\partial y_2}{\partial t_1} & \frac{\partial y_2}{\partial t_2} \end{pmatrix} \text{ from the given conditions.}$$

3.8. Space derivatives. Differentiation in a given direction.

So far we have seen that a differentiable function $f(x, y)$ have partial derivatives in the directions parallel to the x and y axis. Now we shall show that the differentiable function can have derivative in any direction at an angle α to the positive direction of the x axis.

Suppose, the point $(x+h, y+k)$ approaches the point (x, y) along a line through (x, y) making a constant angle α with the positive direction of x axis. Then h and k will not tend to zero independently of another and will satisfy simultaneously the relation

$$h = r \cos \alpha, \quad k = r \sin \alpha$$

so that $r = \sqrt{h^2 + k^2}$.

Then $\underset{r \to 0}{Lt} \dfrac{f(x + r \cos \alpha, y + k \sin \alpha) - f(x, y)}{r}$

if exists is called the derivative of $f(x, y)$ in the direction α and is denoted by $D_{(\alpha)} f(x, y)$

Thus the derivative in the direction α is

$$D_{(\alpha)} f(x, y) = \underset{r \to 0}{Lt} \frac{f(x + r \cos \alpha, y + r \sin \alpha) - f(x, y)}{r}.$$

From this, we can easily deduce the derivative along x axis i.e., for $\alpha = 0$, also the derivative along y axis

i.e., for $\alpha = \dfrac{\pi}{2}$.

Thus $$D_{(0)} f(x, y) = \underset{r \to 0}{Lt} \frac{f(x+r, y) - f(x, y)}{r}$$
$$= \frac{\partial f}{\partial x}.$$

and $$D_{\left(\frac{\pi}{2}\right)} f(x, y) = \underset{r \to 0}{Lt} \frac{f(x, y+r) - f(x, y)}{r}$$
$$= \frac{\partial f}{\partial y}.$$

Note : Since partial derivatives have both magnitude and direction, it is a vector quantity. Thus if $\overrightarrow{G}$ be a vector whose components are f_x and f_y along two rectangular axes x and y respectively, then

$\overrightarrow{G} = i f_x + j f_y$ where i, j are the unit vectors along the axes respectively.

Theorem :

Let f(x, y) be a differentiable function of x and y, $\vec{a}$ a vector whose components are cos ∝, sin ∝, and $\vec{G}$, a vector whose components are f_x, f_y along two perpendicular axes. Then

$$\vec{G} \cdot \vec{a} = \underset{r \to 0}{Lt} \frac{f(x+r\cos\alpha, y+r\sin\alpha)-f(x, y)}{r}$$

we have $\vec{G} = if_x + jf_y$

and $\vec{a} = i\cos\alpha + j\sin\alpha$

$$\therefore \quad \vec{G} \cdot \vec{a} = (if_x + jf_y) \cdot (i\cos\alpha + j\sin\alpha)$$

$$= f_x \cos\alpha + f_y \sin\alpha \qquad \ldots \qquad \ldots \qquad (1)$$

Again since $f(x, y)$ is a differentiable function of (x, y) we must have

$$f(x+r\cos\alpha, y+r\sin\alpha) - f(x, y) = r\cos\alpha \cdot f_x(x, y) + r\sin\alpha \cdot f_y(x, y) + r\cos\alpha \cdot \phi(r\cos\alpha, r\sin\alpha) + r\sin\alpha \cdot \psi(r\cos\alpha, r\sin\alpha)$$

where $\phi(r\cos\alpha, r\sin\alpha) = 0$ and $\psi(r\cos\alpha, r\sin\alpha) = 0$.

or,
$$\frac{f(x+r\cos\alpha, y+r\sin\alpha)-f(x, y)}{r} = \cos\alpha \cdot f_x(x, y) + \sin\alpha \cdot f_y(x, y) + \cos\alpha \cdot \phi(r\cos\alpha, r\sin\alpha) + \sin\alpha \cdot \psi(r\cos\alpha, r\sin\alpha).$$

Now proceeding to the limit as $r \to 0$, we get

$$\underset{r \to 0}{Lt} \frac{f(x+r\cos\alpha, y+r\sin\alpha)-f(x, y)}{r}$$

$$= \cos\alpha \cdot f_x(x, y) + \sin\alpha \cdot f_y(x, y)$$

$$= f_x \cos\alpha + f_y \sin\alpha$$

$$= \vec{G} \cdot \vec{a} \quad \text{by (1).}$$

3.9. Euler's Theorem for a function of three variables.

If $f(x, y, z)$ be a homogeneous function in x, y, z of degree n, possessing continuous partial derivatives, then

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = n \cdot f(x, y, z).$$

Proof. Since $f(x, y, z)$ is a homogeneous function in x, y, z of degree n, we can write

$$f(tx, ty, tz) = t^n \,.\, f(x, y, z) \cdots \qquad \cdots \qquad (1)$$

for all values of t.

Now replacing tx, ty, tz by u, v, w on L.H.S of (1) we get

$$f(u, v, w) = t^n \,.\, f(x, y, z) \cdots \qquad \cdots \qquad (2)$$

where u, v, w are all functions of t,

$\therefore$ Differentiating (2) w. r. to t.

$$\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial t} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial t} = nt^{n-1} \,.\, f(x, y, z)$$

or, $$x\frac{\partial f}{\partial u} + y\frac{\partial f}{\partial v} + z\frac{\partial f}{\partial v} = n \,.\, t^{n-1} . \, f(x, y, z)$$

which is true for all values of t.

$\therefore$ Putting $t=1$ we get

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = n \,.\, f(x, y, z).$$

3.10. Converse of Euler's Theorem for a function of three variables.

If $\mathbf{f(x, y, z)}$ admits of continuous partial derivatives satisfying the relation

$$\mathbf{x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = n.\; f(x, y, z)}$$

where n is a positive integer, then

$\mathbf{f(x, y, z)}$ is a homogeneous function of degree n.

Proof. Let $\xi = \frac{x}{z}, \eta = \frac{y}{z}, \zeta = z \qquad \cdots \qquad \cdots \qquad (1)$

so that $x = \xi z = \xi\zeta,$

$y = \eta z = \eta\zeta$

and $z = \zeta$

Then $f(x, y, z)$ can be expressed as a function ξ, η, ζ

So let $f(x, y, z) = g(\xi, \eta, \zeta)$

Then $$x\frac{\partial f}{\partial x}=x\left(\frac{\partial g}{\partial \xi}.\frac{\partial \xi}{\partial x}+\frac{\partial g}{\partial \eta}.\frac{\partial \eta}{\partial x}+\frac{\partial g}{\partial \zeta}.\frac{\partial \zeta}{\partial x}\right)$$

$$=x\left(\frac{\partial g}{\partial \xi}.\frac{1}{z}+\frac{\partial g}{\partial \eta}.0+\frac{\partial g}{\partial \zeta}.0\right) \qquad \text{(from 1)}$$

$$=\frac{x}{z}\frac{\partial g}{\partial \xi}=\xi\frac{\partial g}{\partial \xi}. \qquad \cdots \quad \cdots \quad (2)$$

$$y\frac{\partial f}{\partial y}=y\left(\frac{\partial g}{\partial \xi}.\frac{\partial \xi}{\partial y}+\frac{\partial g}{\partial \eta}.\frac{\partial \eta}{\partial y}+\frac{\partial g}{\partial \zeta}.\frac{\partial \zeta}{\partial y}\right)$$

$$=y\left(\frac{\partial g}{\partial \xi}.0+\frac{\partial g}{\partial \eta}.\frac{1}{z}+\frac{\partial g}{\partial \eta}.0\right)$$

$$=\frac{y}{z}\frac{\partial g}{\partial \eta}=\eta\frac{\partial g}{\partial \eta} \qquad \cdots \quad \cdots \quad (3)$$

and $$z\frac{\partial f}{\partial z}=z\left(\frac{\partial g}{\partial \xi}.\frac{\partial \xi}{\partial z}+\frac{\partial g}{\partial \eta}.\frac{\partial \eta}{\partial z}+\frac{\partial g}{\partial \zeta}.\frac{\partial \zeta}{\partial z}\right)$$

$$=z\left\{\frac{\partial g}{\partial \xi}\left(-\frac{x}{z^2}\right)+\frac{\partial g}{\partial \eta}\left(-\frac{y}{z^2}\right)+\frac{\partial g}{\partial \zeta}.1\right\}$$

$$=-\frac{x}{z}\frac{\partial g}{\partial \xi}-\frac{y}{z}\frac{\partial g}{\partial \eta}+z\frac{\partial g}{\partial \zeta}$$

$$=-\xi\frac{\partial g}{\partial \xi}-\eta\frac{\partial g}{\partial \eta}+\zeta\frac{\partial g}{\partial \zeta}. \qquad \cdots \quad \cdots \quad (4)$$

Putting (2), (3) and (4) in the given relation

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+z\frac{\partial f}{\partial z}=n.f(x, y, z)$$

we get $$\zeta\frac{\partial g}{\partial \zeta}=n.g(\xi, \eta, \zeta)$$

$$=n.g$$

or, $$\frac{1}{g}\frac{\partial g}{\partial \zeta}=\frac{n}{\zeta}$$

$\therefore$ Integrating w.r. to ζ we get

$$\log g=n\log\zeta+C \text{ (constant)} \qquad \cdots \quad \cdots \quad (5)$$

Hence, the constant C is independent of ζ, but it may depend on ξ and η.

So let $C=\log\phi(\xi, \eta)$.

Then from (5)

$$\log g = n \log \zeta + \log \phi(\xi, \eta)$$

$$= \log \zeta^n . \phi(\xi, \eta)$$

$$\therefore \; g = \zeta^n \phi(\xi, \eta)$$

i.e., $g(\xi, \eta, \zeta) = \zeta^n . \phi(\xi, \eta)$

i.e., $f(x, y, z) = z^n . \phi\left(\frac{x}{z}, \frac{y}{z}\right).$

which shows that $f(x, y, z)$ is a homogeneous function of degree n.

Cor. 1. If $f(x_1, x_2 \cdots x_m)$ be a homogeneous function of m variables of degree n having continuous partial derivatives, then

$$x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} + \cdots + x_m \frac{\partial f}{\partial x_m} = n . f(x_1, x_2 \cdots x_m)$$

Cor. 2. If u be a homogeneous function in x, y, z of degree n, then $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}$ are each a homogeneous function in x, y, z of degree $(n-1)$.

We have $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = nu$

Differentiating this w.r.to x we get

$$x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x \partial y} + z\frac{\partial^2 u}{\partial x \partial z} = n\frac{\partial u}{\partial x}$$

or, $x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} + z\frac{\partial^2 u}{\partial x \partial z} = (n-1)\frac{\partial u}{\partial x}$

or, $x\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) + y\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) + z\frac{\partial}{\partial z}\left(\frac{\partial u}{\partial x}\right) = (n-1)\frac{\partial u}{\partial x}$

$\therefore$ $\frac{\partial u}{\partial x}$ is a homogeneous function of degree $(n-1)$. Similarly, for $\frac{\partial u}{\partial y}$ and $\frac{\partial u}{\partial z}$.

Illustrative Examples :

Ex. 1. If $v = f(u)$ where u is a homogeneous function of degree n in x, y, then $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = nu\frac{dv}{du}$.

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = x\frac{\partial}{\partial x}f(u) + y\frac{\partial}{\partial y}f(u)$$

$$= xf'(u)\frac{\partial u}{\partial x} + yf'(u)\frac{\partial u}{\partial y}$$

$$= \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)f'(u)$$

$$= nu.\frac{dv}{du}.$$

Ex. 2. If $u = v(y-z,\ z-x,\ x-y)$, then prove that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0. \qquad \textbf{(C. H. 1960)}$$

Let $\xi = y - z$, $\eta = z - x$, $\zeta = x - y$.

Then $u = v(\xi, \eta, \zeta)$

$$\therefore\quad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial x} + \frac{\partial u}{\partial \zeta}\,\frac{\partial \zeta}{\partial x}$$

$$= \frac{\partial u}{\partial \zeta}\,.\,0 + \frac{\partial u}{\partial \eta}\,.\,(-1) + \frac{\partial u}{\partial \zeta}\,.\,1 = \frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \eta}.$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} + \frac{\partial u}{\partial \zeta}\cdot\frac{\partial \zeta}{\partial y}$$

$$= \frac{\partial u}{\partial \xi}\,.\,(1) + \frac{\partial u}{\partial \eta}\,.\,0 + \frac{\partial u}{\partial \zeta}(-1) = \frac{\partial u}{\partial \zeta} - \frac{\partial u}{\partial \zeta}.$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial z} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial z} + \frac{\partial u}{\partial \zeta}\cdot\frac{\partial \zeta}{\partial z}$$

$$= \frac{\partial u}{\partial \xi}\,.\,(-1) + \frac{\partial u}{\partial \eta}\,.\,1 + \frac{\partial u}{\partial \zeta}\,.\,0 = \frac{\partial u}{\partial \eta} - \frac{\partial u}{\partial \xi}.$$

So adding, we get

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0.$$

Ex. 3. If $u=f(x^2+2yz,\ y^2+2zx)$, prove that

$$(y^2-zx)\frac{\partial u}{\partial x}+(x^2-yz)\frac{\partial u}{\partial y}+(z^2-xy)\frac{\partial u}{\partial z}=0$$

(C. H. 1947)

Let $\xi=x^2+2yz$ and $\eta=y^2+2zx$, then $u=f(\xi,\eta)$

$$\therefore\quad \frac{\partial u}{\partial x}=\frac{\partial f}{\partial \xi}\cdot\frac{\partial \xi}{\partial x}+\frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial x}=\frac{\partial f}{\partial \xi}(2x)+\frac{\partial f}{\partial \eta}(2z)$$

$$\frac{\partial u}{\partial y}=\frac{\partial f}{\partial \xi}\cdot\frac{\partial \xi}{\partial y}+\frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial y}=\frac{\partial f}{\partial \xi}(2z)+\frac{\partial f}{\partial \eta}(2y)$$

$$\frac{\partial u}{\partial z}=\frac{\partial f}{\partial \xi}\cdot\frac{\partial \xi}{\partial z}+\frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial z}=\frac{\partial f}{\partial \xi}(2y)+\frac{\partial f}{\partial \eta}(2x)$$

$$\therefore\quad \Sigma(y^2-zx)\frac{\partial u}{\partial x}=2\frac{\partial f}{\partial \xi}\{x(y^2-zx)+z(x^2-yz)+y(z^2-xy)\}$$

$$+2\frac{\partial f}{\partial \eta}\{z(y^2-zx)+y(x^2-yz)+x(z^2-xy)\}=0.$$

Ex. 4. If $u=\tan^{-1}\dfrac{x^3+y^3}{x^2+y^2}$, show that

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\sin 2u$$

(C. H. 1963)

Here, $\tan u=\dfrac{x^3+y^3}{x^2+y^2}=\dfrac{x^3\left(1+\dfrac{y^3}{x^3}\right)}{x^2\left(1+\dfrac{y^2}{x^2}\right)}$

$$=x\left\{1+\left(\frac{y}{x}\right)^3\right\}\cdot\left\{1+\left(\frac{y}{x}\right)^2\right\}^{-1}$$

$$=x\phi\left(\frac{y}{x}\right).$$

$\therefore$ $\tan u$ is of degree 1.

Hence by Euler's Theorem

$$x\frac{\partial}{\partial x}(\tan u)+y\frac{\partial}{\partial y}(\tan u)=1\,.\,\tan u.$$

or, $\quad x\sec^2 u\dfrac{\partial u}{\partial x}+y\sec^2 u\dfrac{\partial u}{\partial y}=\tan u$

or, $\quad x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=\dfrac{\tan u}{\sec^2 u}=\dfrac{1}{2}\sin 2u.$

Ex. 5. **If** $u(x, y, z, t)=\frac{1}{r} f(t+r)+\frac{1}{r} g(t-r)$

when $r^2=x^2+y^2+z^2$, **prove that u satisfies the relation** $u_{xx}+u_{yy}+u_{zz}=u_{tt}$ **(C. H. 1961)**

$$\frac{\partial u}{\partial x}=\frac{1}{r} f'(t+r)\frac{\partial r}{\partial x}-\frac{1}{r^2}\frac{\partial r}{\partial x}\cdot f(t+r)$$

$$+\frac{1}{r} g'(t-r)\left(-\frac{\partial r}{\partial x}\right)-\frac{1}{r^2}\frac{\partial r}{\partial x}\cdot g(t-r)$$

$$=\frac{1}{r} f'(t+r)\left(\frac{x}{r}\right)-\frac{1}{r^2}\left(\frac{x}{r}\right) f(t+r) \qquad \because \frac{\partial r}{\partial x}=\frac{x}{r}$$

$$-\frac{1}{r} g'(t-r)\left(\frac{x}{r}\right)-\frac{1}{r^2}\left(\frac{x}{r}\right) g(t-r)$$

$$=\frac{x}{r^2} f'(t+r)-\frac{x}{r^3} f(t+r)-\frac{x}{r^2} g'(t-r)-\frac{x}{r^3} g(t-r)$$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2}=\left\{\frac{1}{r^2} f'(t+r)-\frac{2x}{r^3}\frac{\partial r}{\partial x} f(t+r)+\frac{x}{r^2} f''(t+r)\frac{\partial r}{\partial x}\right\}$$

$$-\left\{\frac{1}{r^3} f(t+r)-\frac{3x}{r^4}\frac{\partial r}{\partial x} f(t+r)+\frac{x}{r^3} f'(t+r)\frac{\partial r}{\partial x}\right\}$$

$$-\left\{\frac{1}{r^2} g'(t-r)-\frac{2x}{r^3}\frac{\partial r}{\partial x} g'(t-r)+\frac{x}{r^2} g''(t-r)\left(-\frac{\partial r}{\partial x}\right)\right\}$$

$$-\left\{\frac{1}{r^3} g(t-r)-\frac{3x}{r^4}\frac{\partial r}{\partial x} g(t-r)+\frac{x}{r^3} g'(t-r)\left(-\frac{\partial r}{\partial x}\right)\right\}$$

$$=\frac{x^2}{r^5} f''(t+r)+\left(\frac{1}{r^2}-\frac{2x^2}{r^4}-\frac{x^2}{r^4}\right) f'(t+r)+\left(\frac{3x^2}{r^5}-\frac{1}{r^3}\right) f(t+r)$$

$$+\frac{x^2}{r^3} g''(t-r)+\left(\frac{2x^2}{r^4}+\frac{x^2}{r^4}-\frac{1}{r^2}\right) g'(t-r)$$

$$+\left(\frac{3x^2}{r^5}-\frac{1}{r^3}\right) g(t-r)$$

Similar expression for $\frac{\partial^2 u}{\partial y^2}$ and $\frac{\partial^2 u}{\partial z^2}$.

$$\therefore \quad u_{xx}+u_{yy}+u_{zz}=\frac{1}{r}\{f''(t+r)+g''(t-r)\}$$

$$=u_{tt}. \text{ from the given relation.}$$

Ex. 6. If $z=xf(x+y)+yg(x+y)$ show that

$$\frac{\partial^2 z}{\partial x^2}-2\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial^2 z}{\partial y^2}=0. \qquad \text{(C. H. 1971)}$$

Put $v=x+y$, Then $\dfrac{\partial v}{\partial x}=1$ and $\dfrac{\partial v}{\partial y}=1.$

Now $z=xf(v)+yg(v)$

$$\therefore\ \frac{\partial z}{\partial x}=f(v)+xf'(v)\frac{\partial v}{\partial x}+yg'(v)\frac{\partial v}{\partial x}$$

$$=f(v)+xf'(v)+yg'(v)$$

$$\frac{\partial^2 z}{\partial x^2}=f'(v)+f'(v)+xf''(v)+yg''(v)$$

$$=2f'(v)+xf''(v)+yg''(v)$$

$$\frac{\partial z}{\partial y}=xf'(v)+g(v)+yg'(v)$$

$$\therefore\ \frac{\partial^2 z}{\partial y^2}=xf''(v)+g'(v)+g'(v)+yg''(v)$$

$$=2g'(v)+xf''(v)+yg''(v)$$

Also $\dfrac{\partial^2 z}{\partial x\partial y}=f'(v)+xf''(v)+g'(v)+yg''(v)$

Hence, $\dfrac{\partial^2 z}{\partial x^2}-2\dfrac{\partial^2 z}{\partial x\partial y}+\dfrac{\partial^2 z}{\partial y^2}=0.$

Ex. 7. If $\dfrac{x^2}{a^2+u}+\dfrac{y^2}{b^2+u}+\dfrac{z^2}{c^2+u}=1$

where u is a function of x, y, z prove that

$$(u_x)^2+(u_y)^2+(u_z)^2=2(xu_x+yu_y+zu_z) \qquad \text{(C. H. 1966)}$$

Differentiating the given relation w. r. to x partially

$$\frac{2x(a^2+u)-x^2u_x}{(a^2+u)^2}-\frac{y^2u_x}{(b^2+u)^2}-\frac{z^2u_x}{(c^2+u)^2}=0$$

or, $$u_x\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right\}=\frac{2x}{a^2+u} \quad \cdots (1)$$

Similarly, differentiating the given relation w. r. to y and z respectively, we get

$$u_y\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right\}=\frac{2y}{a^2+u} \quad \cdots (2)$$

and $$u_z\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right\}=\frac{2z}{a^2+u} \quad \cdots \quad (3)$$

Squaring (1), (2), (3) and adding

$$\left\{(u_x)^2+(u_y)^2+(u_z)^2\right\}\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(a^2+u)^2}+\frac{z^2}{(a^2+u)^2}\right\}^2$$

$$=4\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(a^2+u)^2}+\frac{z^2}{(a^2+u)^2}\right\}$$

or, $$(u_x)^2+(u_y)^2+(u_z)^2=4\bigg/\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(a^2+u)^2}+\frac{z^2}{(a^2+u)^2}\right\} \quad \cdots \quad (4)$$

Again multiplying (1), (2), (3) by x, y, z and adding

$$(xu_x+yu_y+zu_z)\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right\}$$

$$=2\left\{\frac{x^2}{a^2+u}+\frac{y^2}{b^2+u}+\frac{z^2}{c^2+u}\right\}$$

$$=2.$$

$$\therefore \quad xu_x+yu_y+zu_z=2\bigg/\left\{\frac{x^2}{(a^2+u)^2}+\frac{y^2}{(b^2+u)^2}+\frac{z^2}{(c^2+u)^2}\right. \quad \cdots \quad (5)$$

Dividing (4) by (5)

$$\frac{(u_x)^2+(u_y)^2+(u_z)^2}{xu_x+yu_y+zu_z}=2.$$ Hence the result

Ex. 8. If $y_1=f_1(x_1, x_2, x_3)$, $y_2=f_2(x_1, x_2, x_3)$ are differentjal functions of x_1, x_2, x_3 in a region R and $x_1=\phi_1(t_1, t_2)$, $x_2=\phi_2(t_1, t_2)$, $x_2=\phi_3(t_1, t_2)$ are differential functions of t_1, t_2 in a region S, then prove that

$$\frac{\partial(y_1, y_2)}{\partial(t_1, t_2)}=\frac{\partial(y_1, y_2)}{\partial(x_1, x_2)}\cdot\frac{\partial(x_1, x_2)}{\partial(t_1, t_2)}+\frac{\partial(y_1, y_2)}{\partial(x_2, x_3)}\cdot\frac{\partial(x_2, y_3)}{\partial(t_1, t_2)}$$

$$+\frac{\partial(y_1, y_2)}{\partial(x_3, x_1)}\cdot\frac{\partial(x_3, x_1)}{\partial(t_1, t_2)}$$

where $\dfrac{\partial(u, v)}{\partial(p, q)}$ denotes $\begin{vmatrix} \dfrac{\partial u}{\partial p} & \dfrac{\partial u}{\partial q} \\ \dfrac{\partial v}{\partial p} & \dfrac{\partial v}{\partial q} \end{vmatrix}$ (C. H. 1967)

we have $\dfrac{\partial(y_1, y_2)}{\partial(t_1, t_2)} = \begin{vmatrix} \dfrac{\partial y_1}{\partial t_1} & \dfrac{\partial y_1}{\partial t_2} \\ \dfrac{\partial y_2}{\partial t_1} & \dfrac{\partial y_2}{\partial t_2} \end{vmatrix}$

$$= \begin{vmatrix} \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_1} & \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_2}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \\ \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_1} & \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_2}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$= \begin{vmatrix} \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}, & \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_2} \\ \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}, & \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_2} \end{vmatrix}$$

$$+ \begin{vmatrix} \dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_1}, & \dfrac{\partial y_1}{\partial x_2}\dfrac{\partial x_2}{\partial t_2}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \\ \dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_1}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_1}, & \dfrac{\partial y_2}{\partial x_2}\dfrac{\partial x_2}{\partial t_2}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$+ \begin{vmatrix} \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_1}, & \dfrac{\partial y_1}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_1}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \\ \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_1}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_1}, & \dfrac{\partial y_2}{\partial x_1}\dfrac{\partial x_1}{\partial t_2}+\dfrac{\partial y_2}{\partial x_3}\dfrac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$= \begin{vmatrix} \dfrac{\partial y_1}{\partial x_1} & \dfrac{\partial y_1}{\partial x_2} \\ \dfrac{\partial y_2}{\partial x_1} & \dfrac{\partial y_2}{\partial x_2} \end{vmatrix} \begin{vmatrix} \dfrac{\partial x_1}{\partial t_1} & \dfrac{\partial x_2}{\partial t_1} \\ \dfrac{\partial x_1}{\partial t_2} & \dfrac{\partial x_2}{\partial t_2} \end{vmatrix} + \begin{vmatrix} \dfrac{\partial y_1}{\partial x_2} & \dfrac{\partial y_1}{\partial x_3} \\ \dfrac{\partial y_2}{\partial x_2} & \dfrac{\partial y_2}{\partial x_3} \end{vmatrix} \begin{vmatrix} \dfrac{\partial x_2}{\partial t_1} & \dfrac{\partial x_3}{\partial t_1} \\ \dfrac{\partial x_2}{\partial t_2} & \dfrac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$+ \begin{vmatrix} \dfrac{\partial y_1}{\partial x_1} & \dfrac{\partial y_1}{\partial x_3} \\ \dfrac{\partial y_2}{\partial x_1} & \dfrac{\partial y_2}{\partial x_3} \end{vmatrix} \begin{vmatrix} \dfrac{\partial x_1}{\partial t_1} & \dfrac{\partial x_3}{\partial t_1} \\ \dfrac{\partial x_1}{\partial t_2} & \dfrac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$=\begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{vmatrix} \begin{vmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} \end{vmatrix} + \begin{vmatrix} \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \end{vmatrix} \begin{vmatrix} \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} \\ \frac{\partial x_3}{\partial t_1} & \frac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$+\begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_3} \end{vmatrix} \begin{vmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} \\ \frac{\partial x_3}{\partial t_1} & \frac{\partial x_3}{\partial t_2} \end{vmatrix}$$

$$=\frac{\partial(y_1, y_2)}{\partial(x_1, x_2)} \cdot \frac{\partial(x_1, x_2)}{\partial(t_1, t_2)} + \frac{\partial(y_1, y_2)}{\partial(x_2, x_3)} \cdot \frac{\partial(x_2, x_3)}{\partial(t_1, t_2)}$$

$$+\frac{\partial(y_1, y_2)}{\partial(x_1, x_3)} \cdot \frac{\partial(x_1, x_3)}{\partial(t_1, t_2)}.$$

Ex. 9. If $f=f(x, y, z)$, $g=g(x, y, z)$ are differential functions of x, y, z and $x=x(p, q)$, $y=y(p, q)$, $z=z(p, q)$ are differential functions of p, q, then prove that

$$\frac{\partial(f, g)}{\partial(p, q)}=\frac{\partial(f, g)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(p, q)}+\frac{\partial(f, g)}{\partial(y, z)} \frac{\partial(y, z)}{\partial(p, q)}+\frac{\partial(f, g)}{\partial(z, x)} \cdot \frac{\partial(z, x)}{\partial(p, q)}$$

where $\frac{\partial(f, g)}{\partial(p, q)}$ **stands for** $f_p g_q - f_q g_p$ (*C. H. 1969*)

We have $\frac{\partial(f, g)}{\partial(p, q)}=\begin{vmatrix} f_p & f_q \\ g_p & g_q \end{vmatrix}$

$$=\begin{vmatrix} f_x x_p+f_y y_p+f_z z_p & f_x x_q+f_y y_q+f_z z_q \\ g_x x_p+g_y y_p+g_z z_p & g_x x_q+g_y y_q+g_z z_q \end{vmatrix}$$

$$=\begin{vmatrix} f_x x_p+f_y y_p, & f_x x_q+f_y y_q \\ g_x x_p+g_y y_p & g_x x_q+g_y y_q \end{vmatrix}+\begin{vmatrix} f_y y_p+f_z z_p, & f_y y_q+f_z z_q \\ g_y y_p+g_z z_p & g_y y_q+g_z z_q \end{vmatrix}$$

$$+\begin{vmatrix} f_x x_p+f_z z_p, & f_x x_q+f_z z_q \\ g_x x_p+g_z z_p, & g_x x_q+g_z z_q \end{vmatrix}$$

$$=\begin{vmatrix} f_x & f_y \\ g_x & g_y \end{vmatrix} \begin{vmatrix} x_p & y_p \\ x_q & y_q \end{vmatrix}+\begin{vmatrix} f_y & f_z \\ g_y & g_z \end{vmatrix} \begin{vmatrix} y_p & z_p \\ y_q & z_q \end{vmatrix}$$

$$+\begin{vmatrix} f_z & f_x \\ g_z & g_x \end{vmatrix} \begin{vmatrix} z_p & x_p \\ z_q & x_q \end{vmatrix}$$

$$=\begin{vmatrix} f_x & f_y \\ g_x & g_y \end{vmatrix}\begin{vmatrix} x_p & x_q \\ y_p & y_q \end{vmatrix}+\begin{vmatrix} f_y & f_z \\ g_y & g_z \end{vmatrix}\begin{vmatrix} y_p & y_q \\ z_p & z_q \end{vmatrix}$$

$$+\begin{vmatrix} f_z & f_x \\ g_z & g_x \end{vmatrix}\begin{vmatrix} z_p & z_q \\ x_p & x_q \end{vmatrix}$$

$$=\frac{\partial(f,g)}{\partial(x,y)}\cdot\frac{\partial(x,y)}{\partial(p,q)}+\frac{\partial(f,g)}{\partial(y,z)}\cdot\frac{\partial(y,z)}{\partial(p,q)}+\frac{\partial(f,g)}{\partial(z,x)}\cdot\frac{\partial(z,x)}{\partial(p,q)}.$$

Ex. 10. If Pdx+Qdy+Rdz can be made a perfect differential of some function of x, y, z, on multiplication by a factor then

$$\mathbf{P}\left(\frac{\partial Q}{\partial z}-\frac{\partial R}{\partial y}\right)+\mathbf{Q}\left(\frac{\partial R}{\partial x}-\frac{\partial P}{\partial z}\right)+\mathbf{R}\left(\frac{\partial P}{\partial y}-\frac{\partial Q}{\partial x}\right)=0,$$

(C. H. 1949, 54)

Let $u=f(x, y, z)$ so that

$$K(Pdx+Qdy+Rdz = du \quad \ldots \quad \ldots \quad (1)$$

Then K must be some function of x, y, z.

But, since u is a function of x, y, z, we can write

$$du=\frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy+\frac{\partial u}{\partial z}dz \quad \ldots \quad \ldots \quad (2)$$

Comparing (1) and (2) we get

$$\frac{\partial u}{\partial x}=KP, \quad \ldots \quad \ldots \quad (3)$$

$$\frac{\partial u}{\partial y}=KQ. \quad \ldots \quad \ldots \quad (4)$$

$$\frac{\partial u}{\partial z}=KR, \quad \ldots \quad \ldots \quad (5)$$

Differentiating (3) w. r. to y and (4) w. r. to x.

$$\frac{\partial^2 u}{\partial y\partial x}=K\frac{\partial P}{\partial y}+P\frac{\partial K}{\partial y}$$

and
$$\frac{\partial^2 u}{\partial x\partial y}=K\frac{\partial Q}{\partial x}+Q\frac{\partial K}{\partial x}.$$

$\therefore$ Assuming $\dfrac{\partial^2 u}{\partial x\partial y}=\dfrac{\partial^2 u}{\partial y\partial x}$, we get

$$K\frac{\partial P}{\partial v}+P\frac{\partial K}{\partial y}=K\frac{\partial Q}{\partial x}+Q\frac{\partial K}{\partial x}$$

or, $$K\left(\frac{\partial P}{\partial y}-\frac{\partial Q}{\partial x}\right)=Q\frac{\partial K}{\partial x}-P\frac{\partial K}{\partial y}. \quad \ldots \quad (6)$$

Similarly, differentiating (4) w. r to z and (5) w. r. to y

$$K\left(\frac{\partial Q}{\partial z}-\frac{\partial R}{\partial y}\right)=R\frac{\partial K}{\partial y}-Q\frac{\partial K}{\partial z}. \quad \ldots \quad (7)$$

Also differentiating (3) w. r. to z and (5) w. r. to x

$$K\left(\frac{\partial R}{\partial x}-\frac{\partial P}{\partial z}\right)=P\frac{\partial K}{\partial z}-R\frac{\partial K}{\partial x}. \quad \ldots \quad (8)$$

Now multiplying (6) by R, (7) by P and (8) by Q and adding the result follows.

Exercise 3

1. If $u=\tan^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$,

   show that $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=\dfrac{1}{4}\sin 2u$.

2. If $V=f(z)$ where z is a homogeneous function in x, y of degree n, then show that

   $xV_x+yV_y=nz\dfrac{dV}{dz}$.

3. Given $f_1(x, y)=0$ and $f_2(x, z)=0$,

   show that

   $\dfrac{\partial f_2}{\partial z}\cdot\dfrac{\partial f_1}{\partial y}\cdot\dfrac{dy}{dz}=\dfrac{\partial f_1}{\partial x}\cdot\dfrac{\partial f_2}{\partial x}$.

4. If $F(p, v, t)=0$, show that

   $$\left[\frac{dp}{dt}\right]_{v\text{ const.}}\times\left[\frac{dv}{dp}\right]_{t\text{ const.}}\times\left(\frac{dt}{dv}\right)_{p\text{ const.}}=-1.$$

5. Let $z=f(x, y)$ where $x=e^u+e^{-v}$

   and $y=e^{-u}-e^v$,

   show that $\dfrac{\partial z}{\partial u}-\dfrac{\partial z}{\partial v}=x\dfrac{\partial z}{\partial x}-y\dfrac{\partial z}{\partial y}$.

6. If $u=f(x, y, z)$ possess continuous partial derivatives, and if each of x, y, z is a function of t, then prove that,

   $$\frac{du}{dt}=\frac{\partial u}{\partial x}\frac{dx}{dt}+\frac{\partial u}{\partial y}\frac{dy}{dt}+\frac{\partial u}{\partial z}\frac{dz}{dt}.$$

7. V is a homogeneous function in x, y, z and t of order n : prove that $xV_x+yV_y+zV_z+tV_t=nV$.

8. If $u=f(x^2+2yz, y^2+2zx)$, prove that

$$(y^2-zx)\frac{\partial u}{\partial x}+(x^2-yz)\frac{\partial u}{\partial y}+(z^2-xy)\frac{\partial u}{\partial z}=0.$$ (C. H. 1947)

9. If $xyz=\log_e u$, prove that

$$u_{xyz}=(1+3xyz+x^2y^2z^2)e^{xyz}$$

10. Given $V=\log(x^3+y^3+z^3-3xyz)$

show that $\dfrac{\partial^2 V}{\partial x^2}+\dfrac{\partial^2 V}{\partial y^2}+\dfrac{\partial^2 V}{\partial z^2}=-\dfrac{3}{(x+y+z)^2}$.

11. If $u=f(ax^2+2hxy+by^2)$, $v=\phi(ax^2+2hxy+by^2)$,

show that $\dfrac{\partial}{\partial y}\left(u\dfrac{\partial v}{\partial x}\right)=\dfrac{\partial}{\partial x}\left(u\dfrac{\partial v}{\partial y}\right)$ (C. H. 1934)

12. Let $x=x(\xi, \eta, \zeta)$, $y=(\xi, \eta, \zeta)$ are differential function of ξ, η, ζ and $\xi=\xi(t_1\ t_2)$, $\eta=\eta(t_1\ t_2)$ and $\zeta=\zeta(t_1, t_2)$ are differential function of t_1 and t_2.
Prove that

$$\frac{\partial(x, y)}{\partial(t_1, t_2)}=\frac{\partial(x, y)}{\partial(\xi, \eta)}\cdot\frac{\partial(\xi, \eta)}{\partial(t_1, t_2)}+\frac{\partial(x, y)}{\partial(\eta, \zeta)}\cdot\frac{\partial(\eta, \zeta)}{\partial(t_1, t_2)}+\frac{\partial(x, y)}{\partial(\zeta, \xi)}\cdot\frac{\partial(\zeta, \xi)}{\partial(t_1, t_2)}$$

13. Find a function $f(x, y)$ which is a function of x^2+y^2 and is also a product of the form $\Psi(x)$. $\Psi(y)$. (C. H. 1962)

14. If V be a homogeneous function of x, y, z of degree n and if,

$$\log u+\frac{1}{2}(n+1)\log(x^2+y^2+z^2)=0,$$

then show that

$$\frac{\partial}{\partial}\left(V\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(V\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(V\frac{\partial u}{\partial z}\right)=0.$$

15. If $X=r\cos\theta$, $Y=r\sin\theta$, prove that

$$\frac{\partial(x, y)}{\partial(r, \theta)}=r.$$

16. Find differential coefficient of y with respect to x from the relation

$$x^y+y^x=a^b.$$ (C. H. 1944)

17. If $x^2=vw$, $y^2=wu$, $z^2=uv$, and $f(x, y, z)=\phi(u, v, w)$ then show that

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+z\frac{\partial f}{\partial z}=u\frac{\partial \phi}{\partial u}+v\frac{\partial \phi}{\partial v}+w\frac{\partial \phi}{\partial w}.$$

(Maths Tripos 1939)

18. If the independent variables are changed from u, v to x, y and then to p, q, prove that

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial p} & \frac{\partial x}{\partial q} \\ \frac{\partial y}{\partial p} & \frac{\partial y}{\partial q} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial p} & \frac{\partial u}{\partial q} \\ \frac{\partial v}{\partial p} & \frac{\partial v}{\partial q} \end{vmatrix}$$

(C. H. 1953)

19. If $u=f_1(x+by)+f_2(x-by)$, prove that

$$\frac{\partial^2 u}{\partial y^2}=b^2\frac{\partial^2 u}{\partial x^2}.$$

20. Given $u=(x+y)\{1+f\left(\frac{y}{x}\right)\}$, show that

$$x\left(\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y \partial x}\right)=y\left(\frac{\partial^2 u}{\partial y^2}-\frac{\partial^2 u}{\partial x\, \partial y}\right).$$

21. If $f(x, y, z)=0$, show that

$$\frac{\partial z}{\partial x}=-\frac{\partial f}{\partial x}\Big/\frac{\partial f}{\partial z}\,;\quad \frac{\partial z}{\partial y}=-\frac{\partial f}{\partial y}\Big/\frac{\partial f}{\partial z}.$$

22. Let $x=u+\dfrac{\sin u}{e^v}$ and $y=v+\dfrac{\cos u}{e^v}$, where u and v are both functions of x and y.

Show that $\dfrac{\partial u}{\partial y}=\dfrac{\partial v}{\partial x}$.

CHAPTER IV

CHANGE OF VARIABLES

4.1. Choice of Independent Variables.

If the variables x, y, r, θ are connected by the two relations of the type

$$x = r\cos\theta \text{ and } y = r\sin\theta$$

then any one of the variables may be expressed in terms of two of the remaining three. Thus we can express x in terms of (r, θ), (r, y) or (θ, y).

Suppose we are to find $\frac{\partial r}{\partial x}$ from these relations. If x is expressed in terms of θ, y i. e., when $x = y\cos\theta$ we can not find $\frac{\partial x}{\partial r}$, since the independent variable r is absent in the equation. If x is expressed in terms of r, θ i.e., $x = r\cos\theta$, then we can find $\frac{\partial x}{\partial r}$ when θ remains constant. Again if x be expressed in terms of r, y i.e., $x = \sqrt{r^2 - y^2}$ we can find $\frac{\partial x}{\partial r}$ when y remains constant. So there is no reason to believe that the value of $\frac{\partial x}{\partial r}$ when θ remains constant will be equal to the value of $\frac{\partial x}{\partial r}$ when y remains constant.

Let us now write the partial derivatives of x w. r. to r when θ and y remains constant by the notations $\left[\frac{\partial x}{\partial r}\right]_\theta$ and $\left[\frac{\partial x}{\partial r}\right]_y$

Here, it can be easily shown that

$$\left[\frac{\partial x}{\partial r}\right]_\theta \neq \left[\frac{\partial x}{\partial r}\right]_y.$$

From $x=r\cos\theta$ we get

$$\left[\frac{\partial x}{\partial r}\right]_\theta=\cos\theta. \qquad \cdots \qquad (1)$$

Also from $x=\sqrt{r^2-y^2}$

$$\left[\frac{\partial x}{\partial r}\right]_y=\frac{r}{\sqrt{r^2-y^2}}=\frac{r}{x}=\sec\theta \qquad \cdots \qquad (2)$$

Thus form (1) and (2) $\left[\frac{\partial x}{\partial r}\right]_\theta\neq\left[\frac{\partial x}{\partial r}\right]_y$.

4.2. If y be a function of x only we can always change the independent variable x by the relation $\frac{dy}{dx}=1\Big/\frac{dx}{dy}$.

But such a change is not possible if y be a function of several variables. For consider relations $x=r\cos\theta$ and $y=r\sin\theta$

Then $r=\sqrt{x^2+y^2}$ and $\theta=\tan^{-1}\left(\frac{y}{x}\right)$

Now from $x=r\cos\theta$

$$\left[\frac{\partial x}{\partial r}\right]_\theta=\cos\theta$$

And from $r=\sqrt{x^2+y^2}$

$$\left[\frac{\partial r}{\partial x}\right]_y=\frac{x}{\sqrt{x^2+y^2}}=\frac{x}{r}=\cos\theta$$

$$\therefore \quad \left[\frac{\partial x}{\partial r}\right]_\theta\neq 1\Big/\left[\frac{\partial r}{\partial x}\right]_y. \qquad \cdots \qquad (1)$$

Again from $x=r\cos\theta$

$$\left[\frac{\partial x}{\partial \theta}\right]_r=-r\sin\theta$$

And from $\theta=\tan^{-1}\left(\frac{y}{x}\right)$

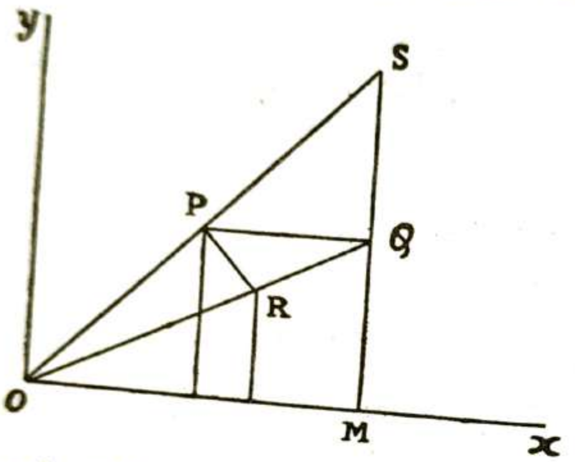
$$\left[\frac{\partial \theta}{\partial x}\right]_y=-\frac{\frac{y}{x^2}}{1+\frac{y^2}{x^2}}=-\frac{y}{x^2+y^2}=-\frac{y}{r^2}=-\frac{\sin\theta}{r}$$

$$\therefore \quad \left[\frac{\partial x}{\partial \theta}\right]_r \neq 1 \Big/ \left[\frac{\partial \theta}{\partial x}\right]_y \quad \ldots \quad (2)$$

In case (1) $\frac{\partial x}{\partial r}$ is the partial differentiation of x, w. r. to r when θ remains constant and $\frac{\partial r}{\partial x}$ is the partial differentiation of r, w. r. to x when y remains constant, so there is no reason that *L. H. S.* should be equal to the *R. H. S.* Similar reasons for (2).

Geometrical interpretaton for inequality

Let P be a point whose cartesian co-ordinates are (x, y) with respect to OX and OY as axes of co-ordinates. Let (r, θ) be its polar co-ordinates.



Let PQ $(=\delta x)$ be the increment of x when y remains constant. Then PQ is parallel to OX.

Join OQ and take a point R in OQ such that $OR=OP$. Then $OR=r$ and RQ $(=\delta r)$ is the increment of r corresponding to the increment PQ $(=\delta x)$ of x.

$$\therefore \quad \frac{\partial x}{\partial r} = Lt\frac{\delta x}{\delta r} = Lt\frac{PQ}{RQ}, \text{ or, } \frac{\partial r}{\partial x} = Lt\frac{RQ}{PQ}.$$

Again let PS $(=\delta r)$ be the increment of r when θ remains constant. Then $\frac{\partial x}{\partial r} = Lt\frac{\partial x}{\partial r} = Lt\frac{PQ}{PS}$.

But $RQ \neq PS$

$$\therefore \quad \left(\frac{\partial x}{\partial r}\right)_y \neq \left(\frac{\partial x}{\partial r}\right)_\theta$$

And $\left(\frac{\partial x}{\partial r}\right)_y \neq 1 \Big/ \left(\frac{\partial r}{\partial x}\right)_\theta.$

4.3. For a function of several variables, the independent variables can be changed to a second set of variables by proper substitution. This is illustrated in the following examples :

Ex. 1. **If v be a function of r alone where $\mathbf{r^2=x^2+y^2}$, show that**

$$\frac{\partial^2 \mathbf{v}}{\partial \mathbf{x}^2}+\frac{\partial^2 \mathbf{v}}{\partial \mathbf{y}^2}=\frac{\partial^2 \mathbf{v}}{\partial \mathbf{r}^2}+\frac{1}{\mathbf{r}}\frac{\partial \mathbf{v}}{\partial \mathbf{r}}$$

Since $v=f(r)$, $\quad \dfrac{dv}{dr}=f'(r)$

Again since r is a function of x, y

$$\frac{\partial v}{\partial x}=f'(r)\frac{\partial r}{\partial x}=\frac{dv}{dr}\cdot\frac{x}{r}$$

$$\therefore \quad \frac{\partial}{\partial x}=\frac{x}{r}\cdot\frac{d}{dr} \qquad \ldots \qquad \ldots \qquad (1)$$

$$\therefore \quad \frac{\partial^2 v}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right)=\frac{\partial}{\partial x}\left\{\frac{x}{r}\frac{dv}{dr}\right\}$$

$$=\frac{1}{r}\frac{dv}{dr}+x.\frac{\partial}{\partial x}\left\{\frac{1}{r}\frac{dv}{dr}\right\}$$

$$=\frac{1}{r}\frac{dv}{dr}+x.\frac{x}{r}\frac{d}{dr}\left\{\frac{1}{r}\frac{dv}{dr}\right\} \quad \text{by (1)}$$

$$=\frac{1}{r}\frac{dv}{dr}+\frac{x^2}{r}\left\{-\frac{1}{r^2}\frac{dv}{dr}+\frac{1}{r}\frac{d^2v}{dr^2}\right\}$$

$$=\frac{1}{r}\frac{dv}{dr}-\frac{x^2}{r^3}\frac{dv}{dr}+\frac{x^2}{r^2}\frac{d^2v}{dr^2}.$$

Similarly, $\dfrac{\partial^2 v}{\partial y^2}=\dfrac{1}{r}\dfrac{dv}{dr}-\dfrac{y^2}{r^3}\dfrac{dv}{dr}+\dfrac{y^2}{r^2}\dfrac{d^2v}{dr^2}.$

$$\therefore \quad \frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}=\frac{2}{r}\frac{dv}{dr}-\frac{(x^2+y^2)}{r^3}\frac{dv}{dr}+\frac{x^2+y^2}{r^2}\frac{d^2v}{dr^2}$$

$$=\frac{2}{r}\frac{dv}{dr}-\frac{1}{r}\frac{dv}{dr}+\frac{d^2v}{dr^2}$$

$$=\frac{1}{r}\frac{dv}{dr}+\frac{d^2v}{dr^2}.$$

Ex. 2. If v is a function of r alone where $r^2=x^2+y^2+z^2$, show that

$$\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}=\frac{d^2 v}{dr^2}+\frac{2}{r}\frac{dv}{dr}.$$

Since v is a function of r alone

$$\frac{\partial v}{\partial x}=\frac{dv}{dr}\frac{\partial r}{\partial x}=\frac{x}{r}\frac{dv}{dr} \qquad \text{(see Ex. 1)}$$

$$\therefore \quad \frac{\partial}{\partial x}=\frac{x}{r}\frac{d}{dr}.$$

$$\therefore \quad \frac{\partial^2 v}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{x}{r}\frac{dv}{dr}\right)$$

$$=\frac{1}{r}\frac{dv}{dr}+x\frac{\partial}{\partial x}\left(\frac{1}{r}\frac{dv}{dr}\right)$$

$$=\frac{1}{r}\frac{dv}{dr}+x.\frac{x}{r}\frac{d}{dr}\left(\frac{1}{r}\frac{dv}{dr}\right)$$

$$=\frac{1}{r}\frac{dv}{dr}+\frac{x^2}{r}\left(\frac{1}{r}\frac{d^2v}{dr^2}-\frac{1}{r^2}\frac{dv}{dr}\right)$$

$$=\frac{1}{r}\frac{dv}{dr}+\frac{x^2}{r^2}\frac{d^2v}{dr^2}-\frac{x^2}{r^3}\frac{dv}{dr}.$$

Similarly, $\dfrac{\partial^2 v}{\partial y^2}=\dfrac{1}{r}\dfrac{dv}{dr}+\dfrac{y^2}{r^2}\dfrac{d^2v}{dr^2}-\dfrac{y^2}{r^3}\dfrac{dv}{dr}$

And $\dfrac{\partial^2 v}{\partial z^2}=\dfrac{1}{r}\dfrac{dv}{dr}+\dfrac{z^2}{r^2}\dfrac{d^2v}{dr^2}-\dfrac{z^2}{r^3}\dfrac{dv}{dr}.$

$$\therefore \quad \frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}=\frac{3}{r}\frac{dv}{dr}+\frac{x^2+y^2+z^2}{r^2}\frac{d^2v}{dr^2}-\frac{x^2+y^2+z^2}{r^3}\frac{dv}{dr}$$

$$=\frac{3}{r}\frac{dv}{dr}+\frac{d^2v}{dr^2}-\frac{1}{r}\frac{dv}{dr}$$

$$=\frac{d^2v}{dr^2}+\frac{2}{r}\frac{dv}{dr}.$$

Ex. 3. If $x=r\cos\theta$, $y=r\sin\theta$ and v is a function of x, y possessing partial derivations of the 2nd order

prove that $\dfrac{\partial^2 v}{\partial x^2}+\dfrac{\partial^2 v}{\partial y^2}=\dfrac{1}{r^2}\left\{r\dfrac{\partial}{\partial r}\left(r\dfrac{\partial v}{\partial r}\right)+\dfrac{\partial^2 v}{\partial \theta^2}\right\}$

Since, x, y are functions of (r, θ), v is a function of (r, θ). So we write $v=f(r, \theta)$

$$\therefore \quad \frac{\partial v}{\partial x}=\frac{\partial v}{\partial r}\cdot\frac{\partial r}{\partial x}+\frac{\partial v}{\partial \theta}\cdot\frac{\partial \theta}{\partial x}$$

$$=\frac{\partial v}{\partial r}\cdot\frac{x}{r}-\frac{\partial v}{\partial \theta}\cdot\frac{y}{r^2}$$

$$=\frac{\partial v}{\partial r}\cos\theta-\frac{\partial v}{\partial \theta}\cdot\frac{\sin\theta}{r} \quad \ldots \quad (1)$$

$$\left[\because \quad r^2=x^2+y^2 \quad \frac{\partial r}{\partial x}=x/r\right.$$

$$\left.\because \quad \theta=\tan^{-1}\frac{y}{x}, \quad \frac{\partial \theta}{\partial x}=-\frac{y}{r^2}\right]$$

$$\frac{\partial v}{\partial y}=\frac{\partial v}{\partial r}\cdot\frac{\partial r}{\partial y}+\frac{\partial v}{\partial \theta}\cdot\frac{\partial \theta}{\partial y}$$

$$=\frac{\partial v}{\partial r}\cdot\frac{y}{r}+\frac{\partial v}{\partial \theta}\cdot\frac{x}{r^2}$$

$$\left[\because \quad \theta=\tan^{-1}\frac{y}{x}, \quad \frac{\partial \theta}{\partial y}=\frac{x}{r^2}\right]$$

$$=\frac{\partial v}{\partial r}\sin\theta+\frac{\partial v}{\partial \theta}\cdot\frac{\cos\theta}{r} \quad \ldots \quad \ldots \quad (2)$$

Squaring (1), (2) and adding

$$\left(\frac{\partial v}{\partial x}\right)^2+\left(\frac{\partial v}{\partial y}\right)^2=\left(\frac{\partial v}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial v}{\partial \theta}\right)^2$$

Now differentiating (1) w. r. to x

$$\frac{\partial^2 v}{\partial x^2}=\frac{\partial^2 v}{\partial r^2}\frac{\partial r}{\partial x}\cos\theta+\frac{\partial^2 v}{\partial \theta \partial r}\cdot\frac{\partial \theta}{\partial x}\cdot\cos\theta-\frac{\partial v}{\partial r}\sin\theta\frac{\partial \theta}{\partial x}$$

$$-\frac{\partial^2 v}{\partial \theta^2}\cdot\frac{\partial \theta}{\partial x}\cdot\frac{\sin\theta}{r}-\frac{\partial^2 v}{\partial r \partial \theta}\frac{\partial r}{\partial \theta}\cdot\frac{\sin\theta}{r}-\frac{\partial v}{\partial \theta}\cdot\frac{\cos\theta}{r}\frac{\partial \theta}{\partial x}$$

$$+\frac{\partial v}{\partial \theta}\frac{\sin\theta}{r^2}\frac{\partial r}{\partial x}$$

$$=\frac{\partial^2 v}{\partial r^2}\cos^2\theta-\frac{\partial^2 v}{\partial\theta\partial r}\frac{\sin\theta\cos\theta}{r}+\frac{\partial v}{\partial r}\frac{\sin^2\theta}{r}$$

$$+\frac{\partial^2 v}{\partial\theta^2}\frac{\sin\theta}{r}\cdot\frac{v}{r^2}-\frac{\partial^2 v}{\partial r.\partial\theta}\frac{\sin\theta\cos\theta}{r}+\frac{\partial v}{\partial\theta}\cdot\frac{\sin\theta\cos\theta}{r^2}$$

$$+\frac{\partial v}{\partial\theta}\frac{\sin\theta\cos\theta}{r^2}.$$

$$=\frac{\partial^2 v}{\partial r^2}\cos^2\theta+\frac{\partial^2 v}{\partial\theta^2}\frac{\sin^2\theta}{r^2}-2\frac{\partial^2 v}{\partial r\partial\theta}\cdot\frac{\sin\theta\cos\theta}{r}$$

$$+\frac{\partial v}{\partial r}\frac{\sin^2\theta}{r}+2\frac{\partial v}{\partial\theta}\frac{\sin\theta\cos\theta}{r^2}$$

Similarly, $\dfrac{\partial^2 v}{\partial y^2}=\dfrac{\partial^2 v}{\partial r^2}\sin^2\theta+\dfrac{\partial^2 v}{\partial\theta^2}\dfrac{\cos^2\theta}{r^2}+2\dfrac{\partial^2 v}{\partial r\,\partial\theta}\dfrac{\cos\theta\sin\theta}{r}$

$$+\frac{\partial v}{\partial r}\frac{\cos^2\theta}{r}-2\frac{\partial v}{\partial\theta}\frac{\cos\theta\sin\theta}{r^2}$$

$$\therefore\quad \frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}=\frac{\partial^2 v}{\partial r^2}+\frac{1}{r^2}\frac{\partial^2 v}{\partial\theta^2}+\frac{1}{r}\frac{\partial v}{\partial r}\quad [C.H.\ 1964\ (old)]$$

$$=\frac{1}{r^2}\left\{r\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right)\right\}+\frac{1}{r^2}\frac{\partial^2 v}{\partial\theta^2}$$

$$=\frac{1}{r^2}\left\{r\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right)+\frac{\partial^2 v}{\partial\theta^2}\right\}$$

Ex. 4. Let $u=f(x, y)$ be a function of two independent variables x, y where $x=\xi\cos\alpha-\eta\sin\alpha$, $y=\xi\sin\alpha+\eta\cos\alpha$.

Prove that $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=\dfrac{\partial^2 u}{\partial\xi^2}+\dfrac{\partial^2 u}{\partial\eta^2}$. (C. H. 1968)

$u=f(x, y)$ also $x=f_1(\xi, \eta)$, $y=f_2(\xi, \eta)$

$$\therefore\quad u=\phi(\xi, \eta)$$

$$\therefore\quad \frac{\partial u}{\partial\xi}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial\xi}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial\xi}$$

$$=\frac{\partial u}{\partial x}\cos\alpha+\frac{\partial u}{\partial y}\sin\alpha\quad\cdots\quad\cdots\quad(1)$$

$$\therefore\quad \frac{\partial u}{\partial\eta}=\frac{\partial u}{\partial x}\frac{\partial x}{\partial\eta}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial\eta}$$

$$=-\sin\alpha\frac{\partial u}{\partial x}+\cos\alpha\frac{\partial u}{\partial y}.\quad\cdots\quad(2)$$

Differentiating (1) w. r. to ξ

$$\frac{\partial^2 u}{\partial \xi^2}=\cos\alpha\ \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial x}\right)+\sin\alpha.\ \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial y}\right)$$

$$=\cos\alpha\left(\frac{\partial^2 u}{\partial x^2}\cdot\frac{\partial x}{\partial \xi}+\frac{\partial^2 u}{\partial y\partial x}\cdot\frac{\partial y}{\partial \xi}\right)$$

$$+\sin\alpha\left(\frac{\partial^2 u}{\partial x\partial y}\ \frac{\partial x}{\partial \xi}+\frac{\partial^2 u}{\partial y^2}\ \frac{\partial y}{\partial \xi}\right)$$

$$=\cos\alpha\left(\cos\alpha\ \frac{\partial^2 u}{\partial x^2}+\sin\alpha\frac{\partial^2 u}{\partial y\partial x}\right)$$

$$+\sin\alpha\left(\cos\alpha\ \frac{\partial^2 u}{\partial x\partial y}+\sin\alpha\ \frac{\partial^2 u}{\partial y^2}\right)\cdots \quad (3)$$

Similarly, differentiating (2) w. r. to η

$$\frac{\partial^2 u}{\partial \eta^2}=-\sin\alpha\left(-\sin\alpha\ \frac{\partial^2 u}{\partial x^2}+\cos\alpha\ \frac{\partial^2 u}{\partial y\partial x}\right)$$

$$+\cos\alpha\left(-\sin\alpha\ \frac{\partial^2 u}{\partial x\partial y}+\cos\alpha\ \frac{\partial^2 u}{\partial y^2}\right)\ \cdots \quad (4)$$

Adding (3) and (4)

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\frac{\partial^2 u}{\partial \xi^2}+\frac{\partial^2 u}{\partial \eta^2}.$$

Ex. 5. If $\mathbf{x+y=(u+v)^3}$, $\mathbf{x-y=(u-v)^3}$, show that

$$\mathbf{9\left(x^2-y^2\right)\left(\frac{\partial^2 f}{\partial x^2}-\frac{\partial^2 f}{\partial y^2}\right)=\left(u^2-v^2\right)\left(\frac{\partial^2 f}{\partial u^2}-\frac{\partial^2 f}{\partial v^2}\right)}$$

It being given that f(x, y) has continuous partial derivatives of the first two orders.

We have $x=u^3+3uv^2$ and $y=v^3+3u^2v$

Now $$\frac{\partial f}{\partial u}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial u}$$

$$=3(u^2+v^2)\frac{\partial f}{\partial x}+6uv\frac{\partial f}{\partial y}$$

$$\frac{\partial f}{\partial v}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial v}.$$

$$=6uv\frac{\partial f}{\partial x}+3(u^2+v^2)\frac{\partial f}{\partial y}$$

$$\therefore \quad \frac{\partial f}{\partial u}+\frac{\partial f}{\partial v}=3(u+v)^2\frac{\partial f}{\partial x}+3(u+v)^2\frac{\partial f}{\partial y}$$

and
$$\frac{\partial f}{\partial u}-\frac{\partial f}{\partial v}=3(u-v)^2\frac{\partial f}{\partial x}-3(u-v)^2\frac{\partial f}{\partial y}$$

So we have the operator

$$\frac{\partial}{\partial u}+\frac{\partial}{\partial y}=3(u+v)^2\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)$$

and
$$\frac{\partial}{\partial u}-\frac{\partial}{\partial y}=3(u-v)^2\left(\frac{\partial}{\partial x}-\frac{\partial}{\partial y}\right).$$

$$\therefore \quad (u+v)\left(\frac{\partial}{\partial u}+\frac{\partial}{\partial v}\right)=3(u+v)^3\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)$$

$$=3(x+y)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)$$

and
$$(u-v)\left(\frac{\partial}{\partial u}-\frac{\partial}{\partial v}\right)=3(u-v)^3\left(\frac{\partial}{\partial x}-\frac{\partial}{\partial y}\right)$$

$$=3(x-y)\left(\frac{\partial}{\partial x}-\frac{\partial}{\partial y}\right)$$

$$\therefore \quad (u+v)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right).(u-v)\left(\frac{\partial f}{\partial u}-\frac{\partial f}{\partial v}\right)$$

$$=3(x+y)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)\left\{3(x-y)\left(\frac{\partial f}{\partial x}-\frac{\partial f}{\partial y}\right)\right\}$$

$$\because \quad \left(\frac{\partial}{\partial u}+\frac{\partial}{\partial v}\right)(u-v)=0.$$

or,
$$(u^2-v^2)\left(\frac{\partial^2 f}{\partial u^2}-\frac{\partial^2 f}{\partial v^2}\right)=9(x^2-y^2)\left(\frac{\partial^2 f}{\partial x^2}-\frac{\partial^2 f}{\partial y^2}\right).$$

Ex. 6. If u be a function of x, y having continuous partial derivatives up to the 2nd order and the variables x, y are changed to ξ, η by the transformation

$$(x+y)=(\xi+\eta)^n,\ (x-y)=(\xi-\eta)^n$$

then prove that

$$(x^2-y^2)\left\{\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}\right\}=\frac{1}{n^2}(\xi^2-\eta^2)\left\{\frac{\partial^2 u}{\partial \xi^2}+\frac{\partial^2 u}{\partial \eta^2}\right\}.$$

[*C. H. 1962*]

Solving the given relations

$$x=\frac{1}{2}\{(\xi+\eta)^n+(\xi-\eta)^n\}, \quad y=\frac{1}{2}\{(\xi+\eta)^n-(\xi-\eta)^n\}.$$

$$\therefore \quad \frac{\partial x}{\partial \xi}=\frac{n}{2}\{(\xi+\eta)^{n-1}+(\xi-\eta)^{n-1}\}$$

and $$\frac{\partial x}{\partial \eta}=\frac{n}{2}\{(\xi+\eta)^{n-1}-(\xi-\eta)^{n-1}\}$$

$$\therefore \quad \frac{\partial^2 x}{\partial \xi^2}=\frac{n(n-1)}{2}\{(\xi+\eta)^{n-2}+(\xi-\eta)^{n-2}\} \quad \cdots \quad (1)$$

and $$\frac{\partial^2 x}{\partial \eta^2}=\frac{n(n-1)}{2}\{(\xi+\eta)^{n-2}+(\xi-\eta)^{n-2}\} \quad \cdots \quad (2)$$

So from (1) and (2) $$\frac{\partial^2 x}{\partial \xi^2}=\frac{\partial^2 x}{\partial \eta^2} \quad \cdots \quad \cdots \quad (3)$$

Also $$\frac{\partial y}{\partial \xi}=\frac{n}{2}\{(\xi+\eta)^{n-1}-(\xi-\eta)^{n-1}\}$$

and $$\frac{\partial y}{\partial \eta}=\frac{n}{2}\{(\xi+\eta)^{n-1}+(\xi-\eta)^{n-1}\}$$

$$\therefore \quad \frac{\partial^2 y}{\partial \xi^2}=\frac{n(n-1)}{2}\{(\xi+\eta)^{n-2}-(\xi-\eta)^{n-2}\} \quad \cdots \quad (4)$$

and $$\frac{\partial^2 y}{\partial \eta^2}=\frac{n(n-1)}{2}\{(\xi+\eta)^{n-2}-(\xi-\eta)^{n-2}\} \quad \cdots \quad (5)$$

So from (4) and (5) $$\frac{\partial^2 y}{\partial \xi^2}=\frac{\partial^2 y}{\partial \eta^2} \quad \cdots \quad \cdots \quad (6)$$

Now $$\frac{\partial u}{\partial \xi}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \xi}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \xi}$$

and $$\frac{\partial u}{\partial \eta}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \eta}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \eta}.$$

$$\therefore \quad \frac{\partial^2 u}{\partial \xi^2}=\frac{\partial^2 u}{\partial x^2}\left(\frac{\partial x}{\partial \xi}\right)^2+\frac{\partial u}{\partial x}\,\frac{\partial^2 x}{\partial \xi^2}+\frac{\partial^2 u}{\partial y^2}\left(\frac{\partial y}{\partial \xi}\right)^2+\frac{\partial u}{\partial y}\,\frac{\partial^2 y}{\partial \xi^2}$$

$$\frac{\partial^2 u}{\partial \eta^2}=\frac{\partial^2 u}{\partial x^2}\left(\frac{\partial x}{\partial \eta}\right)^2+\frac{\partial u}{\partial x}\cdot\frac{\partial^2 x}{\partial \eta^2}+\frac{\partial^2 u}{\partial y^2}\left(\frac{\partial y}{\partial \eta}\right)^2+\frac{\partial u}{\partial y}\,\frac{\partial^2 y}{\partial \eta^2}$$

$$\therefore \quad \frac{\partial^2 u}{\partial \xi^2}-\frac{\partial^2 u}{\partial \eta^2}=\frac{\partial^2 u}{\partial x^2}\left\{\left(\frac{\partial x}{\partial \xi}\right)^2-\left(\frac{\partial x}{\partial \eta}\right)^2\right\}+\frac{\partial^2 u}{\partial y^2}\left\{\left(\frac{\partial y}{\partial \xi}\right)^2-\left(\frac{\partial y}{\partial \eta}\right)^2\right\}$$

using (3) and (6)

$$=\frac{\partial^2 u}{\partial x^2}.n^2(\xi+\eta)^{n-1}.(\xi-\eta)^{n-1}-\frac{\partial^2 u}{\partial y^2}.n^2(\xi+\eta)^{n-1}.(\xi-\eta)^{n-1}$$

$$=\left(\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}\right).n^2(\xi+\eta)^{n-1}.(\xi-\eta)^{n-1}$$

$$=\left(\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}\right).n^2.\frac{(\xi+\eta)^n}{(\xi+\eta)}.\frac{(\xi-\eta)^n}{(\xi-\eta)}$$

$$=\left(\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}\right).n^2\frac{(x+y)(x-y)}{\xi^2-\eta^2}$$

or, $$\frac{1}{n^2}(\xi^2-\eta^2)\left\{\frac{\partial^2 u}{\partial \xi^2}-\frac{\partial^2 u}{\partial \eta^2}\right\}=(x^2-y^2)\left\{\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial y^2}\right\}.$$

Ex. 7. If $\mathbf{x=c\cosh\xi\cos\eta}$, $\mathbf{y=c\sinh\xi\sin\eta}$, show that

$$\frac{\partial^2 \mathbf{u}}{\partial \xi^2}+\frac{\partial^2 \mathbf{u}}{\partial \eta^2}=\tfrac{1}{2}\mathbf{c}^2[\cosh 2\xi-\cos 2\eta]\left(\frac{\partial^2 \mathbf{u}}{\partial \mathbf{x}^2}+\frac{\partial^2 \mathbf{u}}{\partial \mathbf{y}^2}\right).$$

[C. H. 1965 (old)

Since $x=c\cosh\xi\cos\eta,\quad y=c\sinh\xi\sin\eta$

$$\frac{\partial x}{\partial \xi}=c\sinh\xi\cos\eta,\quad \frac{\partial y}{\partial \xi}=c\cosh\xi.\sin\eta.$$

Now $$\frac{\partial u}{\partial \xi}=\frac{\partial u}{\partial x}.\frac{\partial x}{\partial \xi}+\frac{\partial u}{\partial y}.\frac{\partial y}{\partial \xi}$$

$$=c\sinh\xi\cos\eta\frac{\partial u}{\partial x}+c\cosh\xi.\sin\eta\frac{\partial u}{\partial y}.$$

$$\therefore\quad \frac{\partial^2 u}{\partial \xi^2}=c\cos\eta\frac{\partial}{\partial \xi}\left(\sinh\xi.\frac{\partial u}{\partial x}\right)+c\sin\eta\frac{\partial}{\partial \xi}\left(\cosh\xi.\frac{\partial u}{\partial y}\right)$$

$$=c.\cos\eta\cosh\xi\frac{\partial u}{\partial x}+c\cos\eta\sinh\xi$$

$$\left(\frac{\partial^2 u}{\partial x^2}.\frac{\partial x}{\partial \xi}+\frac{\partial^2 u}{\partial y\partial x}.\frac{\partial y}{\partial \xi}\right)+c\sin\eta\sinh\xi\frac{\partial u}{\partial y}$$

$$+c\sin\eta\cosh\xi\left(\frac{\partial^2 u}{\partial x\partial y}.\frac{\partial x}{\partial \xi}+\frac{\partial^2 u}{\partial y^2}.\frac{\partial y}{\partial \xi}\right)$$

$$=c\cos\eta\cosh\xi\frac{\partial u}{\partial x}+c\cos\eta\sinh\xi$$

$$\left(c\sinh\xi.\cos\eta.\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y\partial x}.c\cosh\xi\sin\eta\right)$$

$$+c \sin \eta \sinh \xi \frac{\partial u}{\partial x}+c \sin \eta \cosh \xi \left(c \sinh \xi \cos \eta \times \frac{\partial^2 u}{\partial x \partial y}+\frac{\partial^2 u}{\partial y^2} c \cosh \xi \sin \eta\right). \quad \ldots \quad (1)$$

Again $\frac{\partial x}{\partial \eta}=-c \cosh \xi \sin \eta$ and $\frac{\partial y}{\partial \eta}=c \sinh \xi \cos \eta$.

$$\therefore \quad \frac{\partial u}{\partial \eta}=\frac{\partial u}{\partial x}\frac{\partial x}{\partial \eta}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial \eta}.$$

$$=-c \cosh \xi \sin \eta \frac{\partial u}{\partial x}+c \sinh \xi \cos \eta \frac{\partial u}{\partial y}.$$

or, $$\frac{\partial^2 u}{\partial \eta^2}=-c \cosh \xi \frac{\partial}{\partial \eta}\left(\sin \eta \frac{\partial u}{\partial x}\right)+c \sinh \xi \frac{\partial}{\partial \eta}\left(\cos \eta \frac{\partial u}{\partial y}\right)$$

$$=-c \cosh \xi \cos \eta \frac{\partial u}{\partial x}-c \cosh \xi . \sin \eta \times \left(\frac{\partial^2 u}{\partial x^2}\frac{\partial x}{\partial \eta}+\frac{\partial^2 u}{\partial y \partial x}\frac{\partial y}{\partial \eta}\right)$$

$$-c \sinh \xi \sin \eta \frac{\partial u}{\partial y}+c \sinh \xi \cos \eta \left(\frac{\partial^2 u}{\partial x \partial y}\frac{\partial x}{\partial \eta}+\frac{\partial^2 u}{\partial y^2}\frac{\partial y}{\partial \eta}\right)$$

$$=-c \cosh \xi . \cos \eta \frac{\partial u}{\partial x}-c \cosh \xi \sin \eta \left\{\frac{\partial^2 u}{\partial x^2}(-c \cosh \xi \sin \eta)+\frac{\partial^2 u}{\partial y \partial x} c \sinh \xi \cos \eta\right\}-c \sinh \xi \sin \eta \frac{\partial u}{\partial y}$$

$$+c \sinh \xi \cos \eta \left\{\frac{\partial^2 u}{\partial x \partial y}(-c \cosh \xi . \sin \eta)+\frac{\partial^2 u}{\partial y^2}(c \sinh \xi \cos \eta)\right\} \quad \ldots \quad (2)$$

Adding (and (2)

$$\frac{\partial^2 u}{\partial \xi^2}+\frac{\partial^2 u}{\partial \eta^2}=c^2 \cos^2 \eta \sinh^2 \xi \frac{\partial^2 u}{\partial x^2}+c^2 \cosh^2 \xi \sin^2 \eta \frac{\partial^2 u}{\partial x^2}$$

$$+c^2 \sin^2 \eta . \cosh^2 \xi \frac{\partial^2 u}{\partial y^2}+c^2 \sinh^2 \xi \cos^2 \eta \frac{\partial^2 u}{\partial y^2}.$$

$$=c^2 \cos^2 \eta . \sinh^2 \xi \left\{\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right\}$$

$$+c^2 \sin^2 \eta . \cosh^2 \xi \left\{\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right\}$$

$$=c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)(\cos^2\eta\sinh^2\xi+\sin^2\eta\cosh^2\xi)$$

$$=c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)\{(1-\sin^2\eta)\sinh^2\xi+\sin^2\eta\cosh^2\xi\}$$

$$=c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)\{\sinh^2\xi+\sin^2\eta(\cosh^2\xi-\sinh^2\xi)\}$$

$$=c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)(\sinh^2\xi+\sin^2\eta)$$

$$=\frac{1}{2}c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)\{(\cosh 2\xi-1)+(1-\cos 2\eta)\}$$

$$=\frac{1}{2}c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)(\cosh 2\xi-\cos 2\eta).$$

√**Ex. 8. If f(x, y, z) be a homogeneous function of x, y, z of degree n which satisfies Laplaces equation**

$$\Delta^2 f=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}=0,$$

prove that $\Delta^2(r^{2m}.f)=2m(2n+2m+1)r^{2m-2}.f$ **where** $r^2=x^2+y^2+z^2$. *(C. H. 1962)*

Since $r^2=x^2+y^2+z^2$, $\frac{\partial r}{\partial x}=\frac{x}{r}$, $\frac{\partial r}{\partial y}=\frac{y}{r}$, $\frac{\partial r}{\partial z}=\frac{z}{r}$.

$$\therefore\quad \frac{\partial}{\partial x}(r^{2m}f)=2mr^{2m-1}\frac{\partial r}{\partial x}.f+r^{2m}\frac{\partial f}{\partial x}.$$

$$=2mr^{2m-1}.\frac{x}{r}.f+r^{2m}.f_x$$

$$=2mr^{2m-2}.x.f+r^{2m}.f_x.$$

$$\therefore\quad \frac{\partial^2}{\partial x^2}(r^{2m}.f)=2m(2m-2)r^{2m-3}\frac{\partial r}{\partial x}x.f+2mr^{2m-2}.f$$

$$+2mr^{2m-2}x\frac{\partial f}{\partial x}+2mr^{2m-1}\frac{\partial r}{\partial x}f_x+r^{2m}.f_{xx}$$

$$=2m(2m-2)r^{2m-4}.x^2 f+2mr^{2m-2}.f+4mr^{2m-2}.xf_x \qquad (1)$$
$$+r^{2m}f_{xx}\qquad\ldots\qquad\ldots$$

Similarly,

$$\frac{\partial^2}{\partial y^2}(r^{2m}.\ f)=2m(2m-2)r^{2m-4}.\ y^2f+2mr^{2m-2}.\ f$$

$$+4mr^{2m-2}.\ yf_y+r^{2m}f_{yy} \quad \cdots \quad \cdots (2)$$

$$\frac{\partial^2}{\partial z^2}(r^{2m}.\ f)=2m(2m-2)r^{2m-4}.\ z^2f+2mr^{2m-2}.\ f$$

$$+4mr^{2m-2}.\ zf_z+r^{2m}.\ f_{zz} \quad \cdots \quad (3)$$

Adding (1), (2) and (3)

$$\Delta^2(r^{2m}.\ f)=2m(2m-2)r^{2m-4}(x^2+y^2+z^2)f+6mr^{2m-2}.\ f$$
$$+4mr^{2m-2}(xf_x+yf_y+zf_z)+r^{2m}(f_{xx}+f_{yy}+f_{zz}).$$
$$=2m(2m-2)r^{2m-4}.\ r^2f+6mr^{2m-2}.\ f+4mr^{2m-2}.\ nf$$
$$[\because\ \Sigma f_{xx}=0]$$
$$=2m(2m+2n+1)r^{2m-2}.\ f.$$

Ex. 9. If v_n is a homogeneous functions of x, y, z of degree n and $r^2=x^2+y^2+z^2$, then

$$\nabla^2(r^m v_n)=m(m+2n+1)r^{m-2}.\ v_n+r^m\nabla^2.\ v_n \quad \text{where}$$

$$\nabla^2=\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)$$

Deduce that if further v_n satisties the equation $\nabla^2 v_n=0$, so does $r^{-2n-1}.\ v_n$. **(C. H. 1967)**

Since, $r^2=x^2+y^2+z^2$,

$$\frac{\partial r}{\partial x}=\frac{x}{r},\ \frac{\partial r}{\partial y}=\frac{y}{r},\ \frac{\partial r}{\partial z}=\frac{z}{r}.$$

Now $$\frac{\partial}{\partial x}(r^m v_n)=mr^{m-1}\frac{\partial r}{\partial x}.\ v_n+r^m\frac{\partial v_n}{\partial x}$$

$$=mxr^{m-2}.\ v_n+r^m\frac{\partial v_n}{\partial x},$$

$$\therefore\quad \frac{\partial^2}{\partial x^2}(r^m.\ v_n)=m(m-2)r^{m-3}\frac{\partial r}{\partial x}.x.v_n+mr^{m-2}v_n$$

$$+mxr^{m-2}\frac{\partial v_n}{\partial x}+mr^{m-1}\frac{\partial r}{\partial x}\cdot\frac{\partial v_n}{\partial x}+r^m\frac{\partial^2 v_n}{\partial x^2}$$

$$=m(m-2)x^2r^{m-4}\ v_n+mr^{m-2}v_n+mxr^{m-2}\ \frac{\partial v_n}{\partial x}$$

$$+mxr^{m-2}\ \frac{\partial v_n}{\partial x}+r^m\ \frac{\partial^2 v_n}{\partial x^2}$$

$$=m(m-2)x^2r^{m-4}\ v_n+mr^{m-2}v_n+2mxr^{m-2}\ \frac{\partial v_n}{\partial x}$$

$$+r^m\ \frac{\partial^2 v_n}{\partial x^2} \qquad \ldots \qquad (1)$$

Similar expressions for $\frac{\partial^2}{\partial y^2}\left(r^m v_n\right)$ and $\frac{\partial^2}{\partial z^2}\left(r^m v_n\right)$

$\therefore$ Adding we get

$$\nabla^2(r^m v_n)=m(m-2)(x^2+y^2+z^2)r^{m-4}v_n+3mr^{m-2}v_n$$

$$+2mr^{m-2}\sum\left(x\frac{\partial v_n}{\partial x}\right)+r^m\sum\left(\frac{\partial^2 v_n}{\partial x^2}\right)$$

$$=m(m-2)r^2.\ r^{m-4}.\ v_n+3mr^{m-2}.\ v_n$$

$$+2mr^{m-2}.\ nv_n+r^m\nabla^2 v_n \quad \because \sum\frac{\partial^2 v_n}{\partial x^2}=\nabla^2 v_n$$

$$=m(m+2n+1)r^{m-2}.\ v_n+r^m\nabla^2 v_n.$$

Deduction. If $\nabla^2 v_n=0$.

then $\nabla^2(r^m v_n)=0$

if $m(m+2n+1)r^{m-2}v_n=0$.

i.e., if $m+2n+1=0$

i.e.; if $m=-2n-1$.

So for $m=-2n-1$, we must have

$\nabla^2(r^m v_n)=0$

Hence $\nabla^2(r^{-2n-1}.\ v_n)=0$.

Exercise 4

1. If $x=a\cos u\cosh v$, $y=a\sin u\sinh v$, prove that

$$\frac{\partial(x, y)}{\partial(u, v)}=\frac{1}{2}a^2(\cosh 2v-\cos 2u).$$

2. If $x=r\cos\theta$, $y=r\sin\theta$ where r and θ are functions of t alone, prove that

$$x\frac{dy}{dt}-y\frac{dx}{dt}=r^2\frac{d\theta}{dt}.$$

3. If V is a function of r alone, where $r^2=x_1^2+x_2^2+\ldots+x_n^2$, prove that

$$\frac{\partial^2 V}{\partial x_1^2}+\frac{\partial^2 V}{\partial x_2^2}+\cdots+\frac{\partial^2 V}{\partial x_n^2}=\frac{d^2 V}{dr^2}+\frac{n-1}{r}\cdot\frac{dV}{dr}$$

4. If $x=r\cos\theta$, $y=r\sin\theta$, prove that

(i) $$\frac{\partial^2\theta}{\partial x^2}+\frac{\partial^2\theta}{\partial y^2}=0$$

(ii) $$\frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}=\frac{1}{r}\left\{\left(\frac{\partial r}{\partial x}\right)^2+\left(\frac{\partial r}{\partial y}\right)^2\right\}$$

5. If $V=f(x, y)$ where $x=\xi\cos\alpha-\eta\sin\alpha$ and $y=\xi\sin\alpha+\eta\cos\alpha$, then prove that

$$\frac{\partial^2 V}{\partial x^2}\cdot\frac{\partial^2 V}{\partial y^2}-\frac{\partial^2 V}{\partial x\,\partial y}=\frac{\partial^2 V}{\partial\xi^2}\cdot\frac{\partial^2 V}{\partial\eta^2}-\left(\frac{\partial^2 V}{\partial\xi\,\partial\eta}\right)^2.$$

6. If V be a homogeneous function of x, y, z of degree n and if $u=(x^2+y^2+z^2)^{-\frac{1}{2}(n+1)}$, then prove that

$$\frac{\partial}{\partial x}\left(V\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(V\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(V\frac{\partial u}{\partial z}\right)=0.$$

7. Let $\phi=\phi(x, y)$ where $x=e^v\sec u$, $y=e^v\tan u$, prove that

$$\cos u\left(\frac{\partial^2\phi}{\partial u.\partial v}-\frac{\partial\phi}{\partial u}\right)=xy\left(\frac{\partial^2\phi}{\partial x^2}+\frac{\partial\ \phi}{\partial y^2}\right)+(x^2+y^2)\frac{\partial^2\phi}{\partial x.\partial y}.$$

8. If $\phi=\phi(x, y)$ where $x=e^u\sec v$, $y=e^u\tan v$, prove that

$$\left(\frac{\partial\phi}{\partial x}\right)^2-\left(\frac{\partial\phi}{\partial y}\right)^2=\frac{1}{e^{2u}}\left\{\left(\frac{\partial\phi}{\partial u}\right)^2-\cos^2 v\left(\frac{\partial\phi}{\partial v}\right)^2\right\}$$

9. A function $f(x, y)$ when expressed in terms of the new variables u, v defined by $x=\frac{1}{2}(u+v)$, $y^2=uv$ becomes $g(u, v)$; prove that

$$\frac{\partial^2 g}{\partial u\,\partial v}=\frac{1}{4}\left\{\frac{\partial^2 f}{\partial x^2}+2\frac{x}{y}\frac{\partial^2 f}{\partial x\,\partial y}+\frac{1}{y}\ \frac{\partial^2 f}{\partial y^2}\right\}$$ (C. H. 1965)

10. Let D stands for the operator $x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}$ or, $r\frac{\partial}{\partial r}$ in polar, prove

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\,\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=D(D-1)u.$$

If further $x=r\cos\theta$, $y=r\sin\theta$ and $r=e^z$ then deduce that

$$\frac{\partial^2 u}{\partial\theta^2}+\frac{\partial u}{\partial z}=D(D-1)u.$$

CHAPTER V

EXPANSION OF FUNCTIONS

5.1. Mean Value Theorem for a function of two variables. If $f(x, y)$ be a differentiable function of the two variables x and y defined in a certain domain R containing the points (a, b) and $(a+h, b+k)$ so that the line joining these two points lies wholly in R, then

$$f(a+h, b+k)=f(a, b)+hf_x(a+\theta h, b+\theta k)$$
$$+kf_y(a+\theta h, b+\theta k)$$

where $0<\theta<1$.

Proof. Let $x=a+ht$, $y=b+kt$, so that

$$f(x, y)=f(a+ht, b+kt)$$
$$=F(t), \text{ a composite function of a single variable } t.$$

Then applying Mean Value Theorem of a single variable to the function $F(t)$ between 0 and t, we get

$$F(t)-F(0)=tF'(\theta t), 0<\theta<1. \quad \ldots \quad (1)$$

But $F'(t)=\dfrac{\partial f}{\partial x}\dfrac{dx}{dt}+\dfrac{\partial f}{\partial y}\dfrac{dy}{dt}$

$$=hf_x(a+ht, b+kt)+kf_y(a+ht, b+kt)$$

$\therefore$ from (1)

$$F(t)-F(0)=t\{hf_x(a+h\theta t, b+k\theta t)$$
$$+kf_y(a+h\theta t, b+k\theta t)\}$$

Putting $t=1$

$$F(1)-F(0)=hf_x(a+\theta h, b+\theta k)+kf_y(a+\theta h, b+\theta k)$$

But $F(1)=f(a+h, b+k)$

and $F(0)=f(a, b)$

Hence, we get

$$f(a+h, b+k)=f(a, b)+hf_x(a+\theta h, b+\theta k)$$
$$+kf_y(a+\theta h, b+\theta k)$$

where $0<\theta<1$.

5.2. Taylor's Theorem for a function of two variables.

If $f(x, y)$ possess continuous partial derivatives of nth order in any neighbourhood of a point (a, b) and if $(a+h, b+k)$ be any point of this neighbourhood, then there exists a positive number $\theta<1$ such that

$$f(a+h, b+k)=f(a, b)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a, b)$$

$$+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a, b)+\cdots+\frac{1}{\lfloor n-1}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n-1} f(a, b)$$

$$+\frac{1}{\lfloor n}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(a+\theta h, b+\theta k)$$

where $0<\theta<1$.

Lemma. Let $z=f(x, y)$ and $x=a+ht$, $y=b+kt$, so that z is a composite function of a single variable t.

$$\therefore \quad \frac{dz}{dt}=\frac{\partial z}{\partial x}\frac{dx}{dt}+\frac{\partial z}{\partial y}\frac{dy}{dt}.$$

$$=\frac{\partial z}{\partial x}.h+\frac{\partial z}{\partial y}.k$$

$$=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)z$$

$$\frac{d^2 z}{dt^2}=\frac{d}{dt}\left(\frac{dz}{dt}\right)$$

$$=\frac{d}{dt}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)z$$

$$=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)z$$

$$=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 z.$$

Proceeding in this way, we can write

$$\frac{d^n z}{dt^n}=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n z.$$

Proof of Taylor's Theorem.

We write $f(x, y)=f(a+ht, b+kt)=F(t)$ ... (1)

Now applying Maclaurin's Theorem to the function $F(t)$ of the single variable t,

there exists a positive number $\theta<1$ such that

$$F(t)=F(0)+tF'(0)+\frac{t^2}{\lfloor 2}F''(0)+\cdots+\frac{t^{n-1}}{\lfloor n-1}F^{n-1}(0)$$
$$+\frac{t^n}{\lfloor n}F^n(\theta t). \quad \ldots \quad (2)$$

or, for $t=1$, we get

$$F(1)=F(0)+F'(0)+\frac{1}{\lfloor 2}F''(0)+\cdots+\frac{1}{\lfloor n-1}F^{n-1}(0)$$
$$+\frac{1}{\lfloor n}F^n(\theta) \quad \ldots \quad (3)$$

Now from (1)

$$F(1)=f(a+h, b+k)$$
$$F(0)=f(a, b).$$

Also from the lemma

$$F^n(t)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(x, y)$$

For $t=0$, we have $x=a$, $y=b$

$$\therefore \quad F'(0)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a, b)$$

$$F''(0)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a, b)$$

... ...

$$F^{n-1}(0)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n-1} f(a, b)$$

$$F^n(\theta)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(a+\theta h, b+\theta k)$$

Putting these values in (3) we get

$$f(a+h, b+k)=f(a, b)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a, b)$$

$$+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a, b)+\cdots$$

$$\cdots+\frac{1}{\lfloor n-1}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n-1} f(a, b)$$

$$+\quad\frac{1}{\lfloor n}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(a+\theta h, b+\theta k)$$

$$\text{for } 0<\theta<1.$$

Cor. 1. Replacing a and b by x and y, the Taylor's Formula may be written in the form

$$f(x+h, y+k)=f(x, y)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x, y)$$

$$+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x, y)+\cdots\cdots+\frac{1}{\lfloor n-1}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n-1} f(x, y)$$

$$+\frac{1}{\lfloor n}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(x+\theta h, y+\theta k) \text{ where } 0<\theta<1.$$

This formula may be extended for any number of variables. Thus for three variables x, y, z we can write

$$f(x+h, y+k, z+l)=f(x, y, z)$$

$$+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}+l\frac{\partial}{\partial z}\right)f(x, y, z)+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}+l\frac{\partial}{\partial z}\right)^2 f(x, y, z)$$

$$+\cdots\cdots+\frac{1}{\lfloor n-1}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}+l\frac{\partial}{\partial z}\right)^{n-1} f(x, y, z)$$

$$+\frac{1}{\lfloor n}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}+l\frac{\partial}{\partial z}\right)^n f(x+\theta h, y+\theta k, z+\theta l) \text{ where } 0<\theta<1.$$

5·3. Maclaurin's Theorem for a function of two variables.

Let $f(x, y)$ be a functions of two independent variables x and y and let

(*i*) (x, y) be a point in the neighbourhood of $(0, 0)$

(*ii*) $f(x, y)$ has derivatives up to $(n-1)$th order and all continuous in $0\leqslant h\leqslant x$, $0\leqslant k\leqslant y$

(*iii*) $f(x, y)$ is differentiable up to nth order in

$$0<h<x,\ 0<k<y,$$

then there exists a positive number $\theta<1$ such that

$$f(x, y)=f(0, 0)+\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0, 0)+\frac{1}{\lfloor 2}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^2 f(0, 0)+$$

$$\cdots\frac{1}{\lfloor n-1}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^{n-1} f(0, 0)+\frac{1}{\lfloor n}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^n f_{(\theta x,\, \theta y)}$$

$$\text{for}\quad 0<\theta<1,$$

where $\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0, 0)$ means after calculating partial derivatives of $f(x, y)$ its value taken by putting $x=0$, $y=0$.

By Taylor's Theorem of two variables. we know that if (*i*) $f(x, y)$ has derivatives upto $(n-1)$th order and continuous in $a\leqslant x\leqslant a+h$, $b\leqslant y\leqslant b+k$

(*ii*) $f(x, y)$ is differentiable up to the nth order in $a<x<a+h$, $b<y<b+k$,

then there exists a positive number $\theta<1$ such that

$$f(a+h, b+k)=f(a, b)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a, b)$$

$$+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a, b)+\cdots+\frac{1}{\lfloor n-1}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n-1} f(a, b)$$

$$+\frac{1}{\lfloor n}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(a+\theta h, b+\theta k),\ 0<\theta<1.$$

Putting $a=b=0$, and changing h, k by x and y the above Taylor's statement reduces to Maclaurin's statement and the expansion becomes

$$f(x, y)=f(0, 0)+\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0, 0)+\frac{1}{\lfloor 2}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)f(0, 0)+$$

$$\cdots+\frac{1}{\lfloor n-1}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^{n-1} f(0, 0)+\frac{1}{\lfloor n}\left(x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right)^n f(\theta x, \theta y),$$

$$0<\theta<1.$$

Note. The Maclaurin's expansion can be put in the following convenient form

$$f(x, y)=f(u, v)+\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)f(u, v)+\frac{1}{\lfloor n}\left\{x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right\}^2 f(u, v)+$$
$$\cdots+\frac{1}{\lfloor n-1}\left\{x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right\}^{n-1} f(u, v)+\left\{x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right\}^n f(u, v)$$

where after differentiation u, v are to be replaced by (0, 0) except in the last term, where u, v are to be replaced by θx and θy.

Illustrative Examples :

Ex. 1. If the partial derivatives f_x, f_y of a function f(x, y) exist and have the value zero at every point in its domain, prove that f(x, y) is a constant. (C. H. 1961)

By Mean Value Theorem

$$f(x+h, y+k)-f(x, y)=hf_x(x+\theta h, y+\theta h)$$
$$+kf_y(x+\theta h, y+\theta k),\ 0<\theta<1$$
$$=hf_x(\xi, \eta)+kf_y(\xi, \eta) \quad \cdots \quad (1)$$

where (ξ, η) is an intermediate point on the line joining the two points (x, y) and $(x+h, y+k)$ in the domain.

But by the given condition $f_x(\xi, \eta)=0$, $f_y(\xi, \eta)=0$ for every point in the domain.

$\therefore$ from (1) $f(x+h, y+k)-f(x, y)=0$ in the domain

$\therefore$ $f(x, y)$ is constant in the domain.

Ex. 2. Prove that

$$\sin x \sin y=xy-\frac{1}{6}\{(x^3+3xy^2)\cos\theta x \sin\theta y$$
$$+(y^3+3xy^2)\sin\theta x\cos\theta y\} \quad \text{for } 0<\theta<1.$$

Let $f(x, y)=\sin x \sin y$, then $f(0, 0)=0$.

$$\therefore \left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)f(u, v)=\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)\sin u \sin v$$
$$=x\sin v\cos u+y\sin u\cos v$$
$$=0 \quad \text{for } u=v=0.$$

$$\frac{1}{\lfloor 2}\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)^2 f(u, v)=\frac{1}{\lfloor 2}\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)^2 \sin u \sin v$$

$$=\frac{1}{2}\left\{x^2 \frac{\partial^2}{\partial u^2}(\sin u \sin v)+2xy\frac{\partial^2}{\partial u\, \partial v}(\sin u \sin v)\right.$$

$$\left.+y^2\frac{\partial^2}{\partial v^2}(\sin u \sin v)\right\}$$

$$=\frac{1}{2}\{-x^2 \sin u \sin v+2xy \cos u \cos v-y^2 \sin u \sin v\}$$

$$=xy \text{ for } u=v=0$$

$$\frac{1}{\lfloor 3}\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)^3 f(u, v)$$

$$=\frac{1}{6}\left\{x^3\frac{\partial^3}{\partial u^3}(\sin u \sin v)+3x^2y\frac{\partial^2}{\partial u^2.\partial v}(\sin u \sin v)\right.$$

$$\left.+3xy^2\frac{\partial^3}{\partial u\, \partial v^2}(\sin u \sin v)+y^3\frac{\partial^3}{\partial v^3}(\sin u \sin v)\right.$$

$$=\frac{1}{6}\{-x^3 \cos u \sin v-3x^2y \sin u \cos v$$

$$-3xy^2 \cos u \sin v-y^3 \sin u \cos v\}$$

$$=-\frac{1}{6}\{(x^3+3xy^2) \cos \theta x \sin \theta y$$

$$+(y^3+3x^2y) \sin \theta x \cos \theta y\} \text{ for } u=\theta x,\ v=\theta y.$$

Now putting in the Maclaurin's Theorem, we get

$$\sin x \sin y=xy-\frac{1}{6}\{(x^3+3xy^2) \cos \theta x \sin \theta y$$

$$+(y^3+3x^2y) \sin \theta x \cos \theta\, y\}$$

where $0<\theta<1$

Ex. 3. Show that

$$\mathbf{e^{ax} \sin by=by+abxy}$$

$$\mathbf{+e^{x\theta a}\{a^3x^3-3ab^2xy^2) \sin b\theta y}$$

$$\mathbf{+(3a^2bx^2y-b^3y^3) \cos b\theta y\},\ 0<\theta<1}$$ *(C. H. 1964)*

Let $f(x, y)=e^{ax} \sin by$, then $f(u, v)=e^{au} \sin bv$

$\therefore\ f(0, 0)=0.$

$$\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right) f(u, v)=\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)e^{au} \sin bv$$
$$=xae^{au} \sin bv+ybe^{au} \cos bv$$
$$=by \text{ for } u=v=0.$$

$$\frac{1}{\lfloor 2}\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)^2 e^{au} \sin bv$$
$$=\frac{1}{2}\left\{x^2\frac{\partial^2}{\partial u^2}e^{au} \sin bv+2xy\frac{\partial^2}{\partial u\partial v}e^{au} \sin bv + y^2\frac{\partial^2}{\partial v^2}e^{au} \sin bv\right\}$$
$$=\tfrac{1}{2}(x^2a^2e^{au} \sin bv+2xyabe^{au} \cos bv-y^2b^2e^{au} \sin bv)$$
$$=abxy \quad \text{for } u=v=0.$$

$$\frac{1}{\lfloor 3}\left(x\frac{\partial}{\partial u}+y\frac{\partial}{\partial v}\right)^3 e^{au} \sin bv$$
$$=\frac{1}{6}\left\{x^3\frac{\partial^3}{\partial u^3}(e^{au} \sin bv)+3x^2y\frac{\partial^3}{\partial u^2\partial v}(e^{au} \sin bv) +3xy^2\frac{\partial^3}{\partial u\partial v^2}(e^{au} \sin bv)+y^3\frac{\partial^3}{\partial v^3}(e^{au} \sin bv)\right\}$$
$$=\frac{1}{6}\{x^3a^3e^{au} \sin bv+3x^2ya^2be^{au} \cos bv -3xy^2ab^2e^{au} \sin bv-y^3b^3e^{au} \cos bv\}$$
$$=\frac{1}{6}e^{au}\{(a^3x^3-3ab^2xy^2) \sin bv+(3a^2bx^2y-b^3y^3) \cos bv\}$$
$$=\frac{1}{6}e^{a\theta x}\{(a^3x^3-3ab^2xy^2) \sin b\theta y+(3a^2bx^2y-b^3y^3) \cos b\theta y\}.$$

putting $u=\theta x$ and $v=\theta y$

Now putting in the Maclaurin's Theorem we get the result.

Exercise 5

1. If the partial derivatives f_x, f_y of the function $f(x, y)$ are continuous in R, prove that
$$f(x+h, y+k)-f(x, y)=hf_x(\xi, \eta)+kf_y(\xi, \eta)$$
where (ξ, η) is a point in the line joining (x, y) and $(x+h, y+k)$. *(C. H. 1961)*

2. Prove that for $0<\theta<1$

$$f(x+h,\ y+k)=f(x,\ y)+\frac{1}{\underline{|2}}\left[\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right]f(x,\ y)$$

$$+\frac{1}{\underline{|\ }}\left[x\frac{\partial}{\partial x}+y\frac{\partial}{\partial y}\right](x+\theta h,\ y+\theta k),$$

Also state the conditions under which the above expansion is valid.

3. If $f(x,\ y)=f(a+ht,\ b+kt)=F(t)$ where $a,\ b,\ h,\ k$ are constants, sho[w] that for $0<\theta<1$

$$F^{m}(\theta)=\left[h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right]^{n}f(a+\theta h,\ b+\theta k).$$

4. If $z=f(x,\ y)$ and $x=a+ht,\ y=b+kt$ where $a,\ b,\ h,\ k$ are constants show that

$$\frac{d^n z}{dt^n}=\left\{h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right\}^n_z$$

$$=h^n\frac{\partial^n z}{\partial x^n}+\left\{\begin{matrix}n\\1\end{matrix}\right\}h^{n-1}k\frac{\partial^n z}{\partial x^{n-1}\partial y}+\ldots$$

$$\ldots+\left\{\begin{matrix}n\\r\end{matrix}\right\}h^{n-r}k^r\frac{\partial^n z}{\partial x^{n-r}\partial y^r}+\ldots+k^n\frac{\partial^n z}{\partial y^n}.$$

5. Show that for $0<\theta<1$

$$e^{ax}\cos by=1+ax+\frac{1}{\underline{|2}}(a^2x^2-b^2y^2)$$

$$+\frac{e^{a\theta x}}{\underline{|3}}\{(a^3x^3-3ab^2xy^2)\cos b\theta y-(3a^2bx^2y-b^3y^3)\sin b\theta y\}$$

CHAPTER VI

MAXIMA AND MINIMA

6.1. Extreme value of a function of two variables.

If $f(a, b)$ be a maximum or minimum value of $f(x, y)$ at (a, b), then $f(a, b)$ is said to be an extreme value of the function $f(x, y)$.

If there exist some neighbourhood of (a, b) such that for every point $(a+h, b+k)$ of this neighbourhood

$$f(a, b) > f(a+h, b+k)$$

then $f(a, b)$ is said to be a maximum value of $f(x, y)$ at (a, b).

If there exist some neighbourhood of (a, b) such that for every point $(a+h, b+k)$ of this neighbourhood

$$f(a, b) < f(a+h, b+k)$$

then $f(a, b)$ is said to be minimum value of $f(x, y)$ at (a, b).

6.2. The necessary conditions for f(a, b) to be an extreme value of f(x, y) are that

$$\mathbf{f_x(a, b)=0, \ f_y(a, b)=0}$$

Suppose $f(a, b)$ is an extreme value of the function $f(x, y)$ at (a, b).

Then $f(a, b)$ is also an extreme value of the function $f(x, b)$ at $x=a$

But $f(x, b)$ being a function of a single variable x, the necessary condition for its extreme value is $f_x(x, b)=0$ at $x=a$

$\therefore \ f_x(a, b)=0.$

Similarly, $f_y(a, b)=0.$

Note : $f(a, b)$ is said to be a stationary value of $f(x, y)$ at (a, b) if at this point

$$f_x(a, b)=0, \ f_y(a, b)=0$$

But these are the conditions of an extreme value. Thus every extreme value is a stationary value but the converse is not always true.

6.3. Sufficient condition for the extremum of a function $f(x, y)$ at (a, b).

If $f_x(a, b)=0$, $f_y(a, b)=0$

and $f_{x^2}(a, b)=A$, $f_{xy}(a, b)=B$, $f_{y^2}(a, b)=C$

then (1) $f(a, b)$ is a maximum value of $f(x, y)$ at (a, b) if $AC-B^2>0$ and $A<0$.

(2) $f(a, b)$ is a minimum value of $f(x, y)$ at (a, b) if $AC-B^2>0$ and $A>0$.

(3) $f(a, b)$ is neither a maximum nor a minimum value of $f(x, y)$ at (a, b) if $AC-B^2<0$

(4) the case is doubtful and needs further investigation if $AC-B^2=0$.

Proof. By Taylor's Theorem with remainder after three terms we have

$$f(a+h, b+k)=f(a, b)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(a, b)$$

$$+\frac{1}{\lfloor 2}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(a, b)$$

$$+\frac{1}{\lfloor 3}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^3 f(a+\theta h, b+\theta k)$$

$$=f(a, b)+hf_x(a, b)+kf_y(a, b)$$

$$+\tfrac{1}{2}\{h^2 f_{x^2}(a, b)+2hkf_{xy}(a, b)+k^2 f_{y^2}(a, b)\}$$

$$+\tfrac{1}{6}\{h^3 f_{x^3}(u, v)+3h^2 k f_{x^2 y}(u, v)+3hk^2 f_{xy^2}(u, v)+k^3 f_{y^3}(u, v)\}$$

where $u=a+\theta h$, $v=b+\theta k$.

or, $$f(a+h, b+k)-f(a, b)=hf_x(a, b)+kf_y(a, b)$$

$$+\tfrac{1}{2}\{h^2 f_{x^2}(a, b)+2hkf_{xy}(a, b)+k^2 f_{y^2}(a, b)\}+R_n$$

where R_n contains the terms of third degree in h and k which are very small.

Now by the given conditions, since $f_x(a, b)=0$ and $f_y(a, b)=0$, we get

$$f(a+h, b+k)-f(a, b)=\tfrac{1}{2}(Ah^2+2Bhk+Ck^2)+R_n \quad \cdots \quad (1)$$

Now for sufficiently small values of h and k the value of R_n can be made so small, that the sign of the $R.H.S.$ of (1) will be the same as that of

$$Ah^2+2Bhk+Ck^2.$$

So neglecting R_n in (1) we write

$$f(a+h, b+k)-f(a, b)=\frac{1}{2}(Ah^2+2Bhk+Ck^2)$$

$$=\frac{1}{2A}(A^2h^2+2ABhk+ACk^2)$$

$$=\frac{1}{2A}\{(Ah+Bk)^2+(AC-B^2)k^2\}$$

$$\cdots \quad (2)$$

Case I. Let $AC-B^2>0$, then neither A nor C is zero. So from (2) we get

$f(a+h, b+k)-f(a, b)=\dfrac{1}{2A}\times a$ non zero positive quantity $\quad \cdots \quad \cdots \quad \cdots \quad (3)$

$[\because \quad h\neq 0, k\neq 0.]$

So if besides $AC-B^2>0$, we have $A<0$ then from (3)

$$f(a+h, b+k)-f(a, b)<0$$

i.e., $f(a, b)>f(a+h, b+k)$.

Hence $f(a, b)$ is a maximum value.

Again if besides $AC-B^2>0$ we have $A>0$ then from (3)

$$f(a+h, b+k)-f(a, b)>0$$

i.e., $f(a, b)<f(a+h, b+k)$

Hence, $f(a, b)$ is a minimum value.

Case II. Let $AC-B^2<0$ and suppose $A\neq 0$, then $(AC-B^2)k^2<0$.

Let $(AC-B^2)k^2=-\alpha$ where α is a positive quantity.

Then from (3)

$$f(a+h,\, b+k)-f(a,\, b)=\frac{1}{2A}\{(Ah+Bk)^2-\alpha\}$$

$$=\frac{1}{2A}\times a \text{ positive or a negative}$$

quantity.

So for any sign of A, the $R.H.S.$ may be positive or negative.

i.e., $f(a+h,\, b+k)-f(a,\, b)>0$ or <0 for the same sign of A.

Hence, $f(a,\, b)$ cannot be an extreme value.

Similar is the proof if $C\neq 0$.

Again if both $A=0$ and $C=0$ then from (1)

neglecting R_n, $f(a+h,\, b+k)-f(a,\, b)=\frac{1}{2}\, 2Bhk=Bhk$.

$\therefore$ $f(a+h,\, b+k)-f(a,\, b)$ assumes values with different signs and hence $f(a,\, b)$ is not an extreme value.

Case III. Let $AC-B^2=0$ and suppose $A\neq 0$

Then from (1)

$$f(a+h,\, b+k)-f(a,\, b)=\frac{1}{2}(Ah^2+2Bhk+Ck^2)+R_n$$

$$=\frac{1}{2A}\{(Ah+Bk)^2+(AC-B^2)k^2\}+R_n$$

$$=\frac{1}{2A}(Ah+Bk)^2+R_n$$

$$=R_n \text{ when } Ah+Bk=0.$$

So the nature of the sign of $f(a+h,\, b+k)-f(a,\, b)$ depends on R_n. The case is therefore doubtful.

If $A=0$ then from the condition $AC-B^2=0$ we get $B=0$

$$\therefore\quad f(a+h,\, b+k)-f(a,\, b)=\tfrac{1}{2}Ck^2+R_n$$

$$=R_n \text{ if } k=0 \text{ whether } h \text{ is zero or not.}$$

This case therefore is also doubtful.

Illustrative Examples :

Ex. 1. Find the minimum value of

$$x^2+y^2+(x+y+1)^2 \qquad \textbf{(C. H. 1966)}$$

Let $f(x, y)=x^2+y^2+(x+y+1)^2$

Then $f_x=2x+2(x+y+1)=4x+2y+2$

$f_y=2y+2(x+y+1)=2x+4y+2.$

For $f_x=0$ and $f_y=0$ we have

$4x+2y+2=0$

and $2x+4y+2=0$

Solving these two equations $x=-\frac{1}{3}$, $y=-\frac{1}{3}$.

So $f(x, y)$ may have an extremum at $x=-\frac{1}{3}$, $y=-\frac{1}{3}$.

Now $A=f_{xx}=4,$

$B=f_{xy}=2$

$C=f_{yy}=4$

$\because \ AC-B^2=16-4=12>0$

and $A>0$, the function $f(x,y)$ is minimum at $(-\frac{1}{3}, -\frac{1}{3})$.

The minimum value of $f(x, y)$

$=(-\frac{1}{3})^2+(-\frac{1}{3})^2+(-\frac{1}{3}\cdot \frac{1}{3}+1)^2$

$=\frac{1}{3}.$

Ex. 2. Find the maximum or minimum value of $f(x, y)=x^2+xy+y^2+ax+by$ and determine whether the value you get is maximum or minimum. (C. H. 1964)

Here $f_x=2x+y+a$, $f_y=x+2y+b$.

For $f_x=0$ and $f_y=0$ we get

$2x+y+a=0$

$x+2y+b=0$

$\therefore$ Solving $x=\frac{1}{3}(b-2a)$, $y=\frac{1}{3}(a-2b)$

Now $A=f_{xx}=2,$

$B=f_{xy}=1$

And $C=f_{yy}=2$

Since, $AC-B^2=4-1=3>0$ and $A>0$,

$f(x, y)$ is minimum at $x=\frac{1}{3}(b-2a)$, $y=\frac{1}{3}(a-2b)$.

The minimum value of $f(x, y)$

$$=\tfrac{1}{9}(b-2a)^2+\tfrac{1}{9}(b-2a)(a-2b)+\tfrac{1}{9}(a-2b)^2$$
$$+\frac{a}{3}(b-2a)+\frac{b}{3}(a-2b)$$
$$=\tfrac{1}{9}(-3b^2-3a^2+3ab)$$
$$=\tfrac{1}{3}(ab-a^2-b^2).$$

Ex. 3. Find all the maxima and minima of the function $x^3+y^3-63(x+y)+12xy$. [C. H 1968]

Let $f(x, y)=x^3+y^3-63x-63y+12xy.$

Then
$$f_x=3x^2-63+12y$$
$$f_y=3y^2-63+12x.$$

$\therefore$ $f_x=0$, $f_y=0$ give
$$x^2+4y-21=0$$
$$y^2+4x-21=0.$$

Substrating 2nd from the 1st
$$(x^2-y^2)+4(y-x)=0$$

or,
$$(x-y)(x+y-4)=0$$

$\therefore$
$$x-y=0 \text{ or } x+y-4=0.$$

So we get two set of equations
$$x^2+4y-21=0 \text{ and } \cancel{x}\,y^2+4\cancel{y}\,x-21=0$$
$$x-y=0 \qquad\qquad x+y-4=0.$$

Solving these two set of equations we get the following sets of points as root of $f_x=0$ and $f_y=0$.

$$(3, 3),\ (-7, -7),\ (5, -1) \text{ and } (-1, 5)$$

Now $A=f_{xx}=6x$; $B=12$; $C=6y$.

$\therefore$ At $(3, 3)$ we have $A=18$, $B=12$ and $C=18$

So that $AC-B^2=18^2-12^2>0$ and A is positive.

$\therefore$ $f(x, y)$ is minimum at $(3, 3)$.

At $(-7, -7)$ we have $A=-42$, $B=12$, $C=-42$

So that $AC-B^2=42^2-144>0$ and A is negative

$\therefore$ $f(x,y)$ is maximum at $(-7, -7)$.

At $(5, 1)$ we have $A=30$, $B=12$ and $C=-6$

So that $AC-B^2=-180-144=$negative

$\therefore$ $f(x, y)$ has no extremum at $(5, -1)$.

At $(-1, 5)$, $A=-6$, $B=12$ and $C=30$

So that $AC-B^2=-180-144=$negative

$\therefore$ $f(x, y)$ has no extremum at $(-1, 5)$.

Ex. 4. Show that $u=x^2y^4(1-x-y)^6$ defined in the region $x\geqslant 0$, $y\geqslant 0$, $x+y\leqslant 1$ attain its greatest at an interior point of the region and find this greatest value.

$$u_x=2xy^4(1-x-y)^6-6x^2y^4(1-x-y)^5$$
$$=2xy^4(1-x-y)^5(1-4x-y)$$
$$u_y=4x^2y^3(1-x-y)^6-6x^2y^4(1-x-y)^5$$
$$=2x^2y^3(1-x-y)^5(2-2x-5y).$$

$\therefore$ $u_x=0$ and $u_y=0$ give

$$x=0,\ y=0,\ x+y=1,\ 4x+y=1,\ 2x+5y=2$$

Solving these equations we get the following set of points as solutions of $f_x=0$ and $f_y=0$, $(0, 0)$, $(0, 1)$, $(0, \frac{2}{5})$, $(1, 0)$, $(\frac{1}{4}, 0)$, $(\frac{1}{6}, \frac{1}{3})$, of which $(\frac{1}{6}, \frac{1}{3})$ is an interier point of the given region. So we are to examine the point $(\frac{1}{6}, \frac{1}{3})$ for extremum.

Now $A=u_{xx}=2y^4(1-x-y)^5(1-4x-y)$
$$-10xy^4(1-x-y)^4(1-4x-y)-8xy^4(1-x-y)^5$$
$$=-\frac{1}{54\times 36} \text{ at } x=\frac{1}{6},\ y=\frac{1}{3}.$$

$$B=u_{xy}=8xy^3(1-x-y)^5(1-4x-y)$$
$$-10xy^4(1-x-y)^4(1-4x-y)-2xy^4(1-x-y)^5$$
$$=-\frac{1}{54\times 144} \text{ at } \left(\frac{1}{6}, \frac{1}{3}\right).$$

$$C=u_{yy}=6x^2y^2(1-x-y)^5(2-2x-5y)$$
$$-10x^2y^3(1-x-y)^4(2-2x-5y)$$
$$-10x^2y^3(1-x-y)^5$$

$$=-\frac{10}{36\times 27}\left(1-\frac{1}{6}-\frac{1}{3}\right)^5$$

$$=-\frac{5}{18\times 27}\left(\frac{3}{6}\right)^5 \text{ at } \left(\frac{1}{6},\ \frac{1}{3}\right)$$

$$\therefore\quad AC-B^2=\left(-\frac{1}{54\times 36}\right)\left(-\frac{5}{18\times 27}\cdot\frac{1}{2^5}\right)-\left(\frac{2}{54\times 144}\right)^2$$

$=$ positive i.e., >0 and A is negative.

$\therefore$ $f(x, y)$ is maximum at $\left(\frac{1}{6},\ \frac{1}{3}\right)$.

So that maximum $u=\frac{1}{36}\cdot\frac{1}{81}\left(1-\frac{1}{6}-\frac{1}{3}\right)^6=\frac{1}{36\times 81\times 64}$.

Ex. 5, Investigate the maximum and minimum of

$$\mathbf{U=2(x-y)^2-x^4-y^4}$$

leaving aside any doubtful case that may arise.

$$U_x=4(x-y)-4x^3$$
$$U_y=-4(x-y)-4y^3.$$

$\therefore$ $U_x=0$ and $U_y=0$ give

$(x-y)-x^3=0 \quad \cdots \quad (1)$ and $x-y+y^3=0 \quad \cdots \quad (2)$

Substracting 1st equation from 2nd

$y^3+x^3=0 \qquad \therefore\ y=-x.$

putting $y=-x$ in (1) we get $2x-x^3=0$

$\therefore$ $x=0$ or, $\pm\sqrt{2}$ and $y=0$ or, $\mp\sqrt{2}$.

$\therefore$ The points are $(0, 0)$, $(\sqrt{2}, -\sqrt{2})$, $(-\sqrt{2}, \sqrt{2})$.

Now $A=U_{xx}=4-12x^2$

$B=U_{xy}=-4$

$C=4-12y^2$

$\therefore$ At $(0, 0)$ we have $A=4$, $B=-4$, $C=4$

So that $AC-B^2=16-16=0$, a doubtful case.

At $(\sqrt{2}, -\sqrt{2})$ we have $A=4-24=-20$, $B=-4$ $C=4-24=-20$, so that $AC-B^2=400-16=$positive and A is negative

$\therefore$ $f(x, y)$ is maximum at $(\sqrt{2}, -\sqrt{2})$

Similarly, $f(x, y)$ is maximum at $(-\sqrt{2}, \sqrt{2})$.

6.4. Constrained Maxima and Minima.

1. To find the maximum or minimum value of f(x, y) subject to the additional condition $\phi(x, y)=0$. *(C. H. 1964)*

Let $u=f(x, y)$ and suppose u is a differential function in a given domain.

Now, we have the constrained equation

$$\phi(x, y)=0 \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad (1)$$

If $\phi_y \neq 0$, then we can solve (1) for y.

Suppose, solving (1) we get $y=\psi(x)$.

Then the original function is reduced to $u=f(x, \psi)$ a function of x alone.

Differentiating this w. r. to x

$$\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial \psi}\frac{d\psi}{dx}.$$

Now for an extremum $\frac{du}{dx}=0$

$$\therefore \quad \frac{\partial f}{\partial x}+\frac{\partial f}{\partial \psi}\frac{d\psi}{dx}=0$$

$$\text{or,} \quad \frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{dy}{dx}=0 \quad \cdots \quad \cdots \quad (2)$$

Again from $\phi(x, y)=0$

$$\frac{\partial \phi}{\partial x}+\frac{\partial \phi}{\partial y}\frac{dy}{dx}=0 \quad \cdots \quad \cdots \quad (3)$$

Eliminating $\frac{dy}{dx}$ from (2) and (3) we get

$$\frac{f_x}{\phi_x}=\frac{f_y}{\phi_y} \quad \text{or,} \quad f_x\phi_y-\phi_x f_y=0$$

$$\text{or,} \quad \frac{\partial(f, \phi)}{\partial(x, y)}=0 \quad \ldots \quad \ldots \quad (4)$$

Equations (1) and (4) will now give the systems of values of x, y for which $f(x, y)$ may be a maximum or a minimum.

2. To find the maximum or minimum value of f(x, y, z) subject to the additional condition ϕ(x, y, z)=0.

We have the constrained equation

$$\phi(x, y, z)=0 \quad \ldots \quad \ldots \quad (1)$$

From (1) solving for z, suppose, we get

$$z=\psi(x, y)$$

The problem is then reduced to find the extreme value of $f(x, y, \psi)$.

We now write $F(x, y)=f(x, y, \psi)$ ... (2)

The necessary condition for the extreme of (2) is that F_x and F_y must vanish separately.

$$\therefore \quad \frac{\partial f}{\partial x}+\frac{\partial f}{\partial z}\frac{\partial \psi}{\partial x}=0 \quad \ldots \quad \ldots \quad (3)$$

$$\frac{\partial f}{\partial y}+\frac{\partial f}{\partial z}\frac{\partial \psi}{\partial y}=0 \quad \ldots \quad \ldots \quad (4)$$

Also from (1) we get

$$\frac{\partial \phi}{\partial x}+\frac{\partial \phi}{\partial z}\cdot\frac{\partial \psi}{\partial x}=0 \quad \ldots \quad \ldots \quad (5)$$

$$\frac{\partial \phi}{\partial y}+\frac{\partial \phi}{\partial z}\cdot\frac{\partial \psi}{\partial y}=0 \quad \ldots \quad \ldots \quad (6)$$

Eliminating $\frac{\partial \psi}{\partial x}$ from (3) and (5) we get

$$\frac{\dfrac{\partial f}{\partial x}}{\dfrac{\partial \phi}{\partial x}}=\frac{\dfrac{\partial f}{\partial z}}{\dfrac{\partial \phi}{\partial z}} \quad \text{or,} \quad f_x\phi_z-f_z\phi_x=0$$

$$\text{or,} \quad \frac{\partial(f, \phi)}{\partial(x, z)}=0 \quad \ldots \quad (7)$$

Similarly, eliminating $\dfrac{\partial \psi}{\partial y}$ from (4) and (6) we get

$$\frac{\partial(f, \phi)}{\partial(y, z)}=0 \quad \cdots \quad \cdots \quad (8)$$

Equations (1), (7) and (8) will now give a system of values of x, y, z for which the given function is extremum.

6.5. Lagranges method of finding stationary values of a function of several variables connected by one or more relations. [*C. H. 1948*]

Let $u=f(x_1, x_2, \cdots x_n)$ be a function of n variables $x_1, x_2, \cdots x_n$, connected by the m number of constrained equations

$$\left.\begin{aligned} f_1(x_1, x_2, \cdots x_n) &=0 \\ f_2(x_1, x_2, \cdots x_n) &=0 \\ \cdots \quad \cdots \quad &\cdots \\ f_m(x_1, x_2, \cdots x_n) &=0. \end{aligned}\right\} \cdots \quad \cdots \quad (1)$$

For an extremum $du=0$, so from

$u=f(x_1\ x_2 \cdots x_n)$ we get

$$\frac{\partial f}{\partial x_1}dx_1+\frac{\partial f}{\partial x_2}dx_2 \cdots +\frac{\partial f}{\partial x_n}dx_n=0 \quad \cdots \quad (2)$$

Also from (1)

$$\left.\begin{aligned} \frac{\partial f_1}{\partial x_1}dx_1+\frac{\partial f_1}{\partial x_2}dx_2+\cdots+\frac{\partial f_1}{\partial x_n}dx_n=0 \\ \frac{\partial f_2}{\partial x_1}dx_1+\frac{\partial f_2}{\partial x_2}dx_2+\cdots+\frac{\partial f_2}{\partial x_n}dx_n=0 \\ \cdots \qquad \cdots \qquad \cdots \qquad \cdots \\ \frac{\partial f_m}{\partial x_1}dx_1+\frac{\partial f_m}{\partial x_2}dx_2+\cdots+\frac{\partial f_m}{\partial x_n}dx_n=0 \end{aligned}\right\} \cdots \quad (3)$$

Multiplying (2) by (1) and (3) by $\lambda_1, \lambda_2, \cdots \lambda_m$ respectively and adding we get

$$P_1dx_1+P_2dx_2+\cdots+P_rdx_r+\cdots+P_n dx_n=0 \quad \cdots \quad (4)$$

where $P_r = \dfrac{\partial f}{\partial x_r} + \lambda_1 \dfrac{\partial f_1}{\partial x_r} + \lambda_2 \dfrac{\partial f_2}{\partial x_r} + \cdots + \lambda_m \dfrac{\partial f_m}{\partial x_r}$

Since $\lambda_1, \lambda_2, \cdots \lambda_m$ are arbitrary we can choose them in such a way that

$$P_1 = 0,\ P_2 = 0 \cdots P_m = 0 \qquad \cdots \qquad \cdots \qquad (5)$$

The equation (4) then becomes

$$P_{m+1} dx_{m+1} + P_{m+2} dx_{m+2} + \cdots + P_n dx_n = 0 \qquad \cdots \qquad (6)$$

Now by virtue of (2) and (3) there remains only $n-m$ variables as independent. So the quantities $dx_{m+1}, dx_{m+2}, \cdots dx_n$ in (6) are all independent and their co-efficient should vanish separately.

$\therefore$ Form (6) we

$$P_{m+1} = 0,\ P_{m+2} = 0 \cdots P_n = 0 \qquad \cdots \qquad \cdots \qquad (7)$$

Thus we get the relations (1), (5) and (7) giving $m+m+(n-m)$ i. e., $m+n$ relations viz. $f_1 = 0, f_2 = 0 \cdots f_m = 0$

$$P_1 = 0,\ P_2 = 0 \cdots P_n = 0$$

From these relations we get the m multipliers $\lambda_1, \lambda_2, \cdots \lambda_m$ and the values of $x_1, x_2, \cdots x_n$ for which the given function is extremum.

6.6. If $f(x, y, z, w)$ has an extreme value at (ξ, η, ζ, ρ) subject to the subsidiary conditions

$$\phi(x, y, z, w) = 0,\ \psi(x, y, z, w) = 0$$

and if at that point $\dfrac{\partial(\phi, \psi)}{\partial(z, w)} \neq 0$

then two numbers λ and μ exist such that at the point (ξ, η, ζ, ρ) the equation

$$\begin{aligned} f_x + \lambda\phi_x + \mu\psi_x &= 0 \\ f_y + \lambda\phi_y + \mu\psi_y &= 0 \\ f_z + \lambda\phi_z + \mu\psi_z &= 0 \\ f_w + \lambda\phi_w + \mu\psi_w &= 0 \end{aligned}$$

and also the subsidiary conditions are satisfied.

Let $u = f(x, y, z, w)$ $\qquad \cdots \qquad \cdots \qquad (1)$

For an extremum, $du = 0$

$$\therefore \quad \frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy+\frac{\partial f}{\partial z}dz+\frac{\partial f}{\partial w}dw=0 \quad \ldots \quad (2)$$

Also from the subsidiary conditions,

$$\phi(x, y, z, w)=0 \text{ and } \psi(x, y, z, w)=0$$

We have

$$\frac{\partial \phi}{\partial x}dx+\frac{\partial \phi}{\partial y}dy+\frac{\partial \phi}{\partial z}dz+\frac{\partial \phi}{\partial w}\partial w=0 \quad \ldots \quad (3)$$

And $\dfrac{\partial \psi}{\partial x}dx+\dfrac{\partial \psi}{\partial y}dy+\dfrac{\partial \psi}{\partial z}\partial z+\dfrac{\partial \psi}{\partial w}dw=0 \quad \ldots \quad (4)$

Multiplying (2) by (1), (3) by λ and (4) by μ and adding we get

$$\left(\frac{\partial f}{\partial x}+\lambda\frac{\partial \phi}{\partial x}+\mu\frac{\partial \psi}{\partial x}\right)dx+\left(\frac{\partial f}{\partial y}+\lambda\frac{\partial \phi}{\partial y}+\mu\frac{\partial \psi}{\partial y}\right)dy$$

$$\left(\frac{\partial f}{\partial z}+\lambda\frac{\partial \phi}{\partial z}+\mu\frac{\partial \psi}{\partial z}\right)dz+\left(\frac{\partial f}{\partial w}+\lambda\frac{\partial \phi}{\partial w}+\mu\frac{\partial \psi}{\partial w}\right)dw=0 \quad \ldots \quad (5)$$

Since λ and μ are arbitrary, we can choose them in such a way that

$$\left.\begin{aligned}\frac{\partial f}{\partial z}+\lambda\frac{\partial \phi}{\partial z}+\mu\frac{\partial \psi}{\partial z}=0\\ \text{And}\quad \frac{\partial f}{\partial w}+\lambda\frac{\partial \phi}{\partial w}+\mu\frac{\partial \psi}{\partial w}=0\end{aligned}\right\} \quad \ldots \quad \ldots \quad (6)$$

Solving the pair of equations in (6) we get

$$\frac{\lambda}{\psi_z f_w-\psi_w f_z}=\frac{\mu}{\phi_w f_z-\phi_z f_w}=\frac{1}{\phi_z\psi_w-\psi_z\phi_w}$$

$$\therefore \quad \lambda=\frac{\psi_z f_w-\psi_w f_z}{\phi_z\psi_w-\psi_z\phi_w}, \quad \mu=\frac{\phi_w f_z-\phi_z f_w}{\phi_z\psi_w-\psi_z\phi_w}$$

This shows that the above choice of λ, μ is always possible provided

$$\phi_z\psi_w-\psi_z\phi_w\neq 0$$

i. e., $$\frac{\partial(\phi, \psi)}{\partial(z, w)}\neq 0$$

Now from (5) using (6) we get

$$\left(\frac{\partial f}{\partial x}+\lambda\frac{\partial \phi}{\partial x}+\mu\frac{\partial \psi}{\partial x}\right)dx+\left(\frac{\partial f}{\partial y}+\lambda\frac{\partial \phi}{\partial y}+\frac{\partial \phi}{\partial y}\right)dy=0. \quad (7)$$

Since, the four variables x, y, z, w are connected by two equations

$$\phi(x, y, z, w) = 0 \text{ and } \psi(x, y, z, w) = 0$$

it is always possible to find any two of them in terms of the remaining two. Then u can be regarded as the function of two independent variables x and y which can be made to vary independently of each other. Hence for an extremum the co-efficient of dx and dy in (7) must vanish separately. Thus we get another two relations.

$$\left.\begin{aligned} &\frac{\partial f}{\partial x}+\lambda\frac{\partial \phi}{\partial x}+\mu\frac{\partial \psi}{\partial x}=0 \\ \text{And } &\frac{\partial f}{\partial y}+\lambda\frac{\partial \phi}{\partial y}+\mu\frac{\partial \psi}{\partial y}=0. \end{aligned}\right\} \quad \ldots \quad \ldots \quad (8)$$

Hence for an extremum, equations (1), (6) and (8) determine x, y, z, w, λ, μ.

Hence the theorem.

Ex. 1. Find the maximum or minimum value of $x^m . y^n . z^p$. subject to the condition $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$.

Let $u = x^m y^n z^p$ and $f = \frac{a}{x}+\frac{b}{y}+\frac{c}{z}-1$.

Then by logarithimic differentiation of u

$$\frac{1}{u}du = \frac{m}{x}dx+\frac{n}{y}dy+\frac{p}{z}dz=0$$

Also differentiating f

$$df = -\frac{a}{x^2}dx-\frac{b}{y^2}dy-\frac{c}{z^2}dz=0$$

Thus, we get two equations

$$\frac{m}{x}dx+\frac{n}{y}dy+\frac{p}{z}dz=0 \quad \ldots \quad \ldots \quad (1)$$

$$\text{and } \frac{a}{x^2}dx+\frac{b}{y^2}dy+\frac{c}{z^2}dz=0. \quad \ldots \quad (2)$$

Multiplying (1) and (2) by 1 and λ respectively and adding and then equating co-efficients of dx, dy and dz

$$\frac{m}{x}+\frac{\lambda a}{x^2}=0 \quad \frac{n}{y}+\frac{\lambda b}{y^2}=0, \frac{p}{z}+\frac{\lambda c}{z^2}=0.$$

or, $\quad m+\frac{\lambda a}{x}=0, \quad n+\frac{\lambda b}{y}=0, \quad p+\frac{\lambda c}{z}=0 \quad \cdots \quad (3)$

$\therefore$ adding $(m+n+p)+\lambda\left(\frac{a}{x}+\frac{b}{y}+\frac{c}{z}\right)=0.$

or, $\quad (m+n+p)+\lambda=0$

$\therefore \quad \lambda=-(m+n+p).$

Hence from (3) $\frac{mx}{a}=\frac{ny}{b}=\frac{pz}{c}=(m+n+p)$

$$\therefore \quad x=\frac{a(m+n+p)}{m}, \quad y=\frac{b(m+n+p)}{n},$$

$$z=\frac{c(m+n+p)}{p}. \quad \cdots \quad \cdots \quad (4)$$

$\therefore$ **Maximum or minimum value of** u

$$=\frac{a^m}{m^m}\cdot\frac{b^n}{n^n}\cdot\frac{c^p}{p^p}\cdot(m+n+p)^{m+n+p} \quad \cdots \quad 5)$$

Discrimination : Differentiating $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$, taking z as a function of (x, y) we get

$$-\frac{a}{x^2}-\frac{c}{z^2}\frac{\partial z}{\partial x}=0 \qquad \therefore \quad \frac{\partial z}{\partial x}=-\frac{a}{c}\frac{z^2}{x^2}$$

Now from $\quad u=x^m y^n z^p$

$\log\ u=m\log x+n\log y+p\log z$

$$\therefore \quad \frac{1}{u}\frac{\partial u}{\partial x}=\frac{m}{x}+\frac{p}{z}\frac{\partial z}{\partial x}$$

$$=\frac{m}{x}-\frac{p}{z}\cdot\frac{az^2}{cx^2}$$

$$=\frac{m}{x}-\frac{paz}{cx^2}.$$

Differentiating again w. r. to x

$$-\frac{1}{u^2}\left(\frac{\partial u}{\partial x}\right)^2+\frac{1}{u}\frac{\partial^2 u}{\partial x^2}=-\frac{m}{x^2}+\frac{2paz}{cx^3}-\frac{pa}{cx^2}\frac{\partial z}{\partial x}$$

For maximum or minimum $\frac{\partial u}{\partial x}=0$

So, we get

$$\frac{1}{u}\frac{\partial^2 u}{\partial x^2}=-\frac{m}{x^2}+\frac{2paz}{cx^3}-\frac{pa}{cx^2}\left(-\frac{az^2}{cx^2}\right)$$

$$=-\frac{m}{x^2}+\frac{2pa}{cx^2}\left(\frac{mc}{pa}\right)+\frac{pa^2}{c^2x^2}\left(\frac{mc}{pa}\right)^2 \quad \because \quad \frac{z}{x}=\frac{mc}{pa}$$

$$=-\frac{m}{x^2}+\frac{2m}{x^2}+\frac{am^2}{x^2p}$$

$$=\frac{1}{x^2}\left(m+\frac{am^2}{p}\right)$$

$$\therefore \quad \frac{\partial^2 u}{\partial x^2}>0 \quad \text{for} \quad x=\frac{a(m+n+p)}{m}$$

Hence u is minimum and minimum value of

$$u=\frac{a^m b^n c^p}{m^m \cdot n^n \cdot p^p}(m+n+p)^{m+n+p}.$$

Ex. 2. If $\mathbf{u=x^2+y^2}$ **and if** $\mathbf{ax^2+2hxy+by^2=1}$**, find the extreme value of u.** [*C. H.* 1967]

Here $du=2xdx+2ydy=0$

Also from $f=ax^2+2hxy+by^2-1$

$$df=2ax\,dx+2h(ydx+xdy)+2bydy=0$$

So we get two equations

$$xdx+ydy=0 \qquad \cdots \qquad (1)$$

and $$(ax+hy)dx+(by+hx)dy=0. \qquad \cdots \qquad (2)$$

Multiplying (2) by λ and adding with (1) and then equating co-efficients of dx and dy we get

$$x+\lambda(ax+hy)=0 \qquad \cdots \qquad (3)$$

$$y+\lambda(by+hx)=0. \qquad \cdots \qquad (4)$$

Multiplying these two equations by x and y respectively and adding

$$x^2+y^2+\lambda(ax^2+2hxy+by^2)=0$$

or, $u+\lambda=0$

$\therefore \quad \lambda=-u.$

$\therefore$ from (3) and (4) we get for maximum or minimum

$$x-u(ax+hy)=0$$
$$y-u(by+hx)=0$$

or, $(1-au)x-huy=0$

and $-uhx+(1-bu)y=0.$

$$\therefore \quad \frac{1-au}{uh}=\frac{hu}{1-bu}$$

or, $(h^2-ab)u^2+(a+b)u-1=0$

$\therefore \; u=\dfrac{-(a+b)\pm\sqrt{(a+b)^2+4(h^2-ab)}}{2(h^2-ab)}$ is the extreme value of u.

Ex. 3. If $\mathbf{u=a^3x^2+b^3y^2+c^3z^2}$ where $\mathbf{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1}$ find the maximum or minimum value of u. [*C. H. 1964*]

$\because \quad u=a^3x^2+b^3y^2+c^3z^2$

and $f=\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}-1$

we get, $\quad du=2a^3x\,dx+2b^3y\,dy+2c^3z\,dz=0$

and $\quad df=-\dfrac{1}{x^2}dx-\dfrac{1}{y^2}dy-\dfrac{1}{z^2}dz=0$

Thus we get two equations

$$a^3x\,dx+b^3y\,dy+c^3z\,dz=0 \quad \cdots \quad (1)$$

$$\text{and } \frac{1}{x^2}dx+\frac{1}{y^2}dy+\frac{1}{z^2}dz=0 \quad \cdots \quad \cdots \quad (2)$$

Multiplying (2) by λ and adding with (1) and then equating co-efficients of dx, dy and dz we get

$$a^3x+\frac{\lambda}{x^2}=0, \quad b^3y+\frac{\lambda}{y^2}=0, \quad c^3z+\frac{\lambda}{z^2}=0. \cdots (3)$$

Multiplying these equations by x, y, z respectively and adding

$$(a^3x^2+b^3y^2+c^3z^2)+\lambda\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)=0$$

or, $u+\lambda=0$

$\therefore \quad \lambda=-u.$

So from (3) $c^3x-\frac{u}{x^2}=0$

$\therefore \quad a^3x^3=u$

Similarly, $b^3y^3=u$ and $c^3z^3=u$

$\therefore \quad ax=by=cz=k$ (suppose)

$\therefore \quad x=\frac{k}{a}, \quad y=\frac{k}{b}, \quad z=\frac{k}{c}$

Putting in $\quad \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$

$\frac{1}{k}(a+b+c)=1 \qquad \therefore \quad k=a+b+c$

$\therefore$ for maximum or minimum

$$x=\frac{1}{a}(a+b+c), \quad y=\frac{1}{b}(a+b+c), \quad z=\frac{1}{c}(a+b+c).$$

And the maximum or minimum value of u

$$=a^3\left(\frac{a+b+c}{a}\right)^2+b^3\left(\frac{a+b+c}{b}\right)^2+c^3\left(\frac{a+b+c}{c}\right)^2$$

$$=(a+b+c)^3. \qquad \ldots \qquad \ldots \qquad (4)$$

Discrimination :

Differentiating $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=1$ with respect to x, taking z as a function of x and y

$$-\frac{1}{x^2}-\frac{1}{z^2}\frac{\partial z}{\partial x}=0 \qquad \therefore \quad \frac{\partial z}{\partial x}=-\frac{z^2}{x^2}.$$

From $u=a^3x^2+b^3y^2+c^3z^2$

$$\frac{\partial u}{\partial x}=2a^3x+2c^3z\,\frac{\partial z}{\partial x}$$

$$=2a^3x+2c^3z\left(-\frac{z^2}{x^2}\right)$$

$$=2a^3x-\frac{2c^3z^3}{x^2}$$

$$\therefore\quad A=\frac{\partial^2 u}{\partial x^2}=2a^3-2c^3\left(-\frac{2z^3}{x^3}+\frac{3z^2}{x^2}\,\frac{\partial z}{\partial x}\right)$$

$$=2a^3+\frac{4c^3z^3}{x^3}+\frac{6c^3z^4}{x^4}$$

$$=2a^3+4c^3\,.\,\frac{a^3}{c^3}+6c^3\left(\frac{a}{c}\right)^4 \qquad \left[\because\ \frac{z}{x}=\frac{a}{c}\right]$$

$$=6\left(a^3+\frac{a^4}{c}\right)$$

Similarly, $C=\dfrac{\partial^2 u}{\partial y^2}=6\left(b^3+\dfrac{b^4}{c}\right)$

Also, $B=\dfrac{\partial^2 u}{\partial x\partial y}=-\dfrac{2c^3}{x^2}\,.\,3z^2\,\dfrac{\partial z}{\partial y}$

$$=-\frac{2c^3}{x^2}\,.\,3z^2\left(-\frac{z^2}{y^2}\right)$$

$$=6c^3\left(\frac{z}{x}\right)^2\left(\frac{z}{y}\right)^2$$

$$=6c^3\left(\frac{a}{c}\right)^2\left(\frac{b}{c}\right)^2=\frac{6a^2b^2}{c}$$

Now $AC-B^2=36\left(a^3+\dfrac{a^4}{c}\right)\left(b^3+\dfrac{b^4}{c}\right)-\left(\dfrac{6a^2b^2}{c}\right)^2$

$$=\frac{36a^2b^2}{c^2}\left\{abc(a+b+c)\right\}.$$

>0 if $abc(a+b+c)>0$.

Thus if a, b, c are all positive

$AC-B^2>0$ and $A>0$ and hence u is minimum. The minimum value is given by (4).

Ex. 4. Find the maximum value of $f(x, y, z)=x^2y^2z^2$ subject to the condition $x^2+y^2+z^2=c^2$. (*C. H. 1960*)

Let $u=f(x, y, z)=x^2y^2z^2$

and $f_1=x^2+y^2+z^2-c^2$

then $\log u=2 \log x+2 \log y+2 \log z$

$$\therefore \quad \frac{1}{u}du=\frac{2}{x}dx+\frac{2}{y}dy+\frac{2}{z}dz=0$$

and $df_1=2xdx+2ydy+2zdz=0$

So we get two equations

$$\frac{1}{x}dx+\frac{1}{y}dy+\frac{1}{z}dz=0 \quad \cdots \quad \cdots \quad (1)$$

and $\quad xdx+ydy+zdz=0 \quad \cdots \quad \cdots \quad (2)$

Multiplying (2) by λ and adding with (1) and then equating coefficient of dx, dy and dz, we have

$$\frac{1}{x}+\lambda x=0, \quad \frac{1}{y}+\lambda y=0, \quad \frac{1}{z}+\lambda z=0.$$

Multiplying these equations by x, y, z respectively and adding

$$1+1+1+\lambda(x^2+y^2+z^2)=0$$

$$\text{or,} \quad 3+\lambda c^2=0$$

$$\therefore \quad \lambda=-\frac{3}{c^2}.$$

So from $\frac{1}{x}+\lambda x$ we get $\frac{1}{x}-\frac{3x}{c^2}=0$

or, $3x^2=c^2$ or, $x^2=\frac{1}{3}c^2$.

Similarly, $y^2=z^2=\frac{1}{3}c^2$.

$\therefore$ The maximum or minimum value of u

$$=\frac{1}{3}c^2 \cdot \frac{1}{3}c^2 \cdot \frac{1}{3}c^2=\frac{1}{27}c^6.$$

Discrimination :

$$u = x^2y^2(c^2 - x^2 - y^2) = x^2y^2c^2 - x^4y^2 - x^2y^4$$

$$\therefore \quad \frac{\partial u}{\partial x} = 2xy^2c^2 - 4x^3y^2 - 2xy^4$$

$$\therefore \quad A = \frac{\partial^2 u}{\partial x^2} = 2y^2c^2 - 12x^2y^2 - 2y^4$$

$$= 2 \cdot \frac{1}{3}c^2 \cdot c^2 - 12 \cdot \frac{1}{3}c^2 \cdot \frac{1}{3}c^2 - 2 \cdot \frac{1}{9}c^4$$

$$= -\frac{8c^4}{9}.$$

$$B = \frac{\partial^2 v}{\partial x \partial y} = 4xyc^2 - 8x^3y - 8xy^3$$

$$= 4 \cdot \frac{1}{3}c^4 - 8xy(x^2 + y^2)$$

$$= \frac{4}{3}c^4 - 8\,\frac{1}{3}c^2 \cdot \frac{2}{3}c^2$$

$$= -\frac{4}{9}c^4.$$

Also, $$\frac{\partial u}{\partial y} = 2x^2yc^2 - 2x^4y - 4x^2y^3$$

$$\therefore \quad C = \frac{\partial^2 u}{\partial y^2} = 2x^2c^2 - 2x^4 - 12x^2y^2$$

$$= 2 \cdot \frac{1}{3}c^4 - 2 \cdot \frac{1}{9}c^4 - 12 \cdot \frac{1}{9}c^4$$

$$= -\frac{8}{9}c^4$$

Now, $$\because \quad AC - B^2 = \left(-\frac{8}{9}c^4\right)\left(-\frac{8}{9}c^4\right) - \left(-\frac{4}{9}c^4\right)^2$$

$$= \frac{64}{81}c^8 - \frac{16}{81}c^8$$

$$= \frac{48}{81}c^8 > 0$$

And $A<0$, $f(x, y, z)$ is maximum and its maximum value $=\frac{1}{27}c^6$.

Ex. 5. Find the volume of the greatest rectangular parallelopiped inscribed in the ellipsoid whose equation is $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

Let Volume $V=(2x)(2y)(2z)=8xyz$

And $f=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1$

$\therefore \quad dV=8(yzdx+xzdy+xydz)=0$

$$df=\frac{2x}{a^2}dx+\frac{2y}{b^2}dy+\frac{2z}{c^2}dz=0$$

$\therefore$ For stationary values, we have

$$yz+\lambda\frac{x}{a^2}=0$$

$$xz+\lambda\frac{y}{b^2}=0$$

$$xy+\lambda\frac{z}{c^2}=0$$

$\therefore \quad 3xyz+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\right)=0$

or, $\quad 3xyz+\lambda=0$

$\therefore \quad \lambda=-3xyz=-\frac{3V}{8}$.

Now $\quad yz=-\frac{\lambda x}{a^2}$

or, $\quad xyz=-\frac{\lambda x^2}{a^2}$

or, $\quad \frac{V}{8}=\frac{3V}{8}\cdot\frac{x^2}{a^2}$

or, $\quad \frac{x^2}{a^2}=\frac{1}{3}$.

Similarly, $\dfrac{y^2}{b^2}=\dfrac{z^2}{c^2}=\dfrac{1}{3}$.

$\therefore$ maximum or minimum value of

$$V=8.\frac{a}{\sqrt{3}}.\frac{b}{\sqrt{3}}.\frac{c}{\sqrt{3}}=\frac{8abc}{3\sqrt{3}}.$$

Discrimination :

$\because \quad f=\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}$

$\therefore \quad \dfrac{2x}{a^2}+\dfrac{2z}{c^2}.\dfrac{\partial z}{\partial x}=0$

or, $\quad \dfrac{\partial z}{\partial x}=-\dfrac{xc^2}{za^2}$.

Now from $V=8xyz$

$$\frac{\partial V}{\partial x}=8y\left(z+x\frac{\partial z}{\partial x}\right)=8y\left(z-\frac{x^2c^2}{za^2}\right)$$

$$\frac{\partial^2 V}{\partial x^2}=8y\left\{\frac{\partial z}{\partial x}-\frac{c^2}{a^2}\left(\frac{2xz-\frac{\partial z}{\partial x}.x^2}{z^2}\right)\right\}$$

$$=8y\left\{-\frac{xc^2}{za^2}-\frac{c^2}{a^2z^2}\left(2xz+\frac{xc^2.x^2}{za^2}\right)\right\}$$

$$=-ve \text{ when } \frac{x}{a}=\frac{y}{b}=\frac{z}{c}=\frac{1}{\sqrt{3}}.$$

Hence, volume is maximum and maximum volume $=\dfrac{8abc}{3\sqrt{3}}$.

Ex. 6. Prove that of all rectangular parallelopiped of same volume, the cube has the least surface.

If V be the volume and S be the surface of a parallelopiped of dimensions x, y, z, then

$$V=xyz \text{ and } S=2(xy+yz+zx).$$

$$\therefore \quad dS=2(y+z)dx+2(z+x)dy+2(x+y)dz=0$$
$$dV=yzdx+zxdy+xydz=0.$$

$\therefore$ For stationary points

$(y+z)+\lambda yz=0$

$$(z+x)+\lambda zx=0$$
$$(x+y)+\lambda xy=0.$$

or, $\dfrac{y+z}{yz}=\dfrac{z+x}{zx}=\dfrac{x+y}{xy}=-\lambda$

or, $\dfrac{1}{z}+\dfrac{1}{y}=\dfrac{1}{x}+\dfrac{1}{z}=\dfrac{1}{y}+\dfrac{1}{x}$

$\therefore\ x=y=z.$

Now $\because\ xyz=V$

$\therefore\ x^3=V\ \therefore\ x=V^{\frac{1}{3}}$

$\therefore\ x=y=z=V^{\frac{1}{3}}$

Again from $xyz=V$, differentiating $w.\ r.$ to x and regarding z as a function of x and y

$$yz+xy\frac{\partial z}{\partial x}=0 \qquad \therefore\ \frac{\partial z}{\partial x}=-\frac{z}{x}$$

and $$xz+xy\frac{\partial z}{\partial y}=0 \qquad \therefore\ \frac{\partial z}{\partial y}=-\frac{z}{y}.$$

Now from $S=2(xy+yz+zx)$

$$\frac{\partial S}{\partial x}=2\left(y+y\frac{\partial z}{\partial x}+z+x\frac{\partial z}{\partial x}\right)$$

$$=2y-2y\frac{z}{x}+2z-2x\frac{z}{x}$$

$$=2y-2\frac{yz}{x}.$$

$$\therefore\ \frac{\partial^2 S}{\partial x^2}=-2y\left\{\frac{x\frac{\partial z}{\partial x}-z}{x^2}\right\}=-\frac{2y}{x^2}\left\{-\frac{z}{x}.x-z\right\}$$

$$=\frac{4yz}{x^2}=4 \quad \text{at } x=y=z.$$

Similarly, $\dfrac{\partial^2 S}{\partial y^2}=4$ and $\dfrac{\partial^2 S}{\partial x\,\partial y}=2$

Since, $AC-B^2=4\times4-(2)^2=12>0$ and $A>0$

$\therefore$ S is least when $x=y=z$.

Ex. 7. Find the maximum value of $x^m y^n$ if $x+y=k$, a constant, the quantities being all positive.

Hence, show that

$$a^m b^n < \left(\frac{ma+nb}{m+n}\right)^{m+n}, \text{ except when } a=b \quad (C.\ H.\ 1965)$$

Let $u=x^m y^n$ and $f=x+y-k=0$

Then $\log u = m\log x + n\log y$

or, $\frac{1}{u}du=\frac{m}{x}dx+\frac{n}{y}dy=0$

and $df=dx+dy=0$

So we get two equations

$$\frac{m}{x}dx+\frac{n}{y}dy=0$$

and $dx+dy=0$

$\therefore$ For stationary points

$$\frac{m}{x}+\lambda=0 \quad \text{and} \quad \frac{n}{y}+\lambda=0 \quad \cdots \quad (1)$$

Multiplying (1) by x and y respectively and adding

$$m+n+\lambda(x+y)=0$$

or, $m+n+\lambda k=0$

$$\therefore \quad \lambda=-\frac{m+n}{k}$$

$\therefore$ From (1) $x=-\frac{m}{\lambda}=\frac{mk}{m+n}$

$$y=-\frac{n}{\lambda}=\frac{nk}{m+n}.$$

$\therefore$ Maximum or minimum value of u

$$=x^m\,.\,y^n$$

$$=\left(\frac{mk}{m+n}\right)^m\,.\,\left(\frac{nk}{m+n}\right)^n$$

$$=m^m\,.\,n^n\left(\frac{k}{m+n}\right)^{m+n}$$

Discrimination.

Differentiating $x+y=k$, treating y as a function of x

$$1+\frac{\partial y}{\partial x}=0 \qquad \therefore \quad \frac{\partial y}{\partial x}=-1$$

Again from $u=x^m y^n$,

$$\log u=m\log x+n\log y$$

$$\therefore \quad \frac{1}{u}\frac{\partial u}{\partial x}=\frac{m}{x}+\frac{n}{y}\frac{\partial y}{\partial x}=\frac{m}{x}-\frac{n}{y}$$

Differentiating again w. r. to x

$$-\frac{1}{u^2}\left(\frac{\partial u}{\partial x}\right)^2+\frac{1}{u}\frac{\partial^2 u}{\partial x^2}=-\frac{m}{x^2}+\frac{n}{y^2}\frac{\partial y}{\partial x}$$

$$=-\frac{m}{x^2}-\frac{n}{y^2}$$

or, $\dfrac{1}{u}\dfrac{\partial^2 u}{\partial x^2}=-\left(\dfrac{m}{x^2}+\dfrac{n}{y^2}\right)<0$

$\left[\because \text{ for extreme points } \dfrac{\partial u}{\partial x}=0.\right]$

$\therefore \quad \dfrac{\partial^2 u}{\partial x^2}<0$

Hence u is maximum

Deduction.

Maximum value of $x^m y^n=m^m n^n\left(\dfrac{k}{m+n}\right)^{m+n}$

or, Maximum value of $\left(\dfrac{x}{m}\right)^m\left(\dfrac{y}{n}\right)^n=\left(\dfrac{k}{m+n}\right)^{m+n}$ ... (1)

If we put $\dfrac{x}{m}=a$ and $\dfrac{y}{n}=b$

Then $x=am$ and $y=bn$. and so $am+bn=x+y=k$

$\therefore$ from (1) we get

maximum value of $a^m b^n=\left(\dfrac{am+bn}{m+n}\right)^{m+n}$ except when $a=b$

Exercise 6

1. Show that the maximum value of $u=x^3y^2(1-x-y)$ is at $x=\frac{1}{2}, y=\frac{1}{3}$

2. Prove that the minimum value of $u=xy+c^3\left(\frac{1}{x}+\frac{1}{y}\right)$ is $3c^2$.

3. Show that $x^2y^2-5x^2-8xy-5y^2$ is maximum at the origin.

4. Find the stationary points of the function

$$x^3+3xy^2-15x^2-15y^2+72x$$

examining whether they are maximum or minimum. (D. H. 1953)

5. Find the maximum value of $\cos x.\cos y.\cos z$, where x, y, z are the angles of a plane triangle. [*Ans.* 1/8] (C. H. 1959)

6. Show that $u=\sin x \sin y \sin (x+y)$ is maximum at $x=y=\pi/3$ and the maximum value of u is $3\sqrt{3}/8$.

7. Show that the minimnm value of $x+y+z$ subject to the condition $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$ is $(\sqrt{a}+\sqrt{b}+\sqrt{c})^2$.

8. Let $u=x^2+y^2+z^2$ subject to the condition $ax+by+cz=p$. Show that the minimum value of u is $p^2/(a^2+b^2+c^2)$.

9. If $u=x^2+y^2+z^2$ where $ax^2+by^2+cz^2=1$ show that the extreme values of u are the roots of the equation $(1-au)(1-bu)(1-cu)=0$.

CHAPTER VII

MULTIPLE INTEGRALS

7·1. Double Integral.

Let us consider a function $f(x, y)$ which is bounded and defined in a region A on a plane. Let the region A be sub-divided into sub-regions $A_1, A_2, \cdots A_n$ in any manner, and let w_r be the area of the sub-region A_r. Let us now construct the two sums S and s as follows

$$S = \sum_{r=1}^{n} w_r M_r, \quad s = \sum_{r=1}^{n} w_r m_r$$

where M_r and m_r are the bounds of $f(x, y)$ in A_r.

If $f(x, y)$ is continuous in the region A, then S and s tend to the limit I as the area of the sub-regions w_r tends to zero, and in that case $f(x, y)$ is said to possess a double integral over the field A and is denoted by

$$\iint_A f(x, y) dA \quad or, \quad \iint_A f(x, y)\, dx.dy.$$

Hence $I = Lt \sum w_r M_r = Lt \sum w_r m_r$

$$= \iint_A f(x, y) dx\, dy.$$

7·2. Geometrical meaning of the double Integral.

Suppose the double integral $\iint_A f(x, y) dx\, dy$ exists over the field A. Let the region A be subdivided into sub-regions $A_1, A_2, \cdots A_n$ in any manner and let w_r be the area of the sub-region A_r. Let us now construct a prism on w_r as base and with sides parallel to z-axis cut off an area A_r'

from the surface $z=f(x, y)$. If V_r be the volume of this prism, we can write

$$\sum_1^n w_r m_r \leqslant \sum_1^n V_r \leqslant \sum_1^n w_r M_r.$$

or, $$Lt \sum w_r m_r \leqslant Lt\ \Sigma V_r \leqslant Lt\ \Sigma w_r M_r$$

Since $\iint_A f(x, y) dx dy$ exists

$$\iint_A f(x, y) dx dy = Lt \Sigma w_r m_r = Lt \Sigma w_r M_r$$

Hence $$\iint_A f(x, y) dx dy = \Sigma V_r = V$$

where V is the volume of the space generated by a line parallel to the z axis and moving along the contour of the region A and bounded by the region A below and the surface $z=f(x, y)$ above.

7·3. Theorems on the Evolution of Double Integrals.

I. If f(x, y) be a continuous function defined in the region A and having a double integral over the field A, then

$$\iint_R f(x, y) dx dy = \int_{x_0}^{X} dx \int_{y_0}^{Y} f(x, y) dy$$

when A is the rectangular region R.

Proof. Let the rectangular region R where $f(x, y)$ is continuous be bounded by the lines $x=x_0$, $x=X$ and $y=y_0$, $y=Y$ where x_0, y_0, X, Y are all constants. If the rectangle R be now divided into sub-rectangles $R_1, R_2, \cdots R_n$ by drawing lines parallel to x and y axes, then a small rectangle R_{ik} is formed bounded by the lines $x=x_{i-1}$, $x=x_i$; $y=y_{k-1}$, $y=y_k$.

Let w_{ik} be the area of the rectangle R_{ik},

then $w_{ik}=(x_i-x_{i-1})(y_k-y_{k-1})$ $\cdots$ $\cdots$ (1)

Let (ξ_{ik}, η_{ik}) be a point in R_{ik} and M_{ik}, m_{ik} are the upper and lower bounds of $f(x, y)$ in it. Then clearly

$$m_{ik}w_{ik} \leq f(\xi_{ik}, \eta_{ik})w_{ik} \leq M_{ik}w_{ik}$$

$$\text{or,} \quad \sum_{i=1}^{n}\sum_{k=1}^{n} m_{ik}w_{ik} \leq \sum_{i=1}^{n}\sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})w_{ik} \leq \sum_{i=1}^{n}\sum_{k=1}^{m} M_{ik}w_{ik}$$

$$\text{or,} \quad Lt \sum_{i=1}^{n}\sum_{k=1}^{m} m_{ik}w_{ik} \leq Lt \sum_{i=1}^{n}\sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})w_{ik}$$

$$\leq Lt \sum_{i=1}^{n}\sum_{k=1}^{m} M_{ik}w_{ik} \quad \cdots \quad (2)$$

But, since, the double integral exists

$$\iint_R f(x, y)dxdy = Lt \sum_{i=1}^{n}\sum_{k=1}^{m} m_{ik}w_{ik} = Lt \sum_{i=1}^{n}\sum_{k=1}^{m} M_{ik}w_{ik}$$

Hence from (2)

$$\iint_R f(x, y)dxdy = Lt \sum_{i=1}^{n}\sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})w_{ik}$$

$$= Lt \sum_{i=1}^{n}\sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})(x_i - x_{i-1})(y_k - y_{k-1}) \text{ from (1)} \cdots (3)$$

Let S_r denotes the sum of all the rectangles between the lines $x=x_{i-1}$ and $x=x_i$. Then

$$S_r=(x_i-x_{i-1})[f(\xi_{i1}, \eta_{i1})(y_1-y_0)+f(\xi_{i2}, \eta_{i2})(y_2-y_1)+\cdots$$
$$\cdots\cdots+f(\xi_{ik}, \eta_{ik})(y_k-y_{k-1})+\cdots\cdots]$$

Let $\xi_{i1}=\xi_{i2}=\cdots=\xi_{ik}=x_{i-1}$ (*a* constant quantity) then

$$S_r=(x_i-x_{i-1})[f(x_{i-1}, \eta_{i1})(y_1-y_0)+f(x_{i-1}, \eta_{i2})(y_2-y_1)+$$
$$\cdots\cdots+f(x_{i-1}, \eta_{ik})(y_k-y_{k-1})+\cdots \quad \cdots$$

Let us now choose $\eta_{i1}, \eta_{i2} \cdots, \eta_{ik} \cdots$ in such a way that

$$f(x_{i-1}, \eta_{i1})(y_1 - y_0) + f(x_{i-1}, \eta_{i2})(y_2 - y_1) = \cdots.$$

$$= \int_{y_0}^{Y} f(x_{i-1}, y)dy \text{ where } x_{i-1} \text{ is a constant.}$$

Then from (4)

$$S_r = (x_i - x_{i-1}) \int_{y_0}^{Y} f(x_{i-1}, y)dy.$$

Now considering all such sums of rectangles between the parallel lines $(x = x_0, x = x_1)$, $(x = x_1, x = x_2)$, $(x = x_2, x = x_3) \cdots$ and adding we get the total area of the rectangles.

Hence from (3) we get

$$\iint_R f(x, y)dxdy = Lt\Big\{(x_1 - x_0) \int_{y_0}^{Y} f(x_0, y)dy +$$

$$(x_2 - x_1) \int_{y_0}^{Y} f(x_1, y)dy + \cdots\Big\}$$

$$= Lt\{(x_1 - x_0)\phi(x_0) + (x_2 - x_1)\phi(x_1) + \cdots$$

$$\cdots + (x_i - x_{i-1})\phi(x_{i-1}) + \cdots\}.$$

$$\left[\text{putting } \phi(x) = \int_{y_0}^{Y} f(x, y)dy.\right]$$

$$= Lt\{\delta_0\phi(x_0) + \delta_1\phi(x_1) + \cdots + \delta_r\phi(x_r) + \cdots\}$$

$$= Lt \sum_{r=0}^{n} \delta_r \phi(x_r)$$

$$= \int_{x_0}^{X} \phi(x)dx \text{ in the limit as the greatest } \delta r \to 0 \text{ and}$$

assuming $\phi(x)$ is continuous in the interval.

$$= \int_{x_0}^{X} dx \int_{y_0}^{Y} f(x, y)\, dy$$

Note. The above result gives us a method of evoluting a double integral. Thus to evolute the double integral $f(x, y)$, $f(x, y)$ may be integrated first with respect to y between the limits y_0 and Y keeping x constant, then the resulting function which is a function of x alone may be integrated between the limits x_0 and X.

II. If f(x, y) be a continuous function defined in a region A and having a double integral over the field A, then

$$\iint_R \mathbf{f(x, y)\, dxdy} = \int_{y_0}^{Y} \mathbf{dy} \int_{x_0}^{X} \mathbf{f(x, y)\, dx}$$

where A is the rectangular region R.

Proof. As in the last theorem, from (3)

$$\iint_R f(x, y)\, dx\, dy = Lt \sum_{i=1}^{n} \sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})(x_i - x_{i-1})(y_k - y_{k-1}) \quad \cdots \quad (3)$$

Let S_r denote the sum of all the rectangles between the lines $y = y_{k-1}$ and $y = y_k$, then

$$S_r = (y_k - y_{k-1})[f(\xi_{i1}, \eta_{i1})(x_1 - x_0) + f(\xi_{i2}, \eta_{i2})(x_2 - x_1) + \cdots \\ \cdots + f(\xi_{ik}, \eta_{ik})(x_i - x_{i-1}) + \cdots]$$

Let $\eta_{i1} = \eta_{i2} = \cdots\cdots \eta_{ik} = y_{k-1}$ (a constant quantity) then

$$S_r = (y_k - y_{k-1})[f(\xi_{i1}, y_{k-1})(x_1 - x_0) + f(\xi_{i2}, y_{k-1})(x_2 - x_1) + \\ \cdots + f(\xi_{ik}, y_{k-1})(x_i - x_{i-1}) + \cdots] \quad \cdots \quad (4)$$

Now choose $\xi_{i1}, \xi_{i2}, \cdots \xi_{ik} \cdots$ in such a way that

$$f(\xi_{i1}, y_{k-1})(x_1 - x_0) + f(\xi_{i2}, y_{k-1})(x_2 - x_1) + \cdots$$

$$= \int_{x_0}^{X} f(x, y_{k-1})\, dx, \text{ where } y_{k-1} \text{ is a constant.}$$

Then from (4)

$$S_r=(y_k-y_{k-1})\int_{x_0}^{X} f(x, y_{k-1})dx$$

Now considering all such sums of rectangles between the parallel lines $(y=y_0, y=y_1)$, $(y=y_1, y=y_2)$, $(y=y_2, y=y_3)\cdots$ and adding we get the total sum of rectangles *i.e.*,

$$\sum_{i=1}^{n}\sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})(x_i-x_{i-1}\,(y_k-y_{k-1})$$

Hence from (3) we get

$$\iint_R f(x, y)\,dx\,dy$$

$$=Lt\Big[(y_1-y_0)\int_{x_0}^{X} f(x, y_0)dx+(y_2-y_1)\int_{x_0}^{X} f(x, y_1)dx +\cdots\Big]$$

$$=Lt\{(y_1-y_0)\phi(y_0)+(y_2-y_1)\phi(y_1)+\cdots \cdots+(y_k-y_{k-1})\phi(y_{k-1})+\cdots\}$$

putting $\phi(y)=\int_{x_0}^{X} f(x, y)dx$

$$=Lt\,\{\delta\,\phi(y_0)+\delta_1\phi(y_1)+\cdots+\delta_r\phi(y_r)+\cdots$$

$$=Lt\sum_{r=0}^{n}\delta_r\phi'(y_r)$$

$=\int_{y_0}^{Y}\phi(y)dy.$ assuming $\phi(y)$ is continuous in the range considered and greatest of δ_r tending to zero.

$$=\int_{y_0}^{Y} dy\int_{x_0}^{X} f(x, y)\,dx$$

Note. The second method of evolution of a double integral is that we may first integrate $f(x, y)$ with respect to x between the limits x_0 and X and the resulting function which is a function of y alone may then be integrated with respect to y between the limits y_0 and Y.

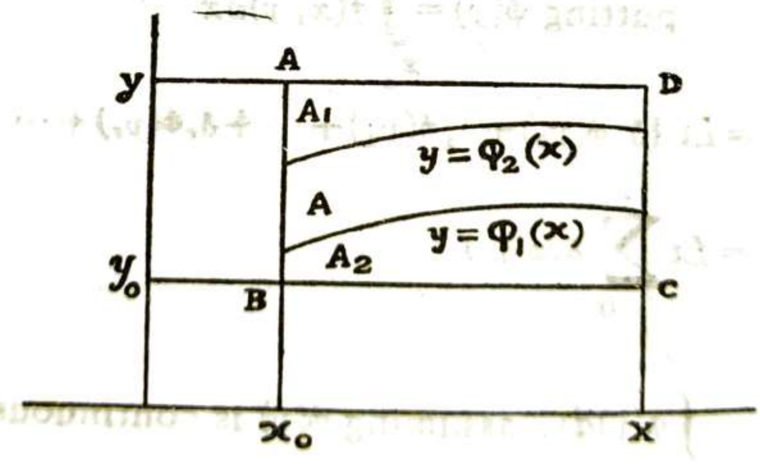
Cor. From the above two theorems, it follows that

$$\int_{x_0}^{X} dx \int_{y_0}^{Y} f(x, y)\,dy = \int_{y_0}^{Y} dy \int_{x_0}^{X} f(x, y)\,dx \quad \text{over the field } A,$$

where $f(x, y)$ is continuous in A and x_0, y_0, X, Y are all constants.

III. Let a function $f(x, y)$ be continuous in a region A which is a plane surface bounded by $x = x_0$, $x = X$, $y = \phi_1(x)$, $y = \phi_2(x)$ and that any line parallel to y-axis meets the bounding curves in two points only. If then $\phi_1(x)$, $\phi_2(x)$ are continuous functions of x in (x_0, X) and $\phi_1(x) \leqslant \phi_2(x)$, then

$$\iint_A f(x, y)\,dx\,dy = \int_{x_0}^{X} dx \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\,dy.$$



Let R be the rectangular region $ABCD$ bounded by $x = x_0$, $x = X$, $y = y_0$, $y = Y$. Also, let A be the region entirely lying in R bounded by $x = x_0$, $x = X$, $y = \phi_1(x)$ and $y = \phi_2(x)$.

Then clearly $R = A + A_1 + A_2$.

Now consider a new function $F(x, y)$ defined as

(1) $F(x, y)=0$ for any point in A_1 and A_2

(2) $F(x, y)=f(x, y)$ for any point in A or on its contour.

Then

$$\iint_R F(x, y)dxdy=\iint_A F(x, y)dxdy+\iint_{A_1} F(x, y)dxdy$$
$$+\iint_{A_2} F(x, y)dxdy.$$

$$=\iint_A f(x, y)\, dxdy. \quad \cdots \qquad \cdots \quad (1)$$

Now the field of the L.H.S. being rectangular

$$\iint_R F(x, y)dxdy=\int_{x_0}^{X} dx\int_{y_0}^{Y} F(x, y)dy$$

$\therefore$ From (1)

$$\int_{x_0}^{X} dx\int_{y_0}^{Y} F(x, y)dy=\iint_A f(x, y)dx.dy \quad \cdots \quad (2)$$

Now $$\int_{y_0}^{Y} F(x, y)dy=\int_{y_0}^{y=\phi_1(x)} F(x, y)dy+\int_{\phi_1(x)}^{\phi_2(x)} F(x, y)dy$$

$$+\int_{\phi_2(x)}^{Y} F(x, y)dy.$$

$$=0+\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)dy+0$$

$$=\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)dy$$

Hence putting in (2) we get

$$\iint_A f(x, y)dxdy = \int_{x_0}^{X} dx \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)dy.$$

Cor. Similarly, as in III.

$$\iint_A f(x, y)dxdy = \int_{y_0}^{Y} dy \int_{\phi_1(y)}^{\phi_2(y)} f(x, y)dx.$$

where A is region bounded by $y=y_0$, $y=Y$, $x=\phi_1(y)$ and $x=\phi_2(y)$ and $f(x, y)$ being continuous in the region considered.

IV. If a continuous function f(x, y), a ⩽ x ⩽ b, c ⩽ y ⩽ d be of the form f(x, y) = φ(x)ψ(y), then

$$\iint_R f(\mathbf{x}, \mathbf{y})\mathbf{dxdy} = \int_a^b \phi(\mathbf{x})\mathbf{dx} . \int_c^d \psi(\mathbf{y})\mathbf{dy}.$$

(C. H. 1960, 63)

As in theorem (I), from (3)

$$\iint_R f(x, y)dxdy = Lt \sum_{i=1}^{n} \sum_{k=1}^{m} f(\xi_{ik}, \eta_{ik})(x_i - x_{i-1})(y_k - y_{k-1})$$

$$= Lt \sum_{i=1}^{n} \sum_{k=1}^{m} \phi(\xi_{ik})\psi(\eta_{ik})(x_i - x_{i-1})(y_k - y_{k-1})$$

... (1)

Let S_r denotes the sum of all the rectangles between the lines $x = x_{i-1}$ and $x = x_i$, then

$$S_r = (x_i - x_{i-1})[\phi(\xi_{i1})\psi(\eta_{i1})(y_1 - y_0) + \phi(\xi_{i2})\psi(\eta_{i2})(y_2 - y_1)$$
$$\cdots + \phi(\xi_{ik})\psi(\eta_{ik})(y_k - y_{k-1}) + \cdots]$$

Let $\xi_{i1} = \xi_{i2} = \cdots = \xi_{ik} = x_{i-1}$ (constant),

Then $S_r = (x_i - x_{i-1})\phi(x_{i-1})[\psi(\eta_{i1})(y_1 - y_0)$

$$+ \psi(\eta_{i2})(y_2 - y_1) + \cdots \quad \ldots(1)$$

We can choose $\eta_{i1}, \eta_{i2}, \eta_{i3} \cdots$ in such away that in the limit

$$\psi(\eta_{i1})(y_1 - y_0) + \psi(\eta_{i2})(y_2 - y_1) + \cdots$$
$$= \int_{y_0}^{Y} \psi(y)dy.$$

So from (1)

$$S_r = (x_i - x_{i-1})\phi(x_{i-1}) \int_{y_0}^{Y} \psi(y)dy.$$

Now considering all such sums of rectangles between the parallel lines $(x = x_0,\ x = x_1)$, $(x = x_1,\ x = x_2)$ etc. and adding we get the total area of the rectangles.

So from (1) we get

$$\iint_R f(x, y)dxdy = \int_{y_0}^{Y} \psi(y)dy \cdot \Big[Lt\Big\{(x_1 - x_0)\phi(x_0)$$
$$+ (x_2 - x_1)\phi(x_1) + \cdots + (x_i - x_{i-1})\phi(x_{i-1}) + \cdots\Big\}\Big]$$
$$= \int_{y_0}^{Y} \psi(y)dy \cdot \int_{x_0}^{X} \phi(x)dx$$

Hence, replacing x_0, X, y_0 and Y by a, b, c and d respectively we get

$$\iint_R f(x, y)dx = \int_a^b \phi(x)dx \int_c^d \psi(y)dy.$$

7.4. Triple Integrals or Volume Integrals.

Let us consider a function $f(x, y, z)$ which is bounded and defined in a three dimensional region $V[a \leq x \leq b, c \leq y \leq d, e \leq z \leq f]$

Let the intervals (a, b), (c, d), (e, f) be divided into p, q, r parts respectively by the points

$$a=x_0, x_1, x_2, \cdots, x_{i-1}, x_i, \cdots x_p=b$$

$$c=y_0, y_1, y_2, \cdots, y_{i-1}, y_i, \cdots y_q=d$$

$$e=z_0, z_1, z_2, \cdots, z_{i-1}, z_i, \cdots z_r=f$$

and consider the ijkth volume denoted by V_{ijk} where i varies from 1 to p, j varies from 1 to q and k varies from 1 to r respectively.

Then $V_{ijk}=(x_i-x_{i-1})(y_j-y_{j-1})(z_k-z_{k-1})$

Let m_{ijk} and M_{ijk} be the lower and upper bound of $f(x, y, z)$ in V_{ijk} and construct the upper and lower sums S and s as,

$$S=\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r}(M_{ijk}V_{ijk})$$

$$s=\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r}(m_{ijk}V_{ijk})$$

If (x, y, z) be any point in V_{ijk}, then evidently

$$m_{ijk}\leqslant f(x, y, z)\leqslant M_{ijk}$$

or, $$m_{ijk}V_{ijk}\leqslant f(x, y, z)V_{ijk}\leqslant M_{ijk}V_{ijk}$$

or, $$\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r}m_{ijk}V_{ijk}\leqslant\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r}f(x, y, z)V_{ijk}$$

$$\leqslant\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r}M_{ijk}V_{ijk}$$

or, $$s\leqslant\sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{r=1}^{r}f(x, y, z)V_{ijk}\leqslant S$$

In the limit if S and s tend to the same limit, then $f(x, y, z)$ is said to be integrable over V and is usually written as

$$\iiint_V f(x, y, z)dxdydz.$$

So we write

$$\iiint_V f(x, y, z)dxdydz = Lt \sum_{i=1}^{p} \sum_{j=1}^{q} \sum_{k=1}^{r} f(x, y, z)V_{ijk}.$$

Also as in the case of double integral we can write

$$\iiint_V f(x, y, z)dxdydz = \int_e^f dz \int_c^d dy \int_a^b f(x, y, z)dx$$

$$= \int_c^d dy \int_e^f dz \int_a^b f(x, y, z)\,dx$$

$$= \int_c^d dy \int_a^b f(x, y, z)dx \int_e^f dz.$$

i e., the order of integration may be interchanged in any maner we like.

7.5. Change of Variables by Jacobians.

Let the variables $x_1, x_2, \cdots x_n$ be functions of $u_1, u_2, \cdots u_n$. Then the jacobians of these variables with respect to $u_1, u_2, \cdots u_n$ is the determinant

$$\begin{vmatrix} \dfrac{\partial x_1}{\partial u_1} & \dfrac{\partial x_2}{\partial u_1} \cdots \dfrac{\partial x_n}{\partial u_1} \\ \dfrac{\partial x_1}{\partial u_2} & \dfrac{\partial x_2}{\partial u_2} \cdots \dfrac{\partial x_n}{\partial u_2} \\ \cdots\cdots & \cdots\cdots\cdots \\ \dfrac{\partial x_1}{\partial u_n} & \dfrac{\partial x_2}{\partial u_n} \cdots \dfrac{\partial x_n}{\partial u_n} \end{vmatrix}$$

This is shortly denoted by J or by $\frac{\partial(x_1, x_2, \cdots x_n)}{\partial(u_1, u_2, \cdots u_n)}$

Thus if $x=f(u_1, u_2)$ $y=\phi(u_1, u_2)$

Then $J=\frac{\partial(x, y)}{\partial(u_1, u_2)}=\begin{vmatrix} \frac{\partial x}{\partial u_1} & \frac{\partial y}{\partial u_1} \\ \frac{\partial x}{\partial u_2} & \frac{\partial y}{\partial u_2} \end{vmatrix}$

Also if $x=f(u_1, u_2, u_3)$, $y=\phi(u_1, u_2, u_3)$, $z=\psi(u_1, u_2, u_3)$

Then $J=\frac{\partial(x, y, z)}{\partial(u_1, u_2, u_3)}=\begin{vmatrix} \frac{\partial x}{\partial u_1} & \frac{\partial y}{\partial u_1} & \frac{\partial z}{\partial u_1} \\ \frac{\partial x}{\partial u_2} & \frac{\partial y}{\partial u_2} & \frac{\partial z}{\partial u_2} \\ \frac{\partial x}{\partial u_3} & \frac{\partial y}{\partial u_3} & \frac{\partial z_3}{\partial u_3} \end{vmatrix}$

Suppose, we now require to transform the double integral $\iint_A f(x, y)dxdy$ to the variables u and v by the substitution $x=\phi(u, v)$, $y=\psi(u, v)$. Then it can be shown

$$\iint_A f(x, y)dxdy = \iint_{A'} f\{\phi(u, v), \psi(u, v)\} \left| \frac{\partial(x, y)}{\partial(u, v)} \right| dudv.$$

where A' is the region of the transformed integral in the (u, v) plane corresponding to the region A in the (x, y) plane.

Similarly, if $x=f(\xi, \eta, \zeta)$, $y=\phi(\xi, \eta, \zeta)$, $z=\psi(\xi, \eta, \zeta)$

Then $\iiint_V F(x, y, z)dxdydz$

$$=\iiint_{V'} F\{f(\xi, \eta, \zeta)\phi(\xi, \eta, \zeta)\psi(\xi, \eta, \zeta)\} \left| \frac{\partial(x, y, z)}{\partial(\xi, \eta, \zeta)} \right| d\xi d\eta d\zeta.$$

where V' is the region of the transformed integral corresponding to the original region V.

Illustrative Examples :

Ex. 1. **Evaluate the integral**

$$\iint \mathbf{x^2 y^2 dxdy} \text{ on } \mathbf{x^2+y^2 \leqslant 1.}$$

Here $f(x, y)=x^2y^2$.

Let $x=r\cos\theta$, $y=r\sin\theta$, so that $x^2+y^2=r^2$

Now $\iint x^2y^2dxdy$ on $x^2+y^2\leqslant 1$

$$=\iint\limits_{r\leqslant 1} f(r\cos\theta, r\sin\theta)\,\frac{\partial(x, y)}{\partial(r, \theta)}drd\theta$$

$$=\iint\limits_{r\leqslant 1} r^2\cos^2\theta \,.\, r^2\sin^2\theta \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial\theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial\theta} \end{vmatrix} drd\theta.$$

$$=\iint\limits_{r\leqslant 1} r^4(\sin\theta\cos\theta)^2 \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} drd\theta$$

$$=\iint\limits_{r\leqslant 1} r^4(\sin\theta\cos\theta)^2 .\, rdrd\theta.$$

$$=\int_0^1 r^5dr\int_0^{2\pi}(\sin\theta\cos\theta)^2\,d\theta.$$

$$=\frac{1}{6}\cdot\frac{1}{4}\int_0^{2\pi}\sin^2 2\theta\,d\theta$$

$$=\frac{1}{48}\int_0^{2\pi} 2\sin^2 2\theta\,d\theta$$

$$=\frac{1}{48}\int_0^{2\pi}(1-\cos 4\theta)d\theta$$

$$=\frac{1}{48}\left[\theta-\frac{\sin 4\theta}{4}\right]_0^{2\pi}=\frac{\pi}{24}.$$

Ex. 2. Evaluate $\iint\limits_R x^2y\ dxdy$ over the positive quadrant of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (C. H. 1971)

Let $x=au$, $y=bv$, then R is transformed into R' where R' is the circle $u^2+v^2=1$.

$$\therefore \iint\limits_R x^2ydxdy=\iint a^2u^2bv\frac{\partial(x,y)}{\partial(u,v)}dudv \text{ on } u^2+v^2=1$$

$$=a^2b\iint u^2v\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}dudv$$

$$=a^2b\iint u^2v\begin{vmatrix}a & 0\\ 0 & b\end{vmatrix}dudv$$

$$=a^3b^2\iint u^2vdudv \quad \text{on} \quad u^2+v^2=1$$

[Put $u=r\cos\theta$, $v=r\sin\theta$]

$$=a^3b^2\iint r^2\cos^2\theta.r\sin\theta\begin{vmatrix}\frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta}\\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial \theta}\end{vmatrix}drd\theta \text{ on } r=1$$

$$=a^3b^2\iint r^3\sin\theta\cos^2\theta\begin{vmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{vmatrix}drd\theta$$

$$=a^3b^2\iint r^4\sin\theta\cos^2\theta\ drd\theta$$

$$=a^3b^2\int_0^1 r^4\,dr\int_0^{\pi/2}\sin\theta(1-\sin^2\theta)d\theta$$

$$=\frac{1}{3}a^3b^2\int_0^1 r^4dr=\frac{1}{15}a^3b^2.$$

Ex. 3. Evaluate $\iint\limits_R \left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)dxdy$ over the first quadrant of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. **(C. H. 1963, 72)**

Let $x=au$, $y=bv$, so that the region R is transformed into R' where R' is the circle $u^2+v^2=1$.

Then $\iint\left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)dxdy$ on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$$=\iint(1-u^2-v^2)\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} dudv, \text{ on } u^2+v^2=1$$

$$=\iint(1-u^2-v^2)\begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} dudv$$

$$=ab\iint(1-u^2-v^2)du\,dv$$

[Put $u=r\cos\theta$, $v=r\sin\theta$]

$$=ab\iint(1-r^2)\begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial \theta} \end{vmatrix} drd\theta, \text{ on } r=1$$

$$=ab\iint(1-r^2)\begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} dr.d\theta$$

$$=ab\iint r(1-r^2)drd\theta$$

$$=ab\int_0^1 r(1-r^2)dr\int_0^{\pi/2} d\theta$$

$$=\frac{\pi ab}{2}\int_0^1 (r-r^3)dr$$

$$=\frac{\pi ab}{8}.$$

Ex. 4. Evaluate $\iint \sqrt{\dfrac{a^2b^2 - b^2x^2 - a^2y^2}{a^2b^2 + b^2x^2 + a^2y^2}}\,dxdy$, **the area of integration being the positive quadrant of the ellipse** $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

(C. H. 1961, 68)

Let $x = au$, $y = bv$ so that the new region becomes the circle $u^2 + v^2 = 1$.

$$\therefore \iint \sqrt{\frac{a^2b^2 - b^2x^2 - a^2y^2}{a^2b^2 + b^2x^2 + a^2y^2}}\,dxdy$$

$$= \iint \sqrt{\frac{1 - u^2 - v^2}{1 + u^2 + v^2}} \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} du\,dv \text{ on } u^2 + v^2 = 1$$

$$= \iint \sqrt{\frac{1 - u^2 - v^2}{1 + u^2 + v^2}} \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} du\,dv$$

[Put $u = r\cos\theta$, $v = r\sin\theta$]

$$= ab \iint \sqrt{\frac{1 - r^2}{1 + r^2}} \begin{vmatrix} \dfrac{\partial u}{\partial r} & \dfrac{\partial u}{\partial \theta} \\ \dfrac{\partial v}{\partial r} & \dfrac{\partial v}{\partial \theta} \end{vmatrix} du\,d\theta, \text{ on } r = 1$$

$$= ab \iint \sqrt{\frac{1 - r^2}{1 + r^2}}\, r\,dr\,d\theta, \text{ on } r = 1$$

$$= ab \int_0^{\pi/2} d\theta \int_0^1 \sqrt{\frac{1 - r^2}{1 + r^2}}\, r\,dr \text{ [On the positive quadrant]}$$

$$= \frac{\pi ab}{2.2} \int_0^{\pi/2} \sqrt{\frac{1 - \sin z}{1 + \sin z}} \cos z\,dz \text{ putting } r^2 = \sin z.$$

$$= \frac{\pi ab}{4} \int_0^{\pi/2} (1 - \sin z)dz$$

$$= \frac{\pi ab}{8}(\pi - 2).$$

Ex. 5. Evaluate the integral $\iint \sqrt{4a^2-x^2-y^2}\,dx\,dy$, taken over the upper half of the circle $x^2+y^2-2ax=0$.

(*C. H. 1966*)

Let $x=r\cos\theta$, $y=r\sin\theta$, so that the new region becomes the circle $r=2a\cos\theta$.

$$\therefore \iint \sqrt{4a^2-x^2-y^2}\,dx\,dy$$

$$=\iint \sqrt{4a^2-r^2}\begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta}\end{vmatrix} dr\,d\theta$$

$$=\iint \sqrt{4a^2-r^2}\; r\,dr\,d\theta, \text{ on } r=2a\cos\theta$$

$$=\int_0^{\pi/2} d\theta \int_0^{2a\cos\theta} \sqrt{4a^2-r^2}\; r\,dr$$

$$=-\frac{1}{2}\int_0^{\pi/2} d\theta \int_{4a^2}^{4a^2\sin^2\theta} \sqrt{z}\,dz \qquad \left[\text{Putting } 4a^2-r^2=z \text{ so that } -2r\,dr=dz.\right]$$

$$=-\frac{1}{2}\int_0^{\pi/2} d\theta\,.\,\frac{2}{3}\Big[z^{\frac{3}{2}}\Big]_{4a^2}^{4a^2\sin^2\theta}$$

$$=-\frac{1.2}{2.3}\int_0^{\pi/2} d\theta.(8a^3\sin^3\theta-8a^3)$$

$$=\frac{8a^3}{3}\int_0^{\pi/2}(1-\sin^3\theta)d\theta=\frac{4a^3}{9}(3\pi-4).$$

Ex. 6. Evaluate the integral $\iint \sqrt{4x^2-y^2}\,dx\,dy$ extended over the triangle formed by the straight lines $y=0$, $x=1$, $y=x$.

(*C. H. 1967, 70*)

Draw the triangle OAB formed by $y=0$, $x=1$ and $y=x$, so that $OB=r$, and $\angle BOA=\theta$. Then $r\cos\theta=1$

$$\therefore \quad r=\frac{1}{\cos\theta}.$$

Now putting $x=r\cos\theta$, $y=r\sin\theta$

$$\iint \sqrt{4x^2-y^2}\,dx\,dy$$

$$=\iint \sqrt{4r^2\cos^2\theta-r^2\sin^2\theta}\; r\,dr\,d\theta \text{ on } r=\frac{1}{\cos\theta}.$$

$$=\int_0^{\pi/4}\int_0^{1/\cos\theta} \sqrt{4\cos^2\theta-\sin^2\theta}\,.\,r^2\,dr\,d\theta$$

$$=\int_0^{\pi/4} \sqrt{4\cos^2\theta-\sin^2\theta}\left[\frac{r^3}{3}\right]_0^{1/\cos\theta} d\theta.$$

$$=\frac{1}{3}\int_0^{\pi/4} \frac{\sqrt{4\cos^2\theta-\sin^2\theta}}{\cos^3\theta}\,d\theta.$$

$$=\frac{1}{3}\int_0^{\pi/4} \sqrt{4-\tan^2\theta}\,\sec^2\theta\,d\theta$$

$$=\frac{1}{3}\int_0^{1} \sqrt{2^2-z^2}\,dz \quad \text{putting } \tan\theta=z$$

$$=\frac{1}{3}\left[\frac{z\sqrt{2^2-z}}{2}+\frac{2^2}{2}\sin^{-1}\frac{z}{2}\right]_0^1=\frac{1}{18}(2\pi+3\sqrt{3}).$$

Ex. 7. Evaluate $\iiint xyz\,dx\,dy\,dz$, the volume of integration being the positive octant of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\leqslant 1$.

Let $x=au$, $y=bv$ and $z=cw$ so that the transferred field of integration becomes the sphere $u^2+v^2+w^2 \leqslant 1$.

$$\therefore \quad \iiint xyz\,dx\,dy\,dz = \iiint abc\,uvw \frac{\partial(x,y,z)}{\partial(u,v,w)} du\,dv\,dw.$$

$$= abc \iiint uvw \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix} du.dv.dw$$

$$= abc \iiint uvw \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} du\,dv\,dw$$

$$= a^2 b^2 c^2 \iiint uvw\,du\,dv\,dw.$$

$$= a^2 b^2 c^2 \iint uv\,du\,dv \int_0^{\sqrt{1-u^2-v^2}} w\,dw$$

$$= \frac{a^2 b^2 c^2}{2} \iint uv(1-u^2-v^2)du\,dv$$

$$= \frac{a^2 b^2 c^2}{2} \iint r^2 \sin\theta \cos\theta (1-r^2) r\,dr\,d\theta$$

[transferring to polar]

$$= \frac{1}{2} a^2 b^2 c^2 \int_0^{\pi/2} \sin\theta \cos\theta \; d\theta \int_0^1 (r^3 - r^5)\,dr$$

$$= \frac{1}{24} a^2 b^2 c^2 \int_0^{\pi/2} \sin\theta \cos\theta \, d\theta$$

$$= \frac{1}{48} a^2 b^2 c^2 \int_0^{\pi/2} \sin \; 2\theta \, d\theta = \frac{1}{48} a^2 b^2 c^2.$$

Ex. 8. Evaluate $\iiint \dfrac{dx\,dy\,dz}{\sqrt{1-x^2-y^2-z^2}}$.

The field of integration being the positive octant of the sphere $x^2+y^2+z^2=1$.

Let $x=r\sin\theta\cos\phi$, $y=r\sin\theta\sin\phi$ and $z=r\cos\theta$. Then in the transformed field r varies from 0 to 1, θ and ϕ varies from 0 to $\pi/2$.

$$\therefore \iiint \frac{dx\,dy\,dz}{\sqrt{1-x^2-y^2-z^2}}$$

$$=\iiint \frac{1}{\sqrt{1-r^2}} \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}\,dr\,d\theta\,d\phi$$

$$=\iiint \frac{1}{\sqrt{1-r^2}} \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} & \frac{\partial z}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \end{vmatrix} dr\,d\theta\,d\phi$$

$$=\iiint \frac{1}{\sqrt{1-r^2}} \begin{vmatrix} \sin\theta\cos\phi, & \sin\theta\sin\phi, & \cos\theta \\ r\cos\theta\cos\phi, & r\cos\theta\sin\phi, & -r\sin\theta \\ -r\sin\theta\sin\phi, & r\sin\theta\cos\phi, & 0 \end{vmatrix} \begin{matrix} dr. \\ d\theta. \\ d\phi. \end{matrix}$$

$$=\iiint \frac{r^2\sin\theta}{\sqrt{1-r^2}}\,dr\,d\theta\,d\phi$$

$$=\int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin\theta\,d\theta \int_0^1 \frac{r^2\,dr}{\sqrt{1-r^2}}=\frac{\pi^2}{8}$$

Ex. 9. Evaluate $\iiint (x^2+y^2+z^2)xyz\,dx\,dy\,dz$ taken throughout the sphere $x^2+y^2+z^2 \leq 1$. (C. H. 1964)

Let $x=r\sin\theta\cos\phi$, $y=r\sin\theta\sin\phi$ and $z=r\cos\theta$ so that in the transformed field r varies from 0 to 1, θ and ϕ varies from 0 to 2π.

$$\therefore \iiint (x^2+y^2+z^2)xyz\, dx\, dy\, dz$$

$$=\iiint (r^2 \sin^2\theta \cos^2\phi + r^2 \sin^2\theta \sin^2\phi + r^2\cos^2\theta)$$

$$\times r^3 \sin^2\theta \cos\theta \sin\phi \cos\phi \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} dr.d\theta.d\phi$$

$$=\iiint r^2.r^3 \sin^2\theta \cos\theta \sin\phi \cos\phi.r^2 \sin\theta\, dr\, d\theta\, d\phi.$$

$$=\int_0^1 r^7\, dr \int_0^{2\pi} \sin^3\theta \cos\theta\, d\theta \int_0^{2\pi} \sin\phi \cos\phi\, d\phi.$$

$$=0 \quad \because \int_0^{2\pi} \sin^3\theta \cos\theta\, d\theta = 0.$$

Exercise 7

1. Evaluate $\iint x^2y^2\, dx\, dy.$

The region of integration being $x \geqslant 0$, $y \geqslant 0$ and $x^2+y^2 \leqslant 1$. Ans. $\pi/96$.

2. Evaluate $\iint \frac{dx\, dy}{(1+x^2+y^2)^2}$.

over a loop of the lemniscate $(x^2+y^2)^2-(x^2-y^2)=0$ (C. H. 1962)

3. Perfrom the integration

$$\iint \sqrt{2(ax+by)-(x^2+y^2)}\, dx\, dy.$$

taken over the circle

$$x^2+y^2-2(ax+by)=0. \qquad \text{Ans. } \frac{2}{3}\left(a^2+b^2\right)^{\frac{3}{2}}\pi.$$

4. Evaluate $\iint (1+x^2+y^2)^{-2} dx\, dy$,

taken over a triangle whose vertices are (0, 0), (2, 0) and $(1, \sqrt{3})$

Ans. $\frac{\sqrt{3}}{2} \tan^{-1}\frac{1}{2}$.

(C. H. 1962)

5. Evaluate

$$\iint \sqrt{\frac{36-9x^2-4y^2}{36+9x^2+4y^2}}\, dx\, dy$$

taken over the positive quadrant of the ellipse

$$\frac{x^2}{2^2}+\frac{y^2}{3^2}=1.$$

Ans. $\frac{9\pi}{2}(\pi-2)$

6. Evaluate $\iint (2a^2-2ax-2ay-x^2+y^2)dx\ dy$, the region of integration being the circle

$$x^2+y^2+2ax+2ay-2a^2=0$$

(C. H. 1962, 63)

7. Evaluate $\iint (x^2+y^2)\ dxdy$ over the region enclosed by the triangle having the vertices (0, 0), (1, 0), (1, 1). (C. H. 1965)

8. Evaluate $\iiint z^2 dxdydz$ extended over the hemisphere $z \geqslant 0$, $x^2+y^2+z^2 \leqslant a^2$. Ans. $\frac{2}{15}\pi a$. (C. H. 1964)

9. Prove that $\iiint (lx+my+nz)^2 dxdydz = \frac{4}{15}\pi(l^2+m^2+n^2)$ taken throughout the sphere $x^2+y^2+z^2=1$.

10. Evaluate $\iiint \frac{dxdydz}{x^2+y^2+(z-2)^2}$ extended over the sphere $x^2+y^2+z^2 \leqslant 1$. Ans. $\pi(2-\frac{3}{2}\log 3)$,